

PURE contactless reader Specifications for PURE Dual-Interface cards and Mobile PURE

Version: 2.1.8

08/06/2016

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Revision status

Revision	Date	Description
1.0	20/11/2012	Initial version dedicated to dual-interface cards
2.0	22/03/2012	Version supporting PURE dual-interface cards and mobile PURE.
2.0.1	27/03/2013	Typographic mistakes
2.1	21/10/2013	• Typographic mistakes • Reduce the number of implementation options (the following functions (long tap, 2nd tap and Confirmation Code support are now mandatory). The former terminal Vendor implementation options are now Acquirer options. • Support of a new mandatory function allowing to recover a torn transaction during GENERATE AC command processing. This function is dedicated to the management of the communication problem with the contactless payment device. For each command sent to the contactless device, the document specifies the consequence of the detection of a communication problem during the processing of the command • Support of a new optional type of transaction (Application Authentication transaction). This function is dedicated to transit industry. It will allow the terminal to authenticate a contactless device. The offline authentication method used is DDA. Its implementation is optional. • Definition of a new CVM code allowing to support Confirmation Code checking with the CVM List method • Support of a two new proprietary data elements allowing the exchange of information between the card and the terminal (Terminal Dedicated Data and Terminal Transaction Data) • Other technical updates: ○ Classification of transaction types in two category: financial transaction types and non-financial transaction types ○ Parameters (limits and flags) used during Preliminary Transaction Processing: the Acquirer should be able to define one set of values for each couple {AID, transaction type}. ○ New information stored inside Previous Kernel Transaction Context' ○ TTPI structure slightly updated ○ The list of data to retrieve using GET DATA (GDDOL) is included inside a template: Memory Slot Read Template – BF71. In the future, this template may contain complementary data ○ The kernel outcome message "End Application with Start different from N/A" used for "Confirmation Code required" and "communication error", has to be processed at reader level and not forwarded to the terminal application. It will simplify the terminal application.
2.1.1	16/12/2013	• Table 19: Add Issuer Authentication Data to the list of data present in Previous Kernel Transaction Context • Improvement of the management of errors in section 9.2.2 (after an online authorization, the kernel shall not request to select the next available application, in case of error. it must send End Application outcome message for requesting the terminal to apply Issuer host decision communicated in Response Code • Clarifications (C1.7, C3.3, R3.2/5/6/7, K5.4, K21.2/3/4/5/11, Table37)
2.1.2	05/02/2014	• Several clarifications ○ All references to Payment Application are related to card Payment application ○ miscellaneous other clarifications ○ Reader and kernel have only to transport the Terminal Transaction Data and Terminal Dedicated Data without analysis their contents

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Revision	Date	Description
		○ The section 7 is renamed in "Terminal Application Requirements related to Reader Interface". This section is outside the scope of the reader/kernel certification. ○ In section 8 related to the Reader, add a sub-section referring to the section 7.5 that is related to the interface terminal/reader. Reader certification is based on section 8. ○ The section 9.1 provides an overview of the PURE kernel. The PURE contactless kernel must comply with the detailed descriptions and the reference requirements that are provided in the other sub-sections of section 9. ○ Clarify the contents of sections 8 and section 9 • Main updates ○ AID partial match is required for all AIDs supported ○ Terminal pre-processing limits may be managed either at terminal application level or at reader level ○ Amount Other is an optional data element. It is the responsibility of the terminal application to provide an Amount Other value in case of purchase with cashback. No check is performed by the reader and by the terminal. ○ Transaction Date may not be part of the information communicated by the terminal if it is managed by the reader. ○ An AID value is considered as wrong not only in case of length strictly less than 5 bytes but also if the length is greater than 16 ○ Introduction of the concept of "Kernel Transaction Database" used for storage of information present in the response of the command sent by the kernel and the information available in the FCI communicated by the kernel. ○ TVR bit 'Transactions exceeds floor limit' is no more set to '1' in case of "cash transaction" and "cashback transaction" (K2.4). ○ After an online authorization process r in case of transaction recovery, the kernel has to compare the FCI value communicated by the reader with the FCI value memorized in the 'Kernel Transaction Database' ○ Support of TVR bit 'SDA performed' ○ During offline CAM checking, PAN value may be retrieved in "Track2 Equivalent Data" data element when PAN data element is not available. ○ During SDA or CDA checking, TVR shall not be updated in case of failure. ○ Definition of minimum length to support for variable length data elements, outcome messages and dynamic transaction parameters ○ Decision to print a receipt is no more control by the kernel. It is only a terminal application decision.
2.1.3	15/03/2014	• Kernel Clarifications: ○ Clarification of the management of the 'Kernel Transaction Database'. An implementation may support a list of referenced data elements but it is an implementation option outside the scope of the kernel certification. When a list of referenced data elements is supported, it is allowed to store in the 'Kernel Transaction Database' only the data elements referenced, the other data elements are ignored and will be considered as "unknown" if they are requested in an Object List. All the data elements referenced in this document as either mandatory or optional, shall be referenced in the list used for kernel certification to avoid kernel certification failure. ○ The contactless reader may provide additional transaction parameters when activating the kernel. These parameters may be used by the kernel if requested in an object list (depending the implementation it may be required to add them to a referenced list of parameters supported). It was not clear in previous version. • Kernel Updates: ○ All Contactless Pre-Processing Indicators must be forced to zero in the case of the Application Authentication Transaction

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Revision	Date	Description
		○ TVR bit 'SDA was performed' must be checked when SDA is the offline CAM selected; Not only in case of successful SDA processing ○ TVR bit 'Offline CAM was not performed' must be set to '0' when an offline CAM is selected independently the result ○ TVR bit 'Online PIN Entered' must be set to 1, each time Online PIN is the method selected (independently if it is a card decision or a terminal decision)
2.1.4	05/06/2014	• Reader Updates: ○ Add a note specifying that Payment Scheme may request an update of the application selection algorithm, for example, for selecting an application based on Merchant-defined priority (not Issuer-defined priority). Definition of the proprietary algorithm is outside the scope of this specification ○ New implementation option (n°5): the terminal application may specify the AID (or part of the AID) to use for processing a single transaction [option dedicated to the Contactless Reader] • Kernel Updates: ○ Any problem on the data elements retrieved in card command response (mandatory data element missing or duplicated data element) result in sending the "Transaction completed on Error" outcome message ○ Add Default DDOL as mandatory kernel parameter when implementation option 4 is supported. Not required in the other cases. • Mistakes correction ○ Length of the data sent to ECHO command is depending on P1 parameter value ○ In K2.3, when transaction type is '78', one of the mandatory data is not 9F70 but BF71. ○ In K3.7, the value of the 6 least significant bits of byte 'CVM Performed' for 'Confirmation Code Verified' is '24' not '21' ○ In K7.4, on the level 5, test on "If the transaction type is '79'" and the action if the test is positive must be removed. It is cover by the block level 3 above. • Clarification ○ In K6.4, GPO command supports only Format 2 ○ Read application data (K8.4 and K8.5) ○ The Acquirer must define consistent values for Terminal Type, Terminal Capabilities and Contactless Kernel Capabilities. Reader and kernel are not required to checks that these data elements are setup with consistent values ○ ATDTOL empty value definition ○ DAC value and IDN value management (9.2.17) ○ Tag value used in GDDOL, ATOL, MTOL, ATDTOL, Memory Slot Update Entry ○ Support of an exception file is not requested and is outside the scope of these specifications
2.1.5	08/08/2014	• Kernel Updates: ○ Clarification and update of the valid 1st GENERATE AC command responses. • Clarification ○ Only format 2 is supported for GENERATE AC command
2.1.6	22/09/2014	• Reader updates ○ In R6.5: Update of the management of implementation option 5 • Mistakes correction ○ In K2.5: Only presence of « Authorization Response Code" in Dynamic Transaction Parameters allows the kernel to identify a kernel call for processing an online authorization response. ○ In K3.5: clarify the consequence of a wrong 2nd GEN AC response format (same for ECHO due to 2nd GEN AC torn) ○ K14.3: Add check to perform on ECHO command response in the case of 1st GEN AC torn.

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Revision	Date	Description
		• Clarification ○ In K2.4: "error other than communication problem" means SW≠ 9000. ○ Section 3.1.1 - possible terminal architectures ○ Add a footnote in K3.4 specifying that CDA is not supported with the 2nd GENERATE AC command as it is defined in section 12.2.5.2. • Other ○ Clarify CCID presence in GET PROCESSING OPTIONS command and in 1st GENERATE AC command in section 3.2.3.
2.1.7	16/02/2015	• Reader updates ○ none • Mistakes correction ○ Add requirement K20.6 removed by mistake in version 1.2.3 and upper • Clarification ○ In K2.4: "error other than communication problem" means SW≠ 9000. ○ In K13.3, clarify the Terminal Action Analysis ○ R8.2/ K2.3 / K20.1 to K20.9: The Unpredictable Number may be generated either by the reader layer or by the PURE contactless kernel ○ K2.3: Add the following note to the data element "Transaction amount - 9F02" and data element "Transaction Type – 9C": When the kernel design mandates the presence of these data elements among the Dynamic Transaction Parameters, checking their presence is not relevant. ○ K2.3: Add the following note to the data element "Unpredictable Number – 9F37": When the kernel design mandates the presence of the Unpredictable Number among the Dynamic Transaction Parameters or when the Unpredictable Number is generated by the kernel, checking its presence is not relevant. ○ K2.3: Add the following note: When the Unpredictable Number is generated by the kernel, new unpredictable number must be generated using a method equivalent to the method used in EMV contact transaction. This value is allocated to the parameter: Unpredictable Number (tag '9F37') • Other ○ None
2.1.8	26/05/2015 08/06/2016	The following Acquirer options become Implementation options • Support of long tap for Issuer Response communication • Support of additional tap for Issuer Response communication • Support of torn transaction recovery using ECHO command The Contactless Application/Kernel Capabilities bits associated to these options becomes RFU (value '0') when the new associated implementation option is not supported. The sections updated are: 1, 3.2.1, 3.2.2, 4, 6,1, 11.2.10 and 11.2.11. Editorial adaptions. Clarification added in footnote for K2.4.

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1 Introduction

EMVco has developed a terminal design for contactless payment with the goal of realizing acceptance of contactless devices compliant with different implementations. The set of documents "EMV™ Contactless Specifications for Payment Systems"[1] describes this terminal design (see section 2).

The design defines a generalized POS System architecture that includes the following modules:

- Terminal Application module that is in charge of all the interfaces except the communication through the contactless interface,
- contactless reader module that performs the initial analysis of a contactless transaction and invokes appropriate kernel module, and
- Several contactless kernel modules, each of which provides processing appropriate to certain contactless transactions.

This document regroups

- a functional and technical specification for a PURE contactless reader and
- a functional and technical specification for a PURE contactless kernel supporting all the functions available in PURE dual-interface cards version 1.5.x and Mobile PURE Payments version 2.x.

The support of additional contactless kernels by the PURE contactless reader is outside of scope of this specification.

The PURE contactless reader specifications support the following PURE main functions:

Mandatory functions to support

- Contactless payment transaction
- Support of the cardholder verification method that consists to request the cardholder to use his mobile phone for entering a confirmation code (function available in mobile application compliant with Mobile PURE specification 2.x)

Optional additional functions

- Application EEPROM memory management
- Support of a new type of transaction dedicated only to authenticate offline the card payment application
- Support of the long tap function allowing to send script commands if the card payment application is not removed from the contactless interface during the online authorization process (function available in cards compliant with PURE specification 1.5.x)
- Support of the 2nd tap function allowing to communicate Issuer Authorization response to the contactless device (mobile phone) by requesting the cardholder to present again his contactless device (function available in PURE specification 2.x)
- Support of the ECHO command allowing to repeat the last card decision in case of torn transaction

[1] EMV™ is a trademark owned by EMVCo LLC.

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Terminal Software Providers may decide to do not support the optional functions in their implementation of the PURE contactless reader and in their implementation of the PURE contactless kernel.

The PURE contactless reader specifications are based on the EMV Entry Point specification document.

An alternative solution for Terminal Software Providers may be to develop contactless reader module compliant with the EMV Entry Point specifications capable to interface with the PURE contactless kernel. Using a contactless reader compliant with EMV Entry Point Specifications, the only functions listed above that may not be supported, is the "Application EEPROM memory management" function. When this option is selected, all requirements related to the PURE contactless reader defined in this document may be ignored. The PURE certification will only check compliance with the section 9 of this document.

2 References

The PURE contactless reader specifications include normative reference to the EMV specification. Consequently, the PURE contactless reader specifications must be read in combination with the EMV specifications.

Each reader of the PURE contactless reader specification is responsible to procure the EMV specification directly from EMVco and all additional reading that is necessary to understand or implement the PURE contactless reader specification.

Reference	Title	Version
EMV-1	EMV. Integrated Circuit Card Specifications for Payment Systems. Book 1 - Application Independent ICC to Terminal Interface Requirements,	Version 4.3. November 2011
EMV-2	EMV. Integrated Circuit Card Specifications for Payment Systems. Book 2 - Security and Key Management,	Version 4.3. November 2011
EMV-3	EMV. Integrated Circuit Card Specifications for Payment Systems. Book 3 - Application Specification	Version 4.3. November 2011
EMV-4	EMV. Integrated Circuit Card Specifications for Payment Systems. Book 4 - Cardholder, Attendant, and Acquirer Interface Requirements	Version 4.3. November 2011
EMV-5	EMV Contactless Specifications for Payment Systems Book A - Architecture and General Requirements	Version 2.3 February 2013
EMV-6	EMV Contactless Specifications for Payment Systems Book B - Entry Point Specification	Version 2.3 February 2013
EMV-7	EMV Contactless Specifications for Payment Systems Book D - EMV Contactless Communication Protocol Specification	Version 2.3 February 2013

Table 1 –Reference Documents

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3 General Overview

3.1 Terminal Architecture Overview

3.1.1 Introduction

There is currently no common EMV contactless transaction flow. Current contactless transaction flows are specific to each contactless implementation.

To facilitate the acceptance of contactless devices supporting different transaction flows, EMVco has defined a terminal design allowing the support of several contactless transaction flows.

The POS System architecture may be represented in a descriptive model composed of three functional components:

- Terminal Application software in charge of the interface with Merchant, with the cardholder and with the Acquirer network,
- Contactless reader module that initializes the contactless transaction and activates appropriate kernel software
- Contactless kernel module dedicated to a specific contactless transaction flow.

Technically, a Terminal Software Provider may implement a solution composed of the three distinct modules or may decide to regroup several functional components inside a single module. For example, regrouping the following functional components in a same module is allowed:

- the contactless reader and the contactless kernel in a single module
- the Terminal Application and the contactless reader in a single module
- the Terminal Application, the contactless reader and the contactless kernel in a single module.

Independently the software architecture supported, the Terminal Software Provider must comply with the following requirements:

S1.1	For the support of contactless payment device (contactless-only card, dual-interface card, Mobile application or any other form factor) supporting a payment application compliant with the PURE specification, it is mandatory to use a contactless kernel module compliant with these specifications
S1.2	For the support of contactless payment device (contactless-only card, dual-interface card, Mobile application or any other form factor) supporting a payment application compliant with the PURE specification, it is mandatory to use a contactless reader module compliant with either • these specifications or • the EMV Contactless Specifications for Payment Systems (reference EMV-5 and EMV-6) or • any contactless reader specifications equivalent to the above specifications and endorsed by the Payment Scheme

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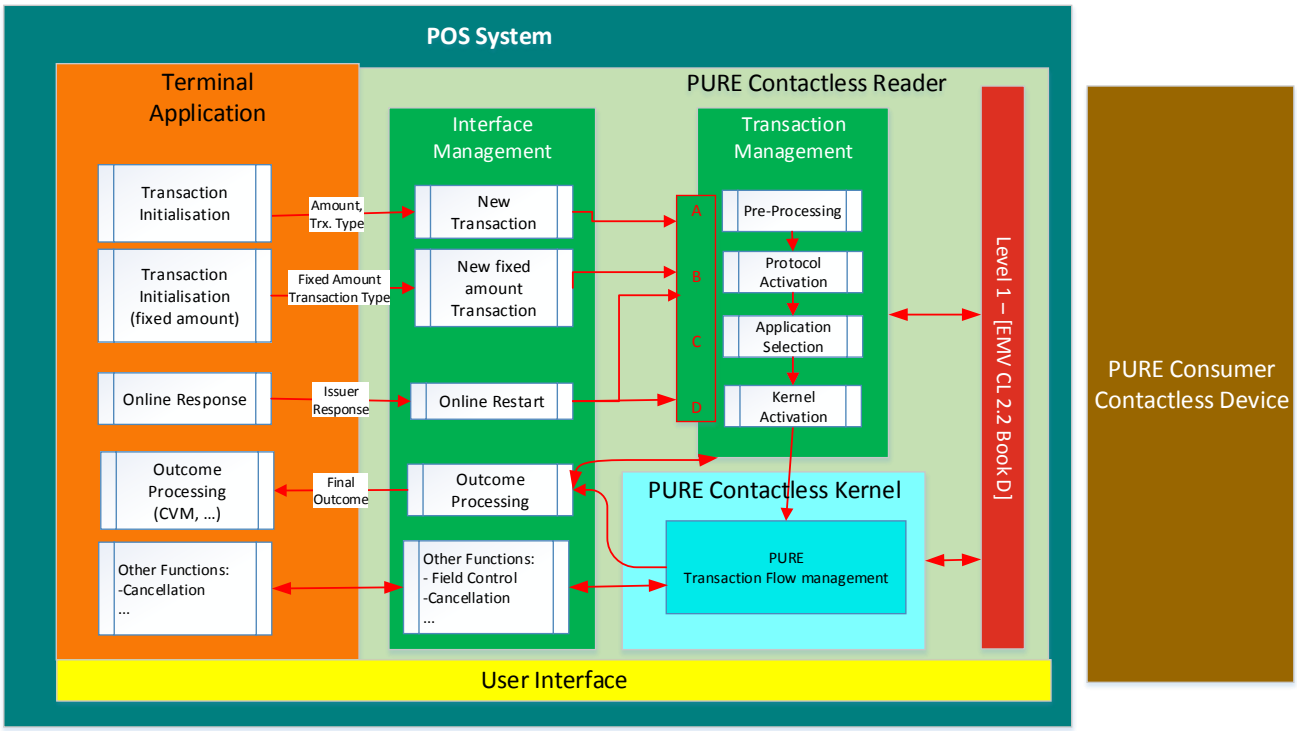


Figure 1 Logical Architecture

3.1.2 Terminal Application software Functions

The Terminal Application software is in charge of the management of the interface with the Merchant, the cardholder and the Acquiring network:

- Management of the data entry by the Merchant
- Management of cardholder verification when required (The PURE contactless reader communicates the method to use)
- Management of the magstripe transactions
- Management of the contact chip transaction
- Processing of the final outcome provided by the contactless reader, including the handling of authorization requests and responses
- Communication for authorization and clearing records
- Other aspect of transaction processing such as communication timeouts and cancellations
- Initialization of new transactions, including card removal and management of the contactless field between two transactions
- Provision of the data included in online authorization response
- Indication of unable to process the transaction online
- ..

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3.1.3 Contactless reader software Functions

The PURE contactless reader software[2] is in charge of the initialization of the contactless transaction. The initialization of the contactless transaction consists in:

- Preliminary transaction processing when terminal knows the transaction amount:

  For each AID supported, terminal checks terminal flags and compares transaction amount with terminal limits associated to the current type of transaction. Terminal flags and terminal limits are specific to each couple {AID, transaction type}. When the terminal supports several currencies, it must support a value per currency supported for each limit supported (depending on the implementation, these limits may be managed either at terminal application level or at reader level).
  - Reader Contactless Transaction Limit
    If the limit is supported and the transaction amount exceeds it, this AID will be removed from the list of applications supported by the terminal in contactless mode for the current transaction
  - Reader Contactless Floor Limit
    If the limit is supported and the transaction amount exceeds it and the AID is selected, the terminal will request to process the transaction online
  - Terminal Floor Limit
    When Reader Contactless Floor Limit is not supported for this AID but Terminal Floor Limit is supported, if the transaction amount exceeds Terminal Floor Limit value and the AID is selected, the terminal will request to process the transaction online
  - Reader CVM Required Limit
    If the limit is supported and the transaction amount exceeds it and the AID is selected, the terminal will request to perform cardholder verification
  - Status Check Support flag
    If the terminal application supports the Status Check function for this AID and the transaction amount is equal to a single unit of currency, the information will be communicated to the card Payment application if this AID is selected. The impact will depend on the card Payment application setting defined by the Issuer.
  - Zero Amount Allowed flag
    If the terminal application supports the Zero Amount check function for this AID and the transaction amount is equal to zero, the information will be communicated to the card Payment application if this AID is selected. The impact will depend on the card Payment application setting defined by the Issuer.

  For each AID supported, the result of the preliminary transaction processing is stored in a contactless reader internal parameter "AID Contactless Pre-Processing Indicators" specific to each AID.

- Initialization of the communication with the consumer contactless device (dual-interface card, mobile phone, sticker…)

[2] See EMVco EMV Contactless Specifications for Payment Systems - Book B - Entry Point Specification (please see reference in section 2)

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- Identification of the applications mutually supported by the consumer contactless device and the terminal. The terminal gets the list of the applications supported by the consumer contactless device, using the PPSE. Only the applications associated to a kernel supported by the terminal, are kept.
- Selection in the resulting list of the application with the highest priority supported by the PURE contactless kernel
- Activation of the PURE contactless kernel
- Analysis of the outcome message sent by the kernel that is the result of the transaction processing by the kernel
- Communication of the kernel outcome message to the terminal
- Management of the "no card response" timeout

Providing more details on these functions is the objective of the section 8 of this document. Terminal Software Providers preferring to develop a contactless reader compliant with the specifications defined by EMV should refer to the EMVco document 'EMV Contactless Specifications for Payment Systems – Book B - Entry Point Specification' referenced in section 2.

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3.1.4 User Interface Management

Depending of the design of the POS, the user interface may be managed either by the Terminal Application software or by the PURE contactless reader software. The user interface covers the following functions:

- Displaying messages to the Merchant
- Displaying messages to the Cardholder
- Audio indication to the cardholder

3.1.5 PURE contactless kernel Functions

The PURE contactless kernel software process contactless transactions using transaction flow described in this document. This transaction flow is compliant with the transaction flow defined in the PURE dual-interface card specifications and in the Mobile PURE Payments specifications.

The current specifications are designed for the support of a single kernel. Therefore the PURE contactless reader may activate only the PURE contactless kernel.

Terminal Software Providers may design their application for the support of several contactless kernels. It is outside the scope of this document.

When several contactless kernels are supported, the contactless reader may rely on an exclusive link between the payment product, defined by its Application Identifier (AID), and the kernel. In some cases, a single payment product may run on different kernels. Therefore, in addition to the AID, the contactless reader needs information on which kernel to activate. The reader may use an EMV-defined parameter, Kernel Identifier, to distinguish between different kernels that may be supported. The combination of AID and Kernel Identifier, both communicated in the PPSE entries, may be used to link the payment product and the kernel.

3.1.6 Communication between the POS modules

Same as in the EMV contactless specifications, these specifications are based on the assumption that each time a POS module activates another POS module, the result of the module processing is communicated to the initial module using an outcome message. The outcome message communicates result of a transaction processing and instruction on how remaining part of the transaction processing should be performed.

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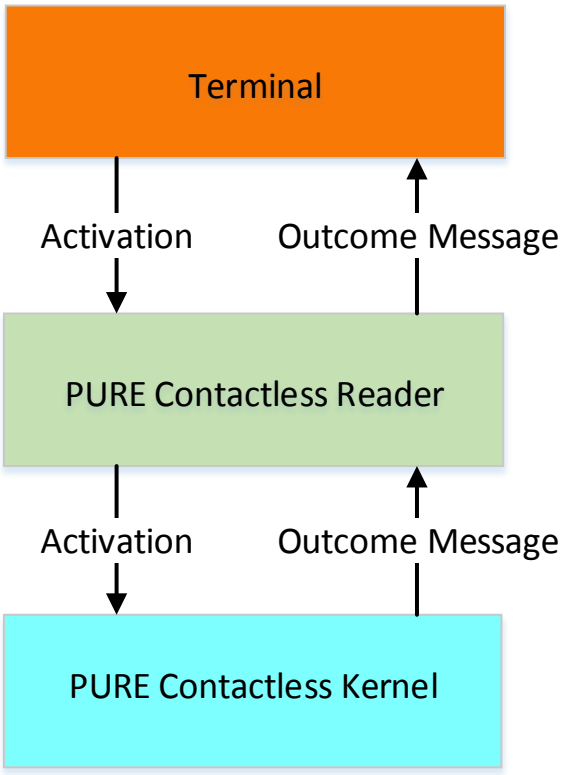


Figure 2 Communication between POS modules

The outcome messages are defined in the EMV specifications [EMV-5: Section 6].

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The PURE contactless kernel may generate the following types of outcomes messages:

- ***Approved***
  The transaction is completed. The transaction is approved.
- ***Declined***
  The transaction is completed. The transaction is declined.
- ***Online Request***
  The transaction is in process. An online authorization is requested.
- ***Try Another Interface***
  The transaction cannot be completed using the contactless interface. The POS should attempt a different interface.
- ***End Application***
  This outcome message may be used in several cases:
  - Completion of a non-financial transaction
  - Communication of specific instructions to the terminal (request Customer to validate Confirmation Code using his contactless device, apply decision communicated by the Issuer Authorization host...)
  - Transaction cancellation

  Depending on the case, the outcome message may instruct the PURE contactless reader to stop the transaction or to restart it.
- ***Select Next***
  The PURE contactless kernel has determinate that the application selected by the contactless reader is unsuitable. The contactless reader should select another card payment application.
- ***Try Again***
  A communication problem happens when processing one of the commands used during a payment transaction. The PURE contactless kernel requests that the Customer contactless device be presented again.

The contactless reader module communicates to the Terminal Application module all the outcome messages generated by the PURE contactless kernel except the ***Select Next*** outcome message, the ***End Application*** outcome message when the kernel has indicated that the transaction need to be restarted and the ***Try Again*** outcome message that are directly processed by the contactless reader module.

In some circumstances, the PURE contactless reader module may generate the ***Try Another Interface*** outcome message dedicated to the Terminal Application without activating the kernel.

Each outcome message supports the following mandatory parameters as described in EMV-5: Section 6]:

- Start
- Online Response Data
- CVM
- UI Request on Outcome Present
- UI Request on Restart Present

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- Data Record Present
- Discretionary Data Present
- Alternate Interface Preference
- Receipt
- Field Off Request
- Removal Timeout

All parameters must be present. Depending on the type of the outcome messages, some parameters are not relevant and should not be checked when the outcome message is processed.

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In the case of PURE contactless kernel, the outcome message parameters that the Terminal Application module (T) and the PURE contactless reader module (R) have to process are defined in the following table:

	Approved	**Declined**	**Online Request**	**End Application**	**Try Another Interface**	**Select Next**	**Try Again**
Start	Ignored	Ignored	**To be processed (R)**	**To be processed (R)**	Ignored	Should be always C (not checked)	Should be always B (not checked)
Online Response Data	Ignored	Ignored	**To be processed (R)**	Ignored	Ignored	Ignored	Ignored
CVM	**To be processed (T)**	Ignored	**To be processed (T)**	Ignored	Ignored	Ignored	Ignored
UI Request on Outcome Present	**To be processed (T or R)**	**To be processed (T or R)**	**To be processed (T or R)**	**To be processed (T or R)**	**To be processed (T or R)**	Ignored	**To be processed (R)**
UI Request on Restart Present	Ignored	Ignored	**To be processed (T or R)**	**To be processed (T or R)**	Ignored	Ignored	Ignored
Data Record Present	**To be processed (T)**	**To be processed (T)**	**To be processed (T)**	Ignored	Ignored	Ignored	Ignored
Discretionary Data Present	**To be processed (T)**	**To be processed (T)**	**To be processed (T)**	**To be processed (T)**	**To be processed (T)**	Ignored	Ignored
Alternate Interface Preference	Ignored	Ignored	Ignored	Ignored	**To be processed (T)**	Ignored	Ignored
Receipt	Ignored	Ignored	Ignored	Ignored	Ignored	Ignored	Ignored
Field Off Request	**To be processed (R)**	**To be processed (R)**	**To be processed (R)**	**To be processed (R)**	**To be processed (R)**	Ignored	**To be processed (R)**
Removal Timeout	Ignored	Ignored	**To be processed (R)**	Ignored	Ignored	Ignored	Ignored

Table 2: Relevant parameters in each outcome message

Depending on the POS architecture, some parameters may be processed either by the Terminal Application or by the PURE contactless reader (example: management of the User interface).

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3.2 PURE Contactless implementation Overview

3.2.1 Set of commands available in PURE contactless mode

The PURE Contactless transaction flow is based on the transaction flow defined by EMVCo for contact transactions.

The set of commands available in PURE contactless mode is a subset of the set of commands available in PURE contact mode.

The commands that the terminal must use, during a contactless payment transaction, are:

- SELECT command
- GET PROCESSING OPTIONS command
- READ RECORD command
  The terminal has to use this command for each record referenced in AFL value communicated in the GET PROCESSING OPTIONS command response.
- GET DATA command (for terminal velocity checking)
- 1st GENERATE AC command

Note: The terminal must use these commands with the sequence defined for EMV contact transactions.

The remaining part of the transaction is depending on the consumer contactless device used, on the implementation options supported by the terminal and on terminal setting defined by the Acquirer.

When decision is to process the transaction online, the PURE card payment application may support the option to reactivate the card payment application when the online authorization is completed, for communicating the Issuer Host decision to the card payment application.

The support of this function at terminal level is an implementation option. The terminal software provider may support the following implementation options (both option may be supported):

- Long Tap support: when the consumer contactless device is not removed from the contactless interface, the terminal may communicate the Issuer Authorization Host decision to the PURE application
- Online Additional Tap support: The terminal may request the end user to tap again his/her consumer contactless device in order to restart a communication with the PURE application for communication of the Issuer Authorization Host decision

When an implementation option is supported, the Acquirer has the option to support it using terminal setting parameters.

When none of these options are commonly supported by the card payment application and by the terminal, transactions processed online are completed based on response code communicated by the Issuer Host in the response message. The card payment application is not involved in the completion of a transaction processed online.

In both cases, in case of communication problem during the online authorization process, the terminal must decline the transaction.

When the communication of Issuer Authorization Host decision to the PURE application is supported and the Issuer Host decision (Issuer Authentication Data (tag 91)) is present in the

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response message, the terminal has to use the following command for communicating its value to the card payment application:

- 2nd GENERATE AC command

When the Issuer Host decision (Issuer Authentication Data (tag 91)) is not present in the response message, the terminal shall not use the 2nd GENERATE AC command.

The PURE contactless dual-interface application and the PURE contactless mobile application are supporting an additional set of commands allowing the storage of information inside the application EEPROM. These commands may be used during a payment transaction.

The commands allowing the application EEPROM memory management are:

- GET DATA command
- PUT DATA (without secure messaging) command

A PURE application may support up to 10 memory slots identified by a tag value (DF0x with x in [1, A]).
The terminal may use:

- GET DATA for retrieval of a memory slot value
- PUT DATA for updating a memory slot value.

During the card payment application personalization, the Issuer has the option to define access control rule protecting these memory slots.
To allow a management through the contactless interface, the access conditions are:

- Free access for reading a memory slot
- Either free access or protection using secure messaging for updating a memory slot

Note: The GET DATA command may be used also for the retrieval of the PURE card risk management parameters.

The capacity to support these commands for reading the memory slots and for updating the memory slots are implementation options for the terminal software provider.
When supported by the terminal, the PURE contactless reader may use these commands either

- Before the GET PROCESSING OPTIONS command or
- After the GET PROCESSING OPTIONS command

The terminal may also send the PUT DATA command after the 1st GENERATE AC command. It is mandatory when secure messaging is required.

The PURE contactless dual-interface application and the PURE contactless mobile application are supporting script commands. The script commands supported are:

- APPLICATION BLOCK command
- APPLICATION UNBLOCK command
- CARD BLOCK command
- PIN CHANGE/UNBLOCK command
- PUT DATA (secure messaging) command
- UPDATE RECORD command

These commands may be send to the PURE application only when both the terminal and the PURE application are setup for the support of either Long Tap and/or Online Additional Tap.

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The PURE contactless dual-interface application and the PURE contactless mobile application may support the ECHO command for limiting the impacts of torn transaction. When the command is supported, the command allows to restart a transaction in case of torn transaction during the processing of the GENERATE AC command (either 1st GENERATE AC command or 2nd GENERATE AC command). The ECHO command allows the retrieval of the previous GENERATE AC response but also restores the transaction context allowing to continue the transaction that was prematurely stopped.
The capacity to send the ECHO command to the PURE application is an implementation option for the terminal software provider.

3.2.2 Special transaction type dedicated to Card Payment Application Authentication

The Application Authentication Transactions are dedicated to terminal that needs only to check offline the authenticity of the card. It may be used for example in transit dedicated terminals. This type of transactions is identified by a Transaction Type value specific to each Payment Scheme.

The PURE Contactless transaction flow for this type of transaction supports the following commands:

- SELECT command
- GET PROCESSING OPTIONS command
- READ RECORD command
  The terminal has to use this command for each record referenced in AFL value communicated in the GET PROCESSING OPTIONS command response.
- INTERNAL AUTHENTICATE command

  The terminal has to check the dynamic RSA signature provided by the INTERNAL AUTHENTICATE command response same as for contact DDA.

The additional set of commands allowing the storage of information inside the application EEPROM are also available with this type of transaction.

The support of this type of transaction is an implementation option for the terminal software provider.

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3.2.3 PURE Proprietary Terminal Data Elements

The PURE contactless specifications have introduced the definition of proprietary terminal data elements:

- Terminal Transaction Processing Information (TTPI) – tag ‘C7’
- Contactless Cryptogram Information Data (CCID) – tag ‘C5’
- ECHO Card Identifier – tag ‘9F75’
- Terminal Transaction Data – tag ‘9F76’
- Terminal Dedicated Data – tag ‘9F77’

The TTPI may be part of the data elements listed in PDOL that needs to be communicated to the card payment application using the GET PROCESSING OPTIONS command.

TTPI provides to the card payment application, information related to the terminal capabilities and result of the PURE contactless reader preprocessing. When the terminal provides no TTPI value (either due a terminal that does not support this PURE proprietary data element or due to a card payment application personalized with a PDOL list containing no reference to TTPI), the PURE card payment application uses a default value defined during card payment application personalization.

The information included in TTPI, are (please refer to Table 43 for TTPI structure):

- Terminal capabilities
  - Support of contactless chip mode
  - Capability to support contact transaction
  - Capability to process a transaction online
  - Cardholder verification methods supported by the terminal (Online PIN / Signature / none / cardholder verification method selection using contact EMV-defined method / confirmation code using the keyboard of the contactless device)
  - Offline CAM supported (SDA / CDA / DDA / none)
  - Level of compatibility with EMV contact transaction flow
    - Support of the cardholder verification method selection using CVM List parameter
    - Support in contactless mode of the terminal checks defined in EMV for contact transaction.
  - Support of the additional tap for Issuer response communication to the card payment application
- Result of the PURE contactless reader preprocessing
  - Request to process the transaction online (based on Reader Pre-Processing check on “Reader Contactless Floor Limit”)
  - Request to perform cardholder verification (based on Reader Pre-Processing check on “Reader CVM Required Limit”)
  - Inform the card payment application that the transaction is a Status Check transaction and that this type of transaction is supported by the terminal
  - Inform the card payment application that the transaction is a zero amount transaction and that this type of transaction is supported by the terminal

The CCID is a PURE proprietary data element that may be part of the information present in the response of the GET PROCESSING OPTIONS command and in the response of the 1st GENERATE AC command.

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The CCID presence is defined during the card payment application personalization. When CCID is present in GET PROCESSING OPTION command response, it must be present in 1st GENERATE AC command response. If it is not the case, the PURE contactless kernel must terminate the transaction. When CCID is not present in GET PROCESSING OPTION command response, it may be present in 1st GENERATE AC command response.

CCID provides (please refer to Table 37 and Table 38 for CCID structure)

- Card payment application decision related to transaction completion
  - preliminary decision in CCID value present in GET PROCESSING OPTIONS command response
  - final decision in CCID value present in 1st GENERATE AC command response
- Card payment application decision to switch to the contact interface and
- Card payment application decision related to cardholder verification
- Card payment application capability related to the support of additional tap for Issuer response communication to the card payment application.

The 'ECHO Card Identifier' is a PURE proprietary data element that may be present in the response of the selection of a card Payment Application (inside FCI Issuer Discretionary Data – 'BF0C'). Its presence indicates to the contactless terminal that the card payment application is supporting the ECHO command allowing the retrieval of the response of the last GENERATE AC command processed by the card payment application. Its value is defined during the card personalization and should be unique to each card payment application. Terminal Transaction Data and Terminal Dedicated Data are two PURE proprietary data elements allowing the card and the terminal to exchange information. Presence of these data elements is optional. Their structure and their usage are specific to each Payment Scheme. The reader and the kernel shall not check their contents.

- Terminal Transaction Data may be present among the Dynamic Transaction Parameters sent by the terminal to the reader and the kernel. When it is present among the Dynamic Transaction Parameters, the kernel must be capable to provide the value to the card payment application if the tag 9F76 is present in a DOL parameter (PDOL, CDOL1 …).
- Terminal Dedicated Data may be part of the data elements read using the READ RECORD command. When Terminal Dedicated Data is present, the kernel must be capable to add it to the list of data elements returned to the terminal using the outcome messages.

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3.2.4 PURE Contactless card payment application decision process

The PURE contactless card payment application decision process is split in two steps

- Preliminary decision during the GET PROCESSING OPTIONS command and
- Final decision during the 1st GENERATE AC command.

During the GET PROCESSING OPTIONS command, the PURE contactless card payment application performs preliminary "card risk management" checks and takes the following decisions based on terminal capability, terminal request, card payment application settings and result of the preliminary "card risk management" checks:

- Selection of the offline CAM (no offline CAM, SDA or CDA)
- Decision to perform cardholder verification. When cardholder verification is required by either the terminal or the card payment application, selection of the method to use for cardholder verification (no cardholder verification required, signature, online PIN, Confirmation Code to be checked using Consumer device, Confirmation Code already successfully checked or verification using EMV-defined CVM List method)
- Preliminary decision related to transaction completion (approved, declined, go online or switch to the contact interface)
  Note: When the card payment application decides either to request a Confirmation Code or request to switch to the contact interface, the preliminary decision is declined.
- Based on the offline CAM selected and card payment application preliminary decision related to transaction completion, definition of the list of records to read during this transaction (AFL value)

The information required for processing an online authorization request or for generating a clearing/settlement message, is communicated to the terminal only in 1st GENERATE AC command response (or ECHO command response in case of torn transaction).

3.2.5 Main Differences with the EMV Contact Transaction Flow

3.2.5.1 General Overview

The PURE contactless transaction flow has many similarities with the transaction flow designed for contact EMV chips and defined in reference [EMV -3]. It is however simplified and adapted for contactless ergonomics. Here are the main features:

- **Online/offline capability:** A transaction can be completed either online or offline. When completed online, the card payment application is not informed about the final transaction outcome.
- **Offline Data Authentication:** When the PURE contactless kernel is offline-capable, it must support SDA method and CDA method. The support of DDA method is required when the terminal support Application Authentication Transactions.
- **Online Data Authentication:** The card payment application generates an ARQC cryptogram that the Issuer host system may verify.
- **Cardholder Verification:** The card payment application or the PURE contactless kernel determines the CVM to use for a payment transaction based on issuer preference and acquirer requirement (terminal setting).

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- **Processing Restriction:** The card payment application determines if processing restrictions is required based on issuer preference and acquirer requirement (terminal setting).
- **Terminal Risk Management checks:** The card payment application determines if EMV-defined terminal risk management checks are required based on acquirer requirement (terminal setting).

3.2.5.2 Cardholder Verification

In PURE contactless transaction flow, the terminal does not systematically perform cardholder verification.

The decision to perform cardholder verification may be either

- A terminal decision based on the result of the PURE contactless reader preliminary preprocessing check using CVM Required Limit
- Or a card payment application decision based on card risk management parameters defined by the Issuer (card payment application must be personalized for returning CCID in GET PROCESSING OPTIONS command and in 1st GENERATE AC command)

When Cardholder verification is decided, the selection of the method to use is either

- A card payment application decision (card payment application should be personalized for returning CCID in GET PROCESSING OPTIONS command and in 1st GENERATE AC command) or
- A terminal decision using the EMV-defined contact algorithm based on CVM List parameter.

  A new CVM code is introduced for the support of Confirmation Code selection using the CVM List method (see section 11.2.5)

Both the Issuer and the Acquirer may decide the method that will be used:

- The Issuer may decide not to support the EMV-defined method: do not personalize CVM List parameter in the card payment application and indicates in AIP that Cardholder Verification is not supported.
- The Acquirer may decide not to support the EMV-defined method: information communicated to the card payment application using TTPI or the default TTPI value personalized in the card payment application.

When Issuer decision is in opposition with Acquirer policy, the terminal considers cardholder verification as failed. Terminal Action Analysis, based on TAC values and IAC values, decides about the consequences on transaction processing.
Note: To avoid acceptance issue, the Payment Scheme should define rules related to the methods to support.

3.2.5.3 Processing Restriction Checks

The PURE contactless kernel supports the following processing restriction checks:

- Application Version Number check
- Application Usage Control check

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- Application Expiration date check
- Application Effective date check

Decision to perform Processing Restriction Checks is a common decision to the Issuer and to the Acquirer.

The Issuer has the option not to personalize in the card payment application the parameters that the terminal needs for the Processing Restriction checks.

When the Acquirer has decided to perform Processing Restriction Checks and the Issuer has decided to do not personalize the parameter required by a check, the PURE contactless kernel has to skip the check.

Note: The Payment Scheme should define rules related to the support of Processing Restriction Checks.

3.2.5.4 Terminal Risk Management Check

The PURE contactless kernel supports only Terminal Velocity checking check.

Decision to perform Terminal Risk Management check is an Acquirer option.

The Issuer has also the option not to personalize in the card payment application, the parameters that the terminal needs for Terminal Velocity checking check.

When the Acquirer has decided to perform Terminal Risk Management check and the Issuer has decided to communicate no information, the PURE contactless kernel has to skip the Terminal Velocity checking check.

3.2.5.5 Transaction completion

At completion of a PURE contactless transaction, the possible decisions may be either:

- Same as in EMV contact transactions:
  - Approve the transaction
  - Request an online authorization
  - Decline the transaction
- Specific to contactless implementation
  - Request to switch to the contact interface.

Decision to switch to the contact interface is either

- a card payment application decision or
- a PURE contactless kernel decision.

The card payment application decision to switch to the contact interface is communicated to the PURE contactless kernel using the CCID information when it is part of the data elements present in the response of the GET PROCESSING OPTIONS command and in the response of the 1st GENERATE AC command.

The PURE contactless kernel has to decide to switch to the contact interface when the card payment application declines a transaction that the PURE contactless kernel has requested to approve offline.

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3.3 Types of transaction supported in PURE contactless kernel

A PURE contactless kernel supports the following types of EMV-defined financial transactions:

- Purchase
- Purchase with cashback
- Cash Advance / cash withdrawal
- Refund
  When processing a Refund transaction, the PURE contactless kernel decision is always to decline the transaction. Declining the transaction has no impact on the card internal counters in charge of the management of offline transactions.
  A Refund transaction may be used for retrieval of card-related information (for example the value of Track2 Equivalent Data, PAN value, Expiration Date value) outside a payment transaction. No online authorization request will be sent to the Issuer Host.

The PURE contactless reader supports also, as an Acquirer option, the following "special" purchase transaction:

- Status Check transaction
  A Status Check transaction is a Purchase Transaction with a transaction Amount of a single unit of currency. A single unit of currency has the value of 1 of the (major) unit of currency as defined in ISO 4217. As an example a single unit of currency for Euro is 1.00 (value 000000000100).Processing of a Status Check transaction is depending on personalization setting of the contactless card payment application.
  When the Status Check option is not selected by the Acquirer, a purchase transaction with an amount of a single unit of currency, is processed as any other purchase transaction.
- Zero Amount Transaction
  A Zero Amount transaction is a Purchase Transaction with a zero transaction Amount. Processing of a Zero Amount transaction is depending on personalization setting of the contactless card payment application.
  When the Zero Amount option is not selected by the Acquirer, a purchase transaction with a zero amount, is processed as any other purchase transaction.

As an implementation option, the PURE contactless kernel may support additional types of transactions specific to PURE:

- Application Authentication Transaction: Special type of transaction allowing the terminal to only authenticate the contactless device
- Retrieval of application proprietary data elements using GET DATA command
- Update of application data proprietary elements using PUT DATA command

The two last types of transactions allow usage the EEPROM of the application for the storage and management of information not linked to the standard EMV payment transaction. It may be used, for example, for the storage of loyalty data or for the storage of information dedicated to a transport application.

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These two types of transaction are processed as defined below:

- "Retrieval of application data elements using GET DATA command" transaction
  After application selection, this transaction is limited to send several times the command GET DATA to the card for retrieval of all the data elements requested by the terminal application. The PURE contactless reader has to communicate to the PURE contactless kernel, the list of data elements to retrieve using a PURE contactless kernel specific parameter Memory Slot Read Template (see section11.2.23). After reading all the data elements requested, the PURE contactless kernel terminates the transaction and provides the value of the data elements read in the outcome message of the transaction.

- "Update application data elements using PUT DATA command" transaction
  This transaction consists to update data elements using PUT DATA command without secure messaging. The reader communicates to the PURE contactless kernel, the list of data elements to update using a PURE contactless kernel specific parameter 'Memory Slot Update Template' (see section 11.2.24). With this type of transaction, the PURE contactless reader may request to update data elements prior or immediately after the GET PROCESSINGS OPTIONS command.
  After updating all the data elements requested, the kernel terminates the transaction and provides the result of the data element update in the outcome message of the transaction.
  The set of commands used during this type of transaction are either
  - SELECT command
  - One or several PUT DATA commands

  Or
  - SELECT command
  - GET PROCESSING OPTIONS command
  - One or several PUT DATA commands

  Or
  - SELECT command
  - One or several PUT DATA commands
  - GET PROCESSING OPTIONS command
  - One or several PUT DATA commands

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4 Implementation Options for Terminal Software Providers

An implementation option denotes functionality that is optional for a Terminal Software Provider to implement.

In the current specifications, the Terminal Software Provider must support all functions in relation with the contactless payment transaction up to the PURE application decision. On the contrary, functions not related to the contactless payment transaction are optional.

The implementation options are:

Implementation options related to the usage of the application EEPROM memory:

- **Implementation option 1 (PURE kernel)**: Retrieval of data element stored in card payment application EEPROM data slots using GET DATA command
- **Implementation option 2 (PURE kernel)**: Update of data element stored in card payment application EEPROM data slots using PUT DATA command

These implementation options may allow an Acquirer to support for example a loyalty program in contactless mode. When a Terminal Software Provider decides to provide this service, it is recommended to support both implementation options.

Implementation option dedicated to terminal supporting fixed-amount transaction (some vending machine)

- **Implementation option 3 (PURE reader)**: fixed-amount transaction support

Implementation option dedicated to terminal requiring the authentication of the card payment application without processing a payment transaction (for example transit terminals)

- **Implementation option 4 (PURE kernel)**: Support of the Application Authentication Transaction for offline card authentication

Implementation option allowing a terminal application to restrict the list of possible applications that may be used for a single transaction (option impacting only the contactless reader).

- **Implementation option 5 (PURE reader)**: Capacity to restrict list of supported application

Note: This option may be useful when implementation 1 and/or implementation 2 are supported. The option allows to select only the application that supports the EEPROM data slots.

Implementation option allowing a terminal application to communicate the Issuer Authorization Host decision to the PURE application if the contactless device was not removed from the contactless field (contactless device not powered off).

- **Implementation option 6 (PURE kernel)**: Long Tap support

Implementation option allowing a terminal application to communicate the Issuer Authorization Host decision to the PURE application when the contactless device was removed from the contactless field (contactless device not powered off), by requesting to tap again the contactless device.

- **Implementation option 7 (PURE kernel)**: Online Additional Tap support

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Implementation option allowing a terminal application to restart a transaction from where the transaction was stopped in case of torn transaction.

- **Implementation option 8 (PURE kernel)**: Support of ECHO command for retrieval of the last GENERATE AC command response in case of torn transactions.

If Terminal Software Provider decides to support an implementation option, it is the Acquirer's choice whether to use the functionality.

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5 POS System General Requirements

5.1 POS General Requirements

The POS System has to support the functions listed in this section. These functions are detailed in the EMVco document "EMV Contactless Specifications for Payment Systems - Book A" [Reference EMV-5] in section 5.5.

The implementation of these functions is specific to each Terminal Software Provider and may also depend on the type of terminal. Depending on the terminal architecture, these functions may be either supported by the Terminal Application software or by the reader software or split in both systems.

Requirement – POS General Requirements	
G1.1	Non-interference of the contactless interface with the Contact Chip Interface
G1.2	A new transaction shall not start prior completion of the previous transaction
G1.3	In POS supporting several interfaces: If the presence of a card with a contact interface is detected, then the POS System shall power down the contactless interface prior to initiating a contact chip transaction and shall keep it powered down for the duration of the transaction. **If** either of the following is true: • the presence of a contactless card is detected, • or a contactless card interaction is not complete (the terminal application has not already received the 1st final outcome message – see section 7.3), • and subsequently a card with a contact interface is detected, **Then** the contactless transaction shall be terminated and the contactless interface shall be powered down for the duration of the transaction. Design consideration should be given to placement of the PURE contactless reader relative to other card interfaces. The PURE contactless reader should not be placed such that it is frequently activated when cardholders attempt to use other interfaces.
G1.4	Once a contactless transaction has started, there must be a means to cancel it in case the cardholder is unable to present a contactless card within a reasonable amount of time that may be a POS configuration setting
G1.5	If transaction type and transaction amount are known before the start, the amount must be indicated to the cardholder.
G1.6	Transaction disposition (approved/declined) must be indicated to the cardholder.

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Requirement – POS General Requirements	
G1.7	For receipt printing, the POS application has to apply rules defined by the Payment Scheme or by local regulation specific to each Acquirer. The kernel is nor involved in the decision to print a receipt.
G1.8	In the case of contactless payment, decision to perform cardholder verification is taken by the kernel based on POS cardholder verification capabilities. Using the outcome message, the kernel communicates to the POS system its request related to cardholder verification. Depending on POS capabilities, the method that the kernel may select, are: o Online PIN: POS system will be in charge to request Online PIN using same procedure as for EMV contact transaction o Signature: POS system will be in charge to request signature using same procedure as for EMV contact transaction o No CVM Required: POS system has to request no verification of the cardholder o Confirmation Code Verified: POS system has to request no verification of the cardholder The Terminal Application must apply PURE contactless kernel decision related to cardholder verification.
G1.9	In some exceptional cases, the PURE contactless kernel may not be capable to select a cardholder verification method due to either a wrong personalization of the card payment application or a wrong setting of the terminal. In that case, if the transaction is approved, no cardholder verification may be performed. The Acquirer or the Payment Scheme should define rules when these exceptional cases happen. Depending on Payment Scheme liability rules, the Merchant may become liable. The POS (terminal application) has to apply rules defined by the Payment Scheme or by local regulation specific to each Acquirer.
G1.10	Depending on type of terminal, a POS System may support a configuration parameter (Autorun parameter) specifying if a contactless transaction is either initialized by a terminal action or starts automatically after the previous contactless transaction has completed. Depending on the type of terminal, the contactless field may be either On or Off between two transactions.
G1.11	Terminal supporting the “long tap” option should support a design allowing the cardholders to deposit their contactless payment device (dual-interface card, mobile phone, sticker …) on the reader during the duration of the transaction.
G1.12	POS must support AID partial match for all AIDs supported.

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5.2 Performance Requirements

Requirement – Performance Requirements	
G2.1	The POS System must comply with the performance requirements defined in the EMVco document “EMV Contactless Specifications for Payment Systems - Book A” [Reference EMV-5] in section 10.

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5.3 Type of transactions to support in contactless mode

Requirement – Transaction Types Management

G3.1

A PURE contactless reader and a PURE contactless kernel must support all the types of financial transaction defined in Table 3.

Type of transaction	Transaction Type Codification	Transaction Type Support
Purchase transaction	‘00’	Mandatory
Cash transaction (cash withdrawn or manual cash)	‘01’	Mandatory
Purchase with cashback	‘09’	Mandatory
Refund transaction	‘20’	Mandatory

Table 3 –Types of financial transactions in PURE contactless reader

Note: Depending on the implementation, value associated to each transaction type may be a POS parameter. Implementation supporting this option shall use the values defined above for reader/kernel certification.

G3.2

A PURE contactless reader and a PURE contactless kernel may support the type of non-financial transaction defined in Table 4. as Terminal Software Provider Implementation Option

Type of transaction	Transaction Type Codification	Transaction Type Support
Retrieve application data elements using GET DATA command	‘78’ (codification specific to PURE)	Terminal Software Provider Implementation Option n°1
Update application data elements using PUT DATA command	‘79’ (codification specific to PURE)	Terminal Software Provider Implementation Option n°2
Application Authentication Transaction	Value defined by Payment Scheme	Terminal Software Provider Implementation Option n°4

Table 4 –Types of non-financial transactions in PURE contactless reader

Note: Depending on the implementation, value associated to each transaction type may be a POS parameter. Implementation supporting this option shall use the values defined above for reader/kernel certification.

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5.4 General requirements related to Implementation Options

Terminal Software Providers have to choose whether to support the implementation options listed in section 4.

The PURE contactless reader and the PURE contactless kernel must comply with the general requirements associated to the implementation options supported. These requirements are listed in this section.

	Requirement – Implementation options
G4.1	When the implementation option 1 is supported, • The PURE contactless reader and the PURE contactless kernel must support a specific transaction type (‘78’ – retrieve data elements) that consists in solely retrieving the values of a list of parameters requested by the Terminal Application and communicate their values to the Terminal Application. • The PURE contactless reader and the PURE contactless kernel must support, for all other types of transaction, the capacity to retrieve the parameters requested by the Terminal Application and communicate their values to the Terminal Application at transaction completion.
G4.2	When the implementation option 2 is supported, • The PURE contactless reader and the PURE contactless kernel must support a specific transaction type (‘79’ – update data elements) that consists in solely updating the values of a list of parameters provided by the Terminal Application at the transaction flow step specified by the Terminal Application (prior or after GET PROCESSING OPTIONS command) and • The PURE contactless reader and the PURE contactless kernel must support, for all other types of transaction, the capacity to update the values of a list of parameters provided by the Terminal Application at the transaction flow step specified by the Terminal Application (prior or after GET PROCESSING OPTIONS command or after 1st GENERATE AC command). Note: In both cases, the PURE contactless kernel communicates the command(s) processing result at transaction completion to the contactless reader and to the Terminal Application.
G4.3	When an implementation option is supported, the POS must support a parameter allowing the Acquirer to deactivate it for all the applications supported by the POS. The POS must support the parameter Contactless POS Implementation Options – see section 11.2.11. This parameter specifies the list of implementation options supported and selected by the Acquirer when the option is implemented.

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6 Requirements related to POS Parameters

The POS System has to support all the parameters required by each part of the module of the terminal architecture. Management of these parameters is specific to each Terminal Software Provider.

6.1 Acquirer Options

Requirement – Acquirer options	
C1.1	A POS must support a parameter for declaration of the options selected by the Acquirer for each AID supported. This parameter is the PURE "Contactless Application Capabilities" parameter defined in Table 39. This parameter must be shared between the PURE contactless reader and the PURE contactless kernel. A value of this parameter must be defined for each AID supported.

Acquirer Options present in the Contactless Application Capabilities parameter	
C1.2	The PURE "Contactless Application Capabilities" parameter (AC_SDA, AC_DDA and AC_CDA) allows the Acquirer to define the list of offline CAM to support. A PURE contactless kernel must be capable to support the following offline CAM for financial transaction: • Static Data Authentication (SDA) • Combined DDA/Application Cryptogram Generation (CDA) When the implementation option 4 is supported, a PURE contactless kernel must be capable to support the following offline CAM for Application Authentication Transaction: • Dynamic Data Authentication (DDA)
C1.3	The PURE "Contactless Application Capabilities" parameter (AC_OfflineCAM) allows the Acquirer to decide to support offline CAM either only in case of transaction offline approval or in all circumstances.
C1.4	In PURE contactless transaction flow, the decision to perform Cardholder Verification may be either a terminal decision (result of contactless reader preliminary checking) or a card decision. When cardholder verification is decided, the PURE contactless transaction flow allows to select the cardholder verification method using either: • an PURE-defined algorithm managed by the card itself or • the EMV-defined method based on CVM List The PURE "Contactless Application Capabilities" parameter (AC_CVM) allows the Acquirer to decide the support or not of the EMV-defined method based on CVM List.

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Acquirer Options present in the Contactless Application Capabilities parameter	
C1.5	The PURE "Contactless Application Capabilities" parameter (AC_OnlinePIN, AC_Signature and AC_CC_Tap) allows the Acquirer to define the list of methods supported by the terminal for cardholder verification. The possible methods are: • Online PIN • Signature • Confirmation Code checked using handset device Note: When cardholder verification is not requested by the terminal, the card payment application may decide to request Confirmation Code even when this method is not supported by the Acquirer as a possible CVM.
C1.6	The PURE "Contactless Application Capabilities" parameter (AC_TRM) allows the Acquirer to decide to support Processing Restriction checks and Terminal Risk Management checks.
C1.7	The PURE "Contactless Application Capabilities" parameter (AC_Contact and AC_Offline) allows the Acquirer to define the following reader capabilities: • Capacity to support contact transaction • Online capacity in contactless mode (the reader may be either online capable [online-only or offline capable with online capacity] or offline-only capable)
C1.8	The PURE "Contactless Application Capabilities" parameter (AC_Balance) allows the Acquirer to request card balance during each transaction. When card balance is available using GET DATA command, the kernel has to communicate its value to the terminal at completion of the transaction.
C1.9	At completion of the GET PROCESSING OPTIONS command, the card may communicate its preliminary decision. The PURE contactless transaction flow may be stopped when card preliminary decision is to reject the transaction. The PURE "Contactless Application Capabilities" parameter (AC_Decline) allows the Acquirer to decide if transaction has to be terminated when card preliminary decision is to reject the transaction.
C1.10	Requirement removed
C1.11	When the kernel supports the implementation option 8 (**IO_Option8**), the PURE "Contactless Application Capabilities" parameter (AC_ECHO) allows the Acquirer to support the recovery of a torn payment transaction using the ECHO command.
C1.12	When the kernel supports the implementation option 4 (**IO_Option4**), the PURE "Contactless Application Capabilities" parameter (AC_Auth_Appli) allows the Acquirer to decide the support of Application Authentication Transaction

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Acquirer Options present in the Contactless Application Capabilities parameter	
C1.13	When the kernel supports the implementation option 7 (**IO_Option7**), the PURE "Contactless Application Capabilities" parameter (AC_Online_Tap) allows the Acquirer to decide the support of the additional tap for communication of Issuer decision after an online authorization request.
C1.14	When the kernel supports the implementation option 6 (**IO_Option6**), the PURE "Contactless Application Capabilities" parameter (AC_Long_Tap) allows the Acquirer to decide the support of the long tap for communication of Issuer decision after an online authorization request.

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6.2 Configuration Parameters in POS

6.2.1 Terminal Application module

Requirement – Configuration Parameters required by Terminal

Terminal implementation must guaranty that transaction completion result is communicated to the cardholder. The terminal must support a parameter specifying how long the result of a transaction completion needs to be displayed to the cardholder.

C2.1

Name	Varies by	Presence[3]	Tag
POS Timeout (Transaction Result Display) - the value is in units of 100ms.	POS	M	-

Table 5: Terminal Configuration Data (in relation with contactless payment)

[3] M = mandatory ; C = conditional ; O = optional

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6.2.2 PURE contactless reader module

Requirement – Configuration Parameters required by contactless reader

C3.1 A contactless POS may support one or more Currency Codes depending on whether multiple currencies are supported. The currency to use for the contactless transaction needs to be communicated to the PURE contactless reader and to the contactless PURE kernel at the beginning of the transaction.

C3.2 A PURE contactless reader must have available at least the configuration parameters defined in Table 6. Some of these parameters must have a value for each AID supported.[4]

Name	Varies by	Presence	Tag
Contactless Application Capabilities	AID	M	-
Contactless POS Implementation Options	POS	M	-

Table 6: Static PURE contactless reader Configuration Data

C3.3 For each AID supported and for each type of financial transaction supported (see Table 3), the PURE contactless reader must be capable to allow the Acquirer to define or not a value for the parameters defined in Table 7. When the POS supports multi-currencies option, the Acquirer must be capable to define, for each limit supported, one value per currency supported.

Name	Varies by	Presence	Tag
Reader Contactless Transaction Limit[5]	AID + Transaction Type + Currency	O	-
Reader Contactless Floor Limit	AID + Transaction Type + Currency	O	-
Reader CVM Required Limit	AID + Transaction Type + Currency	O	-
Status Check Support flag	AID + Transaction Type	O	
Terminal Floor Limit	AID + Transaction Type + Currency	O	
Zero Amount Allowed flag	AID + Transaction Type	O	

Table 7: Static optional PURE contactless reader Configuration Data

Management of parameters values not defined by the Acquirer is specific to each implementation.

[4] Some additional parameters may be supported for satisfying to Payment Scheme specific requirements (example: Merchant Application Priority Indicator associated to each AID supported). Definition of these additional parameters is outside the scope of this document.

[5] In Offline-only readers, the Reader Contactless Transaction Limit, the Reader Contactless Floor Limit and the Terminal Floor Limit should be set to the maximum transaction amount supported by the terminal when these limits are supported.
For transaction type requiring the transaction to be always forced online (cash transaction, purchase with cashback transaction...), the Terminal Floor limit values should be set to 000000000000. It is the responsibility of the Acquirer to satisfy to this recommendation.

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6.2.3 PURE contactless kernel module

Requirement – Configuration Parameters required by PURE contactless kernel	
C4.1	A PURE contactless kernel must have available at least the configuration parameters defined in Table 8. Some of this data depends on the AID selected. These parameters may be shared with the other POS modules.

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Name	Varies by	Presence	Tag
Certification Authority Public Keys	RID	C[6]	-
Additional Tag Object List (ATOL)	RID	M	-
Mandatory Tag Object List (MTOL)	RID	M	-
Authentication Transaction Data Tag Object List (ATDTOL)	RID	O	-
Default DDOL (with the mandatory value "9F3704".)	RID	C[7]	-
Terminal Action Code - Default	AID	O	-
Terminal Action Code - Denial	AID	O	-
Terminal Action Code - Online	AID	O	-
Terminal Application Version Number	AID	M	'9F09'
Contactless Kernel Capabilities[8]	AID	M	-
Contactless POS Implementation Options[9 and 10]	POS	M	-
Acquirer Identifier	POS	M	'9F01'
Merchant Category Code	POS	M	'9F15'
Merchant Name and Location	POS	M	'9F4E'
Terminal Type[10]	POS	M	'9F35'
Terminal Capabilities[10]	POS	M	'9F33'
Terminal Country Code	POS	O[11]	'9F1A'
Transaction Currency Code	POS	O[11]	'5F2A'
POS Timeout (Long tap) (the value is in units of 100ms).	POS	C[12]	-

[6] Presence of this parameter is mandatory when the kernel supports offline transactions

[7] Mandatory when implementation option 4 is supported

[8] This parameter is identical to the Contactless Application Capabilities parameter used by the contactless reader and shall have the same value. Both parameters may be in fact the same parameter shared by the kernel and the contactless reader.

[9] This parameter may be shared with the contactless reader.

[10] The Acquirer must define consistent values for Terminal Type, Terminal Capabilities and Contactless Kernel Capabilities. Reader and kernel are no required to checks that these data elements are setup with consistent values.

[11] When not present, this data element must be part of the Dynamic Transaction Data communicated to the kernel on kernel activation.

[12] Presence of this data element is mandatory when implementation option 3 is supported.

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Name	Varies by	Presence	Tag
Transaction Type value for Application Authentication Transaction (Trans_Type_Auth_Appli)	POS	C[13]	-

Table 8: Static PURE contactless kernel Configuration Data

[13] When Implementation 4 is supported by the kernel.

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7 Terminal Application Requirements related to Reader Interface

7.1 Overview

The terminal application is specific to each Acquirer. Specification of this application is outside the scope of this document. This section is only dedicated to the description of the requirements related to the interface between the Terminal Application Module and the contactless reader Module.

7.2 Transaction processing with the contactless reader

The Transaction processing with the contactless reader may be summarized in the following figure.

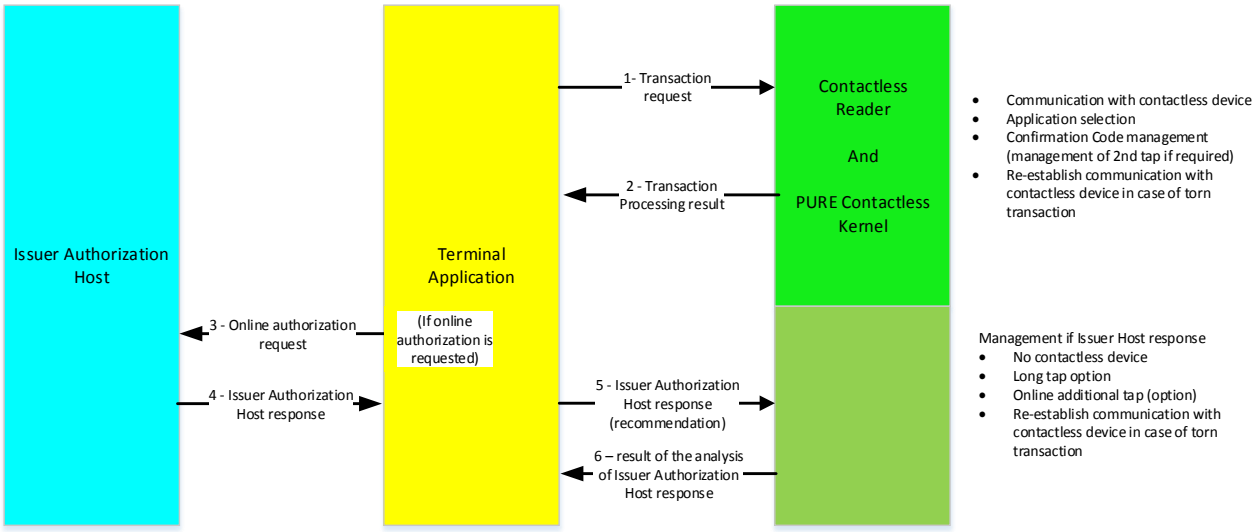


Figure 3 Transaction processing overview

The Terminal Application has to activate the contactless reader, each time the Terminal Application initiates one of the types of transaction defined in section 5.3 using the contactless interface. When activating the contactless reader, the Terminal Application has to specify that it is a "***new***" transaction.

The list of information that the Terminal Application has to communicate to the PURE contactless reader is depending on the type of transaction (see section 7.5.1 and section 7.5.3).

After activation of the contactless reader, the Terminal Application has to wait for the result of the transaction processing. The PURE contactless reader communicates result using final outcome messages. The PURE contactless reader may communicate the following final outcome messages:

- ***Online Request*** final outcome message
- ***Approved*** final outcome message
- ***Declined*** final outcome message
- ***End Application(Start = N/A)*** final outcome message
- ***Try Another Interface*** final outcome message

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The Terminal Application has to process each final outcome message as it is defined in the requirements listed in section 7.3. The processing of a final outcome message may result in the re-activation of the PURE contactless reader.
At completion of an online authorization, the Terminal Application should re-activate the contactless reader, the information that the Terminal Application has to communicate are defined in section 7.5.2.

After re-activation of the PURE contactless reader, the Terminal Application has to wait for the result of the new transaction processing.

When both the POS and the payment device are supporting online additional tap, the contactless reader should request to tap again only when the Issuer has communicated chip data in the online authorization response message. The following table summarizes the different use cases with online additional tap performed:

Chip Data present in field 55	Online additional tap request	Transaction Decision
No data	No additional tap	Based on Response Code value
Issuer Authentication Data	Additional tap	Based on 2nd GENERATE AC response (if problem, based on Response Code value)
Only script command	Additional tap	Based on Response Code value
Issuer Authentication Data and script command	Additional tap	Based on 2nd GENERATE AC response (if problem, based on Response Code value)

Table 9: Online additional tap support

The PURE contactless reader communicates result using second final outcome messages. The PURE contactless reader may communicate the following second final outcome messages:

- ***Approved*** second final outcome message
- ***Declined*** second final outcome message
- ***End Application*** second final outcome message

The Terminal Application has to process each second final outcome message as it is defined in the requirements listed in section 7.4.

At any time, the Terminal Application may request the PURE contactless reader to cancel the transaction.

In some exceptional cases, the PURE contactless kernel may not be capable to select a cardholder verification method due to either a wrong personalization of the card payment application or a wrong setting of the terminal. In that case, if the transaction is approved, no cardholder verification may be performed. The Acquirer or the Payment Scheme should define rules when these exceptional cases happen. Depending on Payment Scheme liability rules, the Merchant may become liable.

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The terminal application should support a parameter specifying rule to apply when the contactless kernel is not capable to define a cardholder verification method. The rule may be, for example, to either ask for the signature or decline the transaction.

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7.3 Contactless reader First final outcome messages processing

Requirement –Online Request final outcome message processing

- The Terminal Application has to check ***CVM*** parameter in the final outcome message.
  - **If** CVM parameter is "No CVM" **then** the Terminal Application has to perform no cardholder verification
  - **If** CVM parameter is "Confirmation Code Verified" **then** the Terminal Application has to perform no cardholder verification
  - **If** CVM parameter is "Obtain Signature" **then** the Terminal Application has to memorize to request a signature if the transaction is approved at transaction completion.
  - **If** CVM parameter is "Online PIN" **then** the Terminal Application has to request the cardholder to enter a PIN value. Management of Online PIN shall comply with rules defined in EMV contact specifications. In case of issue during Online PIN inquiry (either PIN PAD not present or not functioning or PIN entry bypassed), **the transaction must be declined**.
    Note: TVR value shall not be updated.
  - **If** CVM parameter is 'N/A' **then** the Terminal Application has to perform no cardholder verification
- If the Terminal Application supports the implementation option 1 or the implementation option 2, depending on the type of transaction in process, the

T1.1 Terminal Application should check the ***Discretionary Data Present*** parameter in the final outcome message. If data are available, the processing of these data is outside the scope of this document.

- If the final outcome message parameter "***Data Record Present***" is not set to Yes **or** if the Track2 Equivalent Data (tag 57) is not present in the "Data Record" **then the transaction must be declined**.
- **1- If** the transaction is declined as per above processing (e.g. Data record absent, PIN PAD not present or not functioning…)
  1-Then
  - the Terminal Application has to set the EMV parameter Authorization Response Code (tag '8A') to "99" (decline)

  1-Else
  - the Terminal Application has to initiate an online authorization request. The online authorization request message will contain
    - an "encrypted online PIN" field if online PIN was requested in the outcome message and PIN was successfully keyed
    - a "chip data" field if the "chip data" are present in the "***Data Record***" final outcome message parameter.
    - 'Track2' field using the value of Track2 Equivalent Data (tag 57) present in the "***Data Record***" final outcome message parameter.

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Requirement –Online Request final outcome message processing	
	• 'Card Sequence Number' field using the value of PAN Sequence Number (tag 5F34) if it is present in the "***Data Record***" final outcome message parameter. • **2-If** the Terminal Application gets an authorization response • **2- then** • the Terminal Application has to set the EMV parameter 'Authorization Response Code' (tag '8A') using the following rules: **3-If** the "Response Code" field value received in the response message corresponds to an approval (codification is specific to each Acquirer), **3-Then** Set parameter 'Authorization Response Code' (tag '8A') to "00" **3-Else** Set parameter 'Authorization Response Code' (tag '8A') to "99" (or any other value distinct from "00" or "Y3" or "Z3") **3-Endif** • The Terminal Application **should** re-activate the PURE contactless reader for processing the online response. The Terminal Application has to specify that it is for processing an "Online Response". The list of information that the Terminal Application should communicate to the PURE contactless reader is defined in section 7.5.2. Note: Alternative solution is to apply requirement T1.3 but in this case long tap and online additional tap cannot be supported. • **2-else (communication problem, timeout or offline-only terminal)** • **the transaction must be declined** • the Terminal Application has to set the EMV parameter Authorization Response Code (tag '8A') to "Z3" (unable to go online, decline) • **2-endif** **1-Endif**
T1.2	If the Payment Scheme supports partial authorization and the Issuer has decided to authorize only part of the transaction, the value of 'Amount, Authorized (Numeric)' (tag '9F02') and the value of 'Amount, Other (Numeric)' (tag '9F03') need to be set to the updated amounts authorized by the Authorization Host when the Terminal application re-activates the contactless reader after an online authorization. In the other cases, the value of 'Amount, Authorized (Numeric)' (tag '9F02') and the value of 'Amount, Other (Numeric)' (tag '9F03') needs to be equal to the values communicated in the initial request.
T1.3	Alternative solution when the Terminal Application is not reactivating the contactless reader after an online authorization (option allowed when long tap and online additional tap are not supported)

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Requirement –Online Request final outcome message processing

- The Terminal Application has to determine the transaction disposition according to the online authorization result

  The transaction disposition is according to the "Response Code" field value received in the **online authorization response message**

  2-If the "Response Code" field corresponds to a transaction rejection
  2-Then the transaction is declined
  2-Else the transaction is approved
  2-Endif

- The Terminal Application has to check the ***CVM*** field value received in the **1st final outcome message** requesting to process the transaction online to determine if a signature is requested.

  2-If the ***CVM*** field value is 'N/A'
  2-then the Terminal Application has to apply Acquirer or Payment Scheme policy corresponding to the case "the kernel is not capable to select a cardholder verification method". It may consist in requesting signature or to decline the transaction.
  2-Endif

1-If the transaction is approved
1-Then

- Send a **User Interface Request** message using as parameter
  Message Identifier:
  '03' ("Approved") if signature is not requested
  '1A' ("Approved – Please Sign") if signature is requested
  Status: Card Read Successfully (Remove card)
- The Terminal Application prepares a clearing record using the data record provided in the 1st final outcome message requesting to process the transaction online and the chip data received in the online authorization response message.
  - **If** printing a receipt is required by the Payment Scheme
    or if signature is requested (except if the signature might be obtained by other means such as electronic capture)
    Then the Terminal Application prints a receipt with a signature line if signature is requested
  - **If** signature is requested **and** the signature might be obtained by other means such as electronic capture,
    Then activate the signature capture

1-Else

- The Terminal Application has to send a **User Interface Request** message using as parameter
  Message Identifier: '07' ("Not Authorized")
  Status: Card Read Successfully (Remove card)

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Requirement –Online Request final outcome message processing
1-Endif

Requirement – Approved final outcome message processing	
T1.4	• The Terminal Application has to check ***CVM*** parameter. • **If** CVM parameter is "No CVM" **then** the Terminal Application has to perform no cardholder verification • **If** CVM parameter is "Confirmation Code Verified" **then** the Terminal Application has to perform no cardholder verification • **If** CVM parameter is "Obtain Signature" **then** the Terminal Application has to memorize to request a signature if the transaction is approved at transaction completion. • **If** CVM parameter is Online PIN **then the transaction must be declined**. • **If** CVM parameter is 'N/A' **Then** apply Acquirer or Payment Scheme policy corresponding to the case "the kernel is not capable to select a cardholder verification method". It may consist in requesting a signature or to decline the transaction. **Endif** • **If** the final outcome message parameter "***Data Record Present***" is not set to Yes **or if** the Track2 Equivalent Data (tag 57) is not present in the "Data Record", **then** the **transaction must be declined**. • If the Terminal Application supports the implementation option 1 or the implementation option 2, depending on the type of transaction in process, the Terminal Application should check the ***Discretionary Data Present*** parameter in the final outcome message. If data are available, the processing of these data is outside the scope of this document. • **If** the transaction is declined as per above processing (e.g. Data record absent, wrong CVM method selected…) **Then** • The Terminal Application has to send a **User Interface Request** message using as parameter Message Identifier: '07' ("Not Authorized") Status: Card Read Successfully (Remove card) **Else** • The Terminal Application prepares a clearing record using the ***data record*** provided with the final outcome message • **If** printing a receipt is required by the Payment Scheme **or if** signature is requested (except if the signature might be obtained by other means such as electronic capture)

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Requirement – Approved final outcome message processing

Then the Terminal Application prints a receipt with a signature line if signature is requested

- **If** signature is requested **and** the signature might be obtained by other means such as electronic capture,
  Then activate the signature capture

Endif

- The transaction is completed.
- After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must
  - Ensure that the contactless field is off
  - send a **User Interface Request** message using as parameter
    Message Identifier: '14' ("Welcome")
    Status: Idle

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Requirement – Declined final outcome message processing	
T1.5	• If the Terminal Application supports the implementation option 1 or the implementation option 2, depending on the type of transaction in process, the Terminal Application should check the ***Discretionary Data Present*** parameter in the final outcome message. If data are available, the processing of these data is outside the scope of this document. • If the outcome message parameter "***Data Record Present***" is set to Yes. The Terminal Application may process the contents of the "***Data Record***" final outcome message parameter. Usage of the information present in "***Data Record"*** of the final outcome message is outside the scope of this document. For example in the case of a Refund transaction, these data allow to retrieve information related to the card. • The transaction is completed. • After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must: • Ensure that the contactless field is off • send a **User Interface Request** message using as parameter Message Identifier: '14' ("Welcome") Status: Idle

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Requirement –Try Another Interface final outcome message processing	
T1.6	• If the Terminal Application supports the implementation option 1 or the implementation option 2, depending on the type of transaction in process, the Terminal Application should check the ***Discretionary Data Present*** parameter in the final outcome message. If data are available, the processing of these data is outside the scope of this document. • The Terminal Application has to check ***Alternate Interface Preference*** parameter in the final outcome message. • If the parameter value is "Contact Chip", the Terminal Application should recommend either the Merchant or the Cardholder or both to use the chip technology. • If the parameter value is "Mag-stripe", the Terminal Application should recommend either the Merchant or the Cardholder or both to use the magstripe technology. There is no requirement for mandating to use the interface recommended. • The transaction is completed. • After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must: • Ensure that the contactless field is off • send a **User Interface Request** message using as parameter Message Identifier: '14' ("Welcome") Status: Idle

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Requirement –End Application final outcome message processing	
T1.7	• If the Terminal Application supports the implementation option 1 or the implementation option 2, depending on the type of transaction in process, the Terminal Application should check the ***Discretionary Data Present*** parameter in the final outcome message. If data are available, the processing of these data is outside the scope of this document. • The transaction is completed. • After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must • Ensure that the contactless field is off • send a **User Interface Request** message using as parameter Message Identifier: '14' ("Welcome") Status: Idle The remaining part of the transaction is outside the scope of this document Note: The reader communicates this outcome message in the following cases • Completion of a non-financial transaction (transaction type '78', transaction type '79' or card authentication transaction) • Cancellation of the transaction • Oversize of the transaction dependent parameters (maximum buffer size supported by the kernel is exceeded)

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7.4 Contactless reader second final outcome messages processing

Requirement – Approved second final outcome message processing

- The Terminal Application has to check ***CVM*** parameter in the second final outcome message.
  - **If** CVM parameter is "No CVM" **then** the Terminal Application has to perform no cardholder verification
  - **If** CVM parameter is "Confirmation Code Verified" **then** the Terminal Application has to perform no cardholder verification
  - **If** CVM parameter is "Obtain Signature" **then** the Terminal Application has to memorize to request a signature if the transaction is approved at transaction completion.
  - **If** CVM parameter is "Online PIN" **then** the Terminal Application has to perform no cardholder verification.
  - **If** CVM parameter is 'N/A'
    Then apply Acquirer or Payment Scheme policy corresponding to the case "the kernel is not capable to select a cardholder verification method". It may consist in requesting signature or to **decline the transaction**.
    Endif
- If the second final outcome message parameter "***Data Record Present***" is not set to Yes or if the Track2 Equivalent Data (tag 57) is not present in the "Data Record", the **transaction must be declined**.

T2.1

- **If** the transaction is declined as per above processing (e.g. Data record absent, wrong CVM method selected…)
  Then
  - The Terminal Application has to send a **User Interface Request** message using as parameter
    Message Identifier: '07' ("Not Authorized")
    Status: Card Read Successfully (Remove card)

  Else
  - The Terminal Application prepares a clearing record using the ***data record*** provided with the second final outcome message
  - **If** printing a receipt is required by the Payment Scheme
    or if signature is requested (except if the signature might be obtained by other means such as electronic capture)
    Then the Terminal Application prints a receipt with a signature line if signature is requested
  - **If** signature is requested **and** the signature might be obtained by other means such as electronic capture,
    Then activate the signature capture

  Endif

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Requirement – Approved second final outcome message processing

- The transaction is completed.
- After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must:
  - Ensure that the contactless field is off
  - send a **User Interface Request** message using as parameter
    Message Identifier: '14' ("Welcome")
    Status: Idle

Note: The data record field (if present) provided in the second final outcome message may contain Script Result. Usage of this information is outside the scope of this document.

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Requirement – Declined second final outcome message processing	
T2.2	• If the outcome message parameter "***Data Record Present***" is set to Yes. The Terminal Application may process the contents of the "***Data Record***" final outcome message parameter. Usage of the information present in "***Data Record"*** of the final outcome message is outside the scope of this document. For example in the case of a Refund transaction, these data allow to retrieve information related to the card. • The transaction is completed. • After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must: • Ensure that the contactless field is off • send a **User Interface Request** message using as parameter Message Identifier: '14' ("Welcome") Status: Idle Note: the data record field (if present) provided in the second final outcome message may contain Script Result. Usage of this information is outside the scope of this document.

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Requirement –End Application second final outcome message processing

- The Terminal Application has to determine the transaction disposition according to the online authorization result

  The transaction disposition is according to the "Response Code" field value received in the **online authorization response message**

  2-If the "Response Code" field corresponds to a transaction rejection
  2-Then the transaction is declined
  2-Else the transaction is approved
  2-Endif

- The Terminal Application has to check the ***CVM*** field value received in the **1st final outcome message** requesting to process the transaction online to determine if a signature is requested.

  2-If the ***CVM*** field value is 'N/A'
  2-then the Terminal Application has to apply Acquirer or Payment Scheme policy corresponding to the case "the kernel is not capable to select a cardholder verification method". It may consist in requesting signature or to decline the transaction.
  2-Endif

1-If the transaction is approved
1-Then

T2.3

- Send a **User Interface Request** message using as parameter
  Message Identifier:
  '03' ("Approved") if signature is not requested
  '1A' ("Approved – Please Sign") if signature is requested
  Status: Card Read Successfully (Remove card)
- The Terminal Application prepares a clearing record using the data record provided in the 1st final outcome message requesting to process the transaction online and the chip data received in the online authorization response message.
- **If** printing a receipt is required by the Payment Scheme
  or if signature is requested (except if the signature might be obtained by other means such as electronic capture)
  Then the Terminal Application prints a receipt with a signature line if signature is requested
- **If** signature is requested **and** the signature might be obtained by other means such as electronic capture,
  Then activate the signature capture

1-Else

- The Terminal Application has to send a **User Interface Request** message using as parameter
  Message Identifier: '07' ("Not Authorized")
  Status: Card Read Successfully (Remove card)

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Requirement –End Application second final outcome message processing

1-Endif

- The transaction is completed.
- After a timeout (POS Timeout (Transaction Result Display), the Terminal Application must:
  - Ensure that the contactless field is off
  - send a **User Interface Request** message using as parameter
    Message Identifier: '14' ("Welcome")
    Status: Idle

  Note: the data record field (if present) provided in the second final outcome message may contain Script Result. Usage of this information is outside the scope of this document.

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7.5 Terminal Application Information to communicate to the contactless reader

7.5.1 Terminal Application starts a new transaction (other than Application Authentication transaction)

Requirement –Information to communicate to the contactless reader

When the PURE contactless reader supports no implementation option, the Terminal Application must provide at least the following list of transaction-dependent parameters.

T3.1

Data Element	Tag	Requirement
Amount, Authorized (Numeric)	'9F02'	Mandatory
Amount, Other (Numeric)	'9F03'	Optional
Transaction Currency Code	‘5F2A’	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when reader does not support multi-currencies feature)
Terminal Country Code	‘9F1A’	Mandatory(May not be present if it is a static kernel configuration parameter)
Transaction Date	'9A'	Mandatory (May not be present if Transaction Date is managed by the Reader)
Transaction Time	‘9F21’	Optional
Transaction Type	'9C'	Mandatory
Terminal Transaction Data	‘9F76’	Optional
Terminal AID value supported	‘DF7F’	Optional (may be present only when implementation option n°5 is supported)

Table 10 – Transaction Parameters (no implementation option supported)

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Requirement –Information to communicate to the contactless reader

When the PURE contactless reader supports the implementation option 1, the Terminal Application must provide at least the following list of transaction-dependent parameters.

	Data Element	Tag	Requirement when transaction type '78'	Requirement for any other transaction Type
T3.2	Transaction Type	'9C'	Mandatory	Mandatory
	Memory Slot Read Template (List of Data Object to retrieve)	'BF71'	Mandatory	Optional (if present, the reader will use it)
	Terminal AID value supported	'DF7F'	Optional (may be present only when implementation option n°5 is supported)	Optional (may be present only when implementation option n°5 is supported)
	Amount, Authorized (Numeric)	'9F02'	Optional (if present, the kernel will ignore it)	Mandatory
	Amount, Other (Numeric)	'9F03'	Optional (if present, the kernel will ignore it)	Optional
	Transaction Currency Code	'5F2A'	Optional (if present, the kernel will ignore it)	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when reader does not support multi-currencies feature)
	Terminal Country Code	'9F1A'	Optional (if present, the kernel will ignore it)	Mandatory (May not be present if it is a static kernel configuration parameter)
	Transaction Date	'9A'	Optional (if present, the kernel will ignore it)	Mandatory (May not be present if Transaction Date is managed by the Reader)
	Transaction Time	'9F21'	Optional (if present, the kernel will ignore it)	Optional
	Terminal Transaction Data	'9F76'	Optional	Optional
	Terminal AID value supported	'DF7F'	Optional (may be present only when implementation option n°5 is supported)	Optional (may be present only when implementation option n°5 is supported)

Table 11 – Transaction Parameters (implementation option 1 supported)

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Requirement –Information to communicate to the contactless reader

When the PURE contactless reader supports the implementation option 2, the Terminal Application must provide at least the following list of transaction-dependent parameters.

T3.3

Data Element	Tag	Requirement when transaction type '79'	Requirement for any other transaction Type
Transaction Type	'9C'	Mandatory	Mandatory
Memory Slot Update Template	'BF70'	Mandatory	Optional (if present, the kernel will process it)
Terminal AID value supported	'DF7F'	Optional (may be present only when implementation option n°5 is supported)	Optional (may be present only when implementation option n°5 is supported)
Amount, Authorized (Numeric)	'9F02'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory
Amount, Other (Numeric)	'9F03'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Optional
Transaction Currency Code	'5F2A'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when reader does not support multi-currencies feature)
Terminal Country Code	'9F1A'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory (May not be present if it is a static kernel configuration parameter)
Transaction Date	'9A'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory (May not be present if Transaction Date is managed by the Reader)
Transaction Time	'9F21'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Optional
Terminal Transaction Data	'9F76'	Optional	Optional
Terminal AID value supported	'DF7F'	Optional (may be present only when implementation option n°5 is supported)	Optional (may be present only when implementation option n°5 is supported)

Table 12 - Transaction Parameters (implementation option 2 supported)

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Requirement –Information to communicate to the contactless reader	
T3.4	When the PURE contactless reader supports several implementation options, the Terminal Application must comply with all the requirements associated to the implementation options supported.

7.5.2 Terminal Application communicates Online Authorization Result

Requirement –Information to communicate to the contactless reader

T4.1

The Terminal Application must provide at least the following list of transaction-dependent after processing an online authorization.

Data Element	Tag	Requirement
Authorization Response Code Values supported: • "00" = Transaction approved • Other value= Transaction declined	'8A'	Mandatory
Amount, Authorized (Numeric)	'9F02'	Mandatory
Amount, Other (Numeric)	'9F03'	Optional
Transaction Currency Code	'5F2A'	Mandatory
Terminal Transaction Data	'9F76'	Optional
Issuer Authentication Data	'91'	Optional
Critical Issuer Script Command Template	'72'	Optional
Non-critical Issuer Script Command Template	'71'	Optional

Table 13 - Transaction Parameters after completion of an online authorization

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7.5.3 Terminal Application starts an Application Authentication Transaction

Requirement –Information to communicate to the contactless reader

When the PURE contactless reader supports the implementation option 4 and does not support the implementation option 1 or the implementation option 2, the Terminal Application must provide at least the following list of transaction-dependent parameters.

T5.1

Data Element	Tag	Requirement
Transaction Type	'9C'	Mandatory
Terminal AID value supported	‘DF7F’	Optional (may be present only when implementation option n°5 is supported)

Table 14 – Transaction Parameters for Application Authentication Transaction (only implementation option 4 supported)

When the PURE contactless reader supports also the implementation option 1, the Terminal Application must provide at least the following list of transaction-dependent parameters.

T5.2

Data Element	Tag	Requirement for any other transaction Type
Transaction Type	'9C'	Mandatory
Memory Slot Read Template (List of Data Object to retrieve)	‘BF71’	Optional (if present, the reader will use it)
Terminal AID value supported	‘DF7F’	Optional (may be present only when implementation option n°5 is supported)

Table 15 – Transaction Parameters for Application Authentication Transaction (implementation option 1 and option 4 supported)

When the PURE contactless reader supports also the implementation option 2, the Terminal Application must provide at least the following list of transaction-dependent parameters.

T5.3

Data Element	Tag	Requirement for any other transaction Type
Transaction Type	'9C'	Mandatory
Memory Slot Update Template	‘BF70’	Optional (if present, the kernel will process it)
Terminal AID value supported	‘DF7F’	Optional (may be present only when implementation option n°5 is supported)

Table 16 - Transaction Parameters for Application Authentication Transaction (implementation option 2 and option 4 supported)

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Requirement –Information to communicate to the contactless reader	
T5.4	When the PURE contactless reader supports several implementation options, the Terminal Application must comply with all the requirements associated to the implementation options supported.

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8 PURE Contactless Reader Detailed Specifications

8.1 Transaction Flow

The PURE contactless reader is using the following transaction flow:

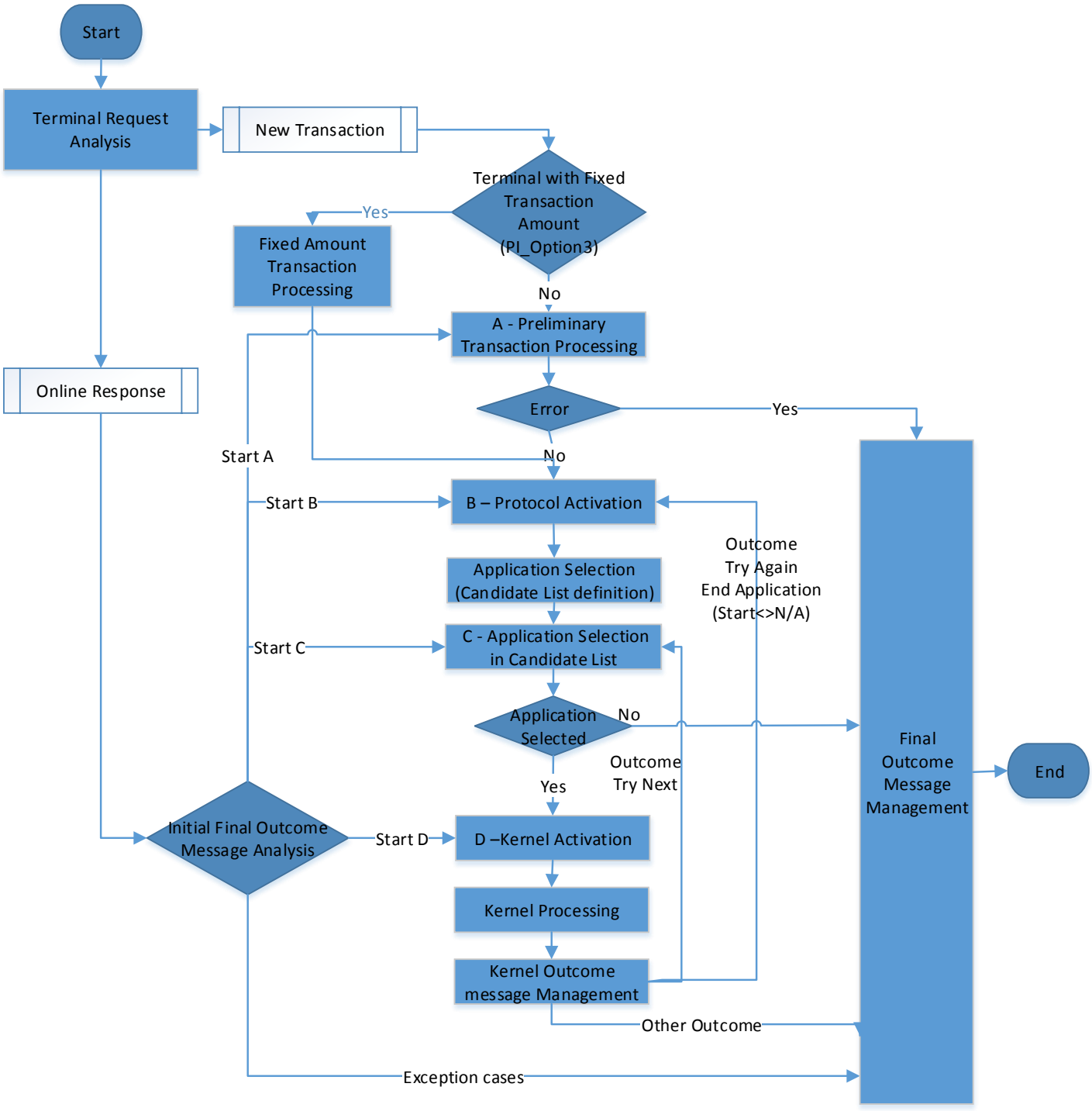


Figure 4 PURE contactless reader transaction flow

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The following section 8.2 provides a detailed description of each step of the transaction flow.

Each section in section 8.2 contains the following sub-sections:

Requirement – Conditions of Execution	
Rx.y	When present, this sub-section provides the pre-requisite for processing the function described in the section. Example: This sub-section may indicate that the processing of this function is mandatory or may depend on the support of an implementation option

Requirement – Sequence of Execution	
Rx.y	This sub-section provides what will be the next step at the completion of the function.

Requirement – <Name of the function>	
Rx.y	This sub-section provides a detailed description of the function.

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8.2 Detailed Description of the transaction flow

8.2.1 Contactless Reader Database

Requirement – Contactless Reader Database

R1.1

For processing contactless transactions requested by the Terminal Application, the PURE contactless reader needs availability of the following types of information:

- Static Configuration Parameters (see section 6.2.2)
- Previous Reader Transaction Context Parameters(information memorized by the contactless reader related to the current terminal request or to the previous terminal request) (see Table 17)
- Transaction-transient parameters(information memorized by the contactless reader only during processing a terminal request) (see Table 18)
- Dynamic Transaction Parameters communicated by the Terminal Application for the current terminal request (see section 7.5)

Management of these data is specific to each contactless reader implementation.

R1.2

The contactless reader has to support the following Previous Reader Transaction Context Parameters

Name
Restart Flag
Application Candidate List
Currently Selected Application
Last Kernel Outcome Message
Contactless Reader Timeout Indicator

Table 17: PURE contactless reader Previous Reader Transaction Context Parameters

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Requirement – Contactless Reader Database

R1.3

A PURE contactless reader has to support the transaction-transient parameters defined in Table 18 or equivalent other parameters. Some of these parameters are depending on the AID selected.

Name	Varies by
AID Contactless Pre-Processing Indicators (It is a static configuration parameter when the implementation 3 (Fixed-amount transaction terminal) is supported)	AID
Current Contactless Pre-Processing Indicators Value of Contactless Pre-Processing Indicators associated to the application AID selected	POS
Current Process Indicator This indicator specifies the type of request communicated by the Terminal Application. The possible values are: • New: new transaction • Online Response: Terminal Application has communicated the online response.	POS

Table 18: Transaction-transient Parameters in PURE contactless reader

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8.2.2 Terminal Application Request Analysis

Requirement – Conditions of Execution	
R2.1	This function is the first function processed, each time the Terminal Application activates the contactless reader. The Terminal Application calls this function in two distinct cases: • Initialization of a new transaction • Communication of the response of an online authorization request Note: At any time, the Terminal Application may request the PURE contactless reader to cancel the transaction (see section 8.2.12).

Requirement – Sequence of Execution	
R2.2	The next step is depending on the Terminal Application request • In case of initialization of a new transaction ○ if the implementation option 3 is supported and is selected by the Acquirer (terminal supporting fixed-amount transaction) (**IO_Option3** = 1), the next step is defined in section 8.2.4 ○ In the other case, the next step is defined in section 8.2.3 • In the case of communication of an online authorization response, the next step is depending on the step that the kernel has specified in the last kernel outcome message. • In case of error the next step is defined in section 8.2.10.

Requirement – Terminal Application Request Analysis	
R2.3	**1-If** the Terminal Application has activated the PURE contactless reader for a "new transaction" **1-Then** • Set 'Current Process Indicator' to 'New' • The PURE contactless reader has to reset some internal parameters: • Set 'Restart Flag' to 'No' • The 'Application Candidate List' must be cleared (set to NULL). • Reset 'Currently Selected Application' to NULL • Reset the contents of the "Last Kernel Outcome Message" parameter (set to NULL) • Set 'Current Contactless Pre-Processing Indicators' to zero • The PURE contactless reader has to stop the timeout function if it was running, and set 'Contactless Reader Timeout Indicator' to '0' • **2- If** the implementation option 3 is supported and is selected by the Acquirer (terminal supporting fixed-amount transaction) (**IO_Option3** = 1), **2-Then** the next step is defined in section 8.2.4 **2-Else**, the next step is defined in section 8.2.3

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Requirement – Terminal Application Request Analysis	
	2-Endif **1-Else**
R2.4	Set 'Current Process Indicator' to 'Online Response'
R2.5	*'Check if the conditions for restarting the transaction are satisfied* **2-If** 'Last Kernel Outcome Message' is not available (NULL) **2-Or** The timeout limit is reached ('Contactless Reader Timeout Indicator' equal to '1') **2-Then** error = 1 **2-Else** The PURE contactless reader has to stop the timeout function if it was running, and set 'Contactless Reader Timeout Indicator' to '0' **3-If *Start*** parameter in the 'Last Kernel Outcome Message' available is equal to 'N/A' **3-Or [** ***[Online Response Data*** parameter in the 'Last Kernel Outcome Message' available is equal '**N/A**' ***3-Or*** ***Online Response Data*** parameter in the 'Last Kernel Outcome Message' available is equal to '***EMV Data***' **and** none of the following tag values ('91', '71' or '72') is present in the list of Dynamic Transaction Parameters communicated by the Terminal Application] **3-Then** error = 1 **3-Else** Depending on ***Start*** parameter in the 'Last Kernel Outcome Message' available: Case ***Start = B:*** **4-If** [No 'Application Candidate List' defined (value NULL) **4-Or** Currently Selected Application not defined (value NULL)] **4- Then** error = 1 **4-Endif** Case ***Start = C:*** **4-If** No 'Application Candidate List' defined (value NULL) **4- Then** error = 1 **4-Endif** Case ***Start = D:*** **4-If** No 'Application Candidate List' defined (value NULL) **4-Or** Currently Selected Application not defined (value NULL) **4- Then** error = 1 **4-Endif** **3-Endif** **2-Endif**
R2.6	**2-If error = 1** **2-Then**

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Requirement – Terminal Application Request Analysis	
	The PURE contactless reader must stop the transaction and must communicate to the Terminal Application the "**End Application**" final outcome message with the following parameters: **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** No **UI Request on Restart Present:** No **Data Record Present:** No **Discretionary Data Present:** No **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero The next step is defined in section 8.2.10.
	2-Else • Depending ***Start*** parameter in the 'Last Kernel Outcome Message' available: Case ***Start* = *A:*** **3- If** the implementation option 3 is supported and is selected by the Acquirer (terminal supporting fixed-amount transaction) (**IO_Option3** = 1), **3-Then** the next step is defined in section 8.2.4 **3-Else**, the next step is defined in section 8.2.3 **3-Endif** Case ***Start* = *B:*** The next step is defined in section 8.2.5.
R2.7	Case ***Start* = *C:*** The next step is defined in section 8.2.7 Case ***Start* = *D:*** The reader has to check ***UI Request on Restart Present*** parameter in the "Last Kernel Outcome Message". If ***UI Request on Restart Present*** parameter is equal to 'Yes' then the reader has to send a **User Interface Request** message using as parameter values, the sub-parameters associated to ***UI Request on Restart Present*** parameter The next step is defined in section 8.2.8.[14] **2-Endif** **1-Endif**

[14] When several kernels may be supported (outside scope of this document), the Contactless reader shall ensure that the correct kernel will be used. Information related to the kernel to use may be added in PURE contactless reader Previous Reader Transaction Context Parameters.

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8.2.3 Step A – Preliminary Transaction Processing

Requirement – Sequence of Execution	
R3.1	When at least one of the application supported may be used for the current contactless transaction, the next step is defined in section 8.2.5. In the other case, the next step is defined in section 8.2.10.

Preliminary Transaction Processing Overview
For each AID supported, it is required to perform the following checks • Reset transaction dependent indicators • Check if Status Check Transaction • Check if Zero Amount transaction • Check Reader Contactless Transaction limit • Check Reader Contactless Floor limit • Check CVM Required limit The result is stored in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. If the "Contactless Application Not Allowed" bit value in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to each AID, is equal to '1', the PURE contactless reader must request to use another interface.

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - *Reset transaction dependent indicators*	
R3.2	Reset (force to zero) the value of the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to each AID supported by the terminal. **If** Transaction Type ('9C') is equal to Trans_Type_Auth_Appli **and** Implementation option 4 is supported **then** the next step is defined in section 8.2.5. **I**n the other cases, continue the review of the other requirements

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - *Check if the transaction is a Status Check Transaction*	
R3.3	**If** all the following are true: • Status Check Flag associated to the AID and the transaction type ('9C') is present and is set

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Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - *Check if the transaction is a Status Check Transaction*	
	• Amount, Authorized ('9F02') is equal to a single unit of the currency used by the terminal (single unit value associated to Transaction Currency Code ('5F2A')[15]) **Then** Set to 1 the "Status Check Requested Indicator" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Endif**

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - *Check if the transaction is a Zero Amount Transaction*	
R3.4	**If** all the following are true: • Zero Amount Allowed flag associated to the AID and the transaction type ('9C') is present and is set • Transaction Type ('9C') is equal to '00' • Amount, Authorized ('9F02') is equal to '000000000000' **Then** Set to 1 the "Zero Amount Indicator" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Endif**

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - Check Reader Contactless Transaction limit	
R3.5	**If** the following conditions are both true: • Reader Contactless Transaction Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') is present • Amount, Authorized ('9F02') is greater than or equal to Reader Contactless Transaction Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') **Then** Set to 1 the "Contactless Application Not Allowed" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Endif**

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - Check Reader Contactless Floor limit	
R3.6	**If** the following conditions are both true:

[15] In the case of Euro, a single unit of currency corresponds to the transaction amount 000000000100 (1.00€).

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Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - Check Reader Contactless Floor limit	
	• Reader Contactless Floor Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') is present • Amount, Authorized ('9F02') is greater than or equal to Reader Contactless Floor Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') **Then** **S**et to 1 the "Reader Contactless Floor Limit Exceeded Indicator" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Else** **If** the following conditions are both true: • Terminal Floor Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') is present • Amount, Authorized ('9F02') is greater than or equal to Terminal Floor Limit associated to the AID, the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') **Then** **S**et to 1 the "Reader Contactless Floor Limit Exceeded Indicator" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Endif** **Endif**

Requirement –Preliminary Transaction Processing Overview – Loop on AIDs supported - Check CVM Required limit	
	If the following conditions are both true: • Reader CVM Required Limit associated to the AID the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') is present • Amount, Authorized ('9F02') is greater than or equal to Reader CVM Required Limit associated to the AID the transaction type ('9C') and to the currency used by the terminal (Transaction Currency Code – '5F2A') •
R3.7	**Then**, Set to 1 the "CVM Required Limit Exceeded Indicator" bit in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID. **Endif**

Requirement –Preliminary Transaction Processing Overview – Completion	
R3.8	**If** for all the AIDs supported:

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Requirement –Preliminary Transaction Processing Overview – Completion

- the "Contactless Application Not Allowed" bit value in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the AID, is equal to '1'

Then

The PURE contactless reader must stop the transaction and must communicate to the Terminal Application the "**Try Another Interface**" final outcome with the following parameters:.

Start: N/A
Online Response Data: N/A
CVM: N/A
UI Request on Outcome Present: Yes
Message Identifier: '18' ("Please insert or swipe card")
Status: Ready to Read
UI Request on Restart Present: No
Data Record Present: No
Discretionary Data Present: No
Alternate Interface Preference: Contact Chip
Receipt: N/A
Field Off Request: N/A
Removal Timeout: Zero

The next step is defined in section 8.2.10

Else

The next step is defined in section 8.2.5.

Endif

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8.2.4 Fixed-amount transaction Processing

Requirement – Conditions of Execution	
R4.1	Support of this function is required only when the implementation option 3 is supported.

Requirement – Sequence of Execution	
R4.2	The next step is defined in section 8.2.5.

Requirement – Fixed-amount transaction Processing	
R4.3	For transaction other than Application Authentication transaction (Transaction type not equal to equal to Trans_Type_Auth_Appli when Implementation option 4 is supported), when the implementation option 3 (fixed transaction amount" is supported and selected(**IO_Option3** = 1), the PURE contactless reader must know for each AID supported and each type of financial transaction supported, the value of the transient transaction parameter 'AID Contactless Pre-Processing Indicators' corresponding to the fixed transaction amount value. For Application Authentication transaction all the 'AID Contactless Pre-Processing Indicators' are set to '0'.

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8.2.5 Step B – Protocol Activation

Requirement – Sequence of Execution	
R5.1	When a card is in communication with the reader, the next step is depending on the value of "Current Process Indicator"

Requirement – Protocol Activation	
R5.2	**If** 'Restart Flag' is set to 'No' *'New transaction* **Then** • Set 'Current Contactless Pre-Processing Indicators' to zero • The 'Application Candidate List' must be cleared (set to NULL). • Reset 'Currently Selected Application' to NULL **Endif**
R5.3	**If** all the following are true: • 'Restart Flag' is set to 'No' **OR** • the value of the "UI Request on Restart Present" parameter in the 'Last Kernel Outcome Message' available is 'No' **Then** Send the **User Interface Request** message with the following parameters: • Message Identifier: '15' ("Present Card") • Status: Ready to Read **Else** Send a **User Interface Request** message using as parameter values, the sub-parameters associated to ***UI Request on Restart Present*** parameter in the 'Last Kernel Outcome Message' **Endif**
R5.4	The field shall be powered up and polling performed as defined in the EMV specification (Reference EMV-7)
R5.5	**If** a collision as defined in the EMV specification (Reference EMV-7) is detected, **Then** send a **User Interface Request** message with the following parameters: • Message Identifier: '19' ("Please Present One Card Only") • Status: Processing Error **Endif**
R5.6	**When** the collision condition is no longer indicated, **Then** send a **User Interface Request** message with the following parameters: • Message Identifier: '19' ("Please Present One Card Only") • Status: Ready to Read **End**
R5.7	When a card is in communication with the contactless reader: **If** 'Current Process Indicator' is set to "Online Response" **Then** the next step is defined in section 8.2.7 **Else** the next step is defined in section 8.2.6. **Endif**

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Requirement – Protocol Activation – Communication Errors	
R5.8	**If** at any time during Protocol Activation, a communication errors as defined in EMV Specifications (Reference EMV-7) is reported to the contactless reader, **Then** The PURE contactless reader shall restart Protocol Activation function from the first requirement R5.2. **Endif**

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8.2.6 Application Selection (Application Candidate List Definition)

Requirement – Sequence of Execution

R6.1	The next step is defined in section 8.2.7

Application Selection – Application Candidate List Definition Overview

Note: It is supposed that all the applications (AIDs) supported by the terminal are using the PURE contactless kernel. The support of additional contactless kernels is allowed but is outside the scope of this specification. If an application on the card needs the usage of a specific kernel, Kernel ID may be personalized in Directory Entry associated to that application in the response of the PPSE selection. Mechanism to link a contactless kernel with a Kernel ID value retrieved in the Directory Entry associated to that application is outside the scope of this specification.
Also the management of Extended Selection parameter (tag '9F29') defined in EMV specification (EMV-6) is outside the scope of this specification.

Selection of the card payment application in contactless mode is similar to the EMV contact application selection process. The PURE contactless reader has to determine the list of card payment applications mutually supported by the contactless device and by the terminal. In case of multiple applications mutually supported, the PURE contactless reader selects automatically the application with the highest priority.

- The PPSE provides to the terminal the list of contactless applications available in the contactless device. The list is available in the response of the selection of the PPSE (see section 11.2.34).
- When the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F'), the list of applications supported by the terminal is restricted to the applications associated to an AID value with most significant bytes equal to the value communicated by the Terminal Application
- The applications supported by the terminal that are associated to a "Contactless Application Not Allowed" bit set to '1' in the parameter 'AID Contactless Pre-Processing Indicators', are excluded from the list of applications that may be selected. The list of applications mutually supported composes the 'Application Candidate List' that will be used in the next step for selecting the application to use for the current transaction.

Example:

Let suppose that the AIDs supported by the terminal are:

- D9 99 99 99 99 10 10
- D9 99 99 99 99 20 20
- D8 88 88 88 88 10 10
- D8 88 88 88 88 20 20

Let suppose that the AIDs present in the card n°1 are:

- D9 99 99 99 99 10 10 with priority 1

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Application Selection – Application Candidate List Definition Overview

- D8 88 88 88 88 10 10 with priority 2
- D7 77 77 77 77 10 10 with priority 3

Let suppose that the AIDs present in the card n°2 are

- D7 77 77 77 77 10 10 with priority 1

The result of application selection when the card n°1 is in communication with the terminal in contactless mode, is:

1. If the implementation option 5 is not supported
   The application selected is D9 99 99 99 99 10 10
2. If the implementation option 5 is supported but the terminal application does not communicate a 'Terminal AID values supported' (tag 'DF7F') value,
   The application selected is D9 99 99 99 99 10 10
3. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D8 88 88 88 88
   The application selected is D8 88 88 88 88 10 10
4. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D7 77 77 77 77
   The application selected is D9 99 99 99 99 10 10
5. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D6 66 66 66 66
   The application selected is D9 99 99 99 99 10 10

The result of application selection when the card n°2 is in communication with the terminal in contactless mode, is

1. If the implementation option 5 is not supported
   The candidate list is empty
2. If the implementation option 5 is supported but the terminal application does not communicate a 'Terminal AID values supported' (tag 'DF7F') value,
   The candidate list is empty
3. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D8 88 88 88 88
   The candidate list is empty
4. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D7 77 77 77 77
   The candidate list is empty

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Application Selection – Application Candidate List Definition Overview

5. If the implementation option 5 is supported and the terminal application communicates a 'Terminal AID values supported' (tag 'DF7F') with the value D6 66 66 66 66
   The candidate list is empty

The PURE contactless reader shall process all the requirements defined in this section in the sequential order defined below.

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Requirement – Application Selection - Application Candidate List Definition	
R6.2	The PURE contactless reader shall send the SELECT command for selecting the PPSE using AID '2PAY.SYS.DDF01'
R6.3	**If** the SELECT command returns a SW1SW2 = '9000' **Then** continue to the next requirement R6.4 **Else** the PURE contactless reader has to process the next step defined in section 8.2.7 **Endif**
R6.4	**If** PPSE Selection Response is badly structured **Then** the PURE contactless reader has to process the next step defined in section 8.2.7 **Else** continue to the next requirement R6.5 **Endif** Note: PPSE Selection Response is badly structured when at least one of the following errors exists: • Response not encapsulated in data element tag '6F' • Response not structured as defined in section 11.2.34 (not TLV structured) • Data element tag 'A5' is not present • Data element tag 'BF0C' is not present • No Directory Entry tag '61' present inside FCI Issuer Discretionary Data tag 'BF0C'
R6.5	The PURE contactless reader must analyze sequentially each of the Directory Entry present in FCI Issuer Discretionary Data tag 'BF0C', starting by the first directory entry. **1- For each** Directory Entry, the checks to perform are: **2-If** ADF Name (tag '4F') is not present **OR** has a format error (its value has a length strictly less than 5 bytes or more than 16 bytes) **2-Then** the PURE contactless reader shall proceed to the next Directory Entry **2-Else** **3-If** [[ADF Name (tag '4F') fully matches (same length and same value) with one of the AID supported by the terminal **OR** ADF Name (tag '4F') partially matches (full value of one of the AID supported by the terminal matches with most significant bytes of ADF Name)[16]] **AND** [('Terminal AID values supported' (tag 'DF7F') is not communicated by the terminal application) **OR** (**IO_Option5** = 0) 'implementation option 5 is not supported **OR** **[** (IO_Option5 = 1) 'implementation option 5 is supported

[16] Example: ADF AID 'A000987755101001' partially matches with terminal AID 'A00098775510'.

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Requirement – Application Selection - Application Candidate List Definition

AND
('Terminal AID values supported' (tag 'DF7F') is communicated by the terminal application
AND
[ADF Name (tag '4F') fully matches (same length and same value) with 'Terminal AID value supported' **OR**
ADF Name (tag '4F') partially matches ('Terminal AID value supported' matches with most significant bytes of ADF Name)[17]]
]
]
]
3-Then
4-If the "Contactless Application Not Allowed" bit value in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the terminal AID that matches with ADF Name, is equal to '1'
4-Then
The PURE contactless reader shall proceed to the next Directory Entry
4-Else
The PURE contactless reader shall add the application referenced in the Directory Entry to the 'Application Candidate List' [ADF Name, Application Priority Indicator(if present) and Kernel ID (if present)][18]
The PURE contactless reader shall proceed to the next Directory Entry

Note: If the PURE contactless reader supports several contactless kernels, some additional checks may be supported for example on Kernel ID if it is present in the Director Entry, (outside the scope of this document).
4-Endif
3-Else
The PURE contactless reader shall process to the next Directory Entry
3-Endif
2-Endif
1- EndFor
1- IF the 'Application Candidate List' is empty **AND** ('Terminal AID values supported' (tag 'DF7F') is communicated by the terminal application AND (**IO_Option5** = 1)

1-THEN

2- For each Directory Entry, the checks to perform are:

[17] Example: ADF AID 'A000987755101001' partially matches with 'Terminal AID value supported' 'A00098'.

[18] Some additional parameters may be added to the entry of the Application Candidate List for satisfying to Payment Scheme specific requirements related to application selection (for example a Merchant Application Priority Indicator allowing to sort application per Merchant preference). Definition of these additional parameters is outside the scope of this specification. The presence of additional parameters has no impact on the reader certification process but it is not checked.

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Requirement – Application Selection - Application Candidate List Definition

3-If ADF Name (tag '4F') is not present **OR** has a format error (its value has a length strictly less than 5 bytes or more than 16 bytes)

3-Then

the PURE contactless reader shall proceed to the next Directory Entry

3-Else

4-If [ADF Name (tag '4F') fully matches (same length and same value) with one of the AID supported by the terminal **OR**

ADF Name (tag '4F') partially matches (full value of one of the AID supported by the terminal matches with most significant bytes of ADF Name)[19]]

4-Then

5-If the "Contactless Application Not Allowed" bit value in the transient transaction parameter 'AID Contactless Pre-Processing Indicators' associated to the terminal AID that matches with ADF Name, is equal to '1'

5-Then

The PURE contactless reader shall proceed to the next Directory Entry

5-Else

The PURE contactless reader shall add the application referenced in the Directory Entry to the 'Application Candidate List' [ADF Name, Application Priority Indicator(if present) and Kernel ID (if present)][20]

The PURE contactless reader shall proceed to the next Directory Entry

Note: If the PURE contactless reader supports several contactless kernels, some additional checks may be supported for example on Kernel ID if it is present in the Director Entry, (outside the scope of this document).

5-Endif

4-Else

[19] Example: ADF AID 'A000987755101001' partially matches with terminal AID 'A00098775510'.

[20] Some additional parameters may be added to the entry of the Application Candidate List for satisfying to Payment Scheme specific requirements related to application selection (for example a Merchant Application Priority Indicator allowing to sort application per Merchant preference). Definition of these additional parameters is outside the scope of this specification. The presence of additional parameters has no impact on the reader certification process but it is not checked.

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Requirement – Application Selection - Application Candidate List Definition

The PURE contactless reader shall process to the next Directory Entry

4-Endif

3-Endif

2- EndFor

1-ENDIF

The PURE contactless reader has to process the next step defined in section 8.2.7.

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8.2.7 Step C – Application Selection using the Application Candidate List

Requirement – Sequence of Execution	
R7.1	The next step is depending on the processing of the 'Application Candidate List'. • If it is possible to select an application, the next step is defined in section 8.2.8 • In case of communication error, the next step is defined in section 8.2.5 • If no application can be selected, the next step is defined in section 8.2.10.

Requirement – Application Selection using the Application Candidate List	
R7.2	**1-If** 'Current Process Indicator' is set to "Online Response" **1-Then** Select the application using the SELECT command and the AID memorized in 'Currently Selected Application' parameters. **2-If** the response to the SELECT command includes a SW1SW2 other than '9000' **2-Then** The PURE contactless reader must stop the transaction and must communicate to the Terminal Application the "**End Application**" final outcome message with the following parameters: **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** No **UI Request on Restart Present:** No **Data Record Present:** No **Discretionary Data Present:** No **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero The next step is defined in section 8.2.10. **2-Else** • The next step is defined in section 8.2.8. **2-Endif** **1-Else**
R7.3	**2-If** the 'Application Candidate List' is empty **2-Then** The PURE contactless reader must stop the transaction and must communicate to the Terminal Application the "**End Application**" final outcome with the following parameters **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** Yes Message Identifier: '1C' ("Insert, Swipe or Try Another Card")

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Requirement – Application Selection using the Application Candidate List

Status: Ready to Read
UI Request on Restart Present: No
Data Record Present: No
Discretionary Data Present: No
Alternate Interface Preference: N/A
Receipt: N/A
Field Off Request: N/A
Removal Timeout: Zero

The next step is defined in section 8.2.10

2-Else

3-If the 'Application Candidate List' contains more than one application

3-Then

The PURE contactless reader has to determine the application to select as follow[21]

- Consider each application that has an Application Priority Indicator with the value 0 or no application priority defined to be of equal lowest priority
- Ignore the value of the most significant nibble of Application Priority Indicator byte. Only the least significant nibble must be used.[22]
- If an application has a higher priority than any other applications in the 'Application Candidate List', select that application
- If several applications in the 'Application Candidate List' have the highest priority, then select the application that occurs first in the PPSE

3-Else

The application to select is the application present in the 'Application Candidate List'

3-Endif

2-Endif

Select the application using the SELECT command and the AID associated to the application selected based on 'Application Candidate List'.

2-If the response to the SELECT command includes a SW1SW2 other than '9000'

2-Then

- Remove the application selected from the 'Application Candidate List'

[21] Payment Scheme may require an alternative method for selecting one of the applications present in the candidate List. For example, the application selection may be based on merchant preference and not on card issuer preference. Algorithm to use for satisfying Payment Scheme specific requirements is outside the scope of this document but it is permitted to support it. When the reader supports a specific application selection process, the reader configuration used for reader certification must allow a selection only based on Issuer preference for complying with requirement R7.3 (for example, Merchant Application Priority Indicator[18] allocated to applications may be the same for all applications in order to do a selection only based on Application priority defined by the card Issuer)

[22] Application Priority Indicator most significant bit allowing to mandate cardholder confirmation before selecting the application, is ignored in contactless mode.

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Requirement – Application Selection using the Application Candidate List	
	• Process again application selection (R7.3) using the updated 'Application Candidate List' **2-Else** • Set 'Currently Selected Application' to the value of the AID selected. • Initialize the value of 'Current Contactless Pre-Processing Indicators' to the value of the 'AID Contactless Pre-Processing Indicators' associated to the AID selected • The next step is defined in section 8.2.8. **2-Endif** **1-Endif**

Requirement – Application Selection using the 'Application Candidate List'– Communication Errors	
R7.4	**If** at any time during Application Selection using the 'Application Candidate List', a communication errors as defined in EMV Specifications (Reference EMV-7) is reported to the contactless reader, **Then** the PURE contactless reader shall restart Protocol Activation (see section 8.2.5) function from the first requirement. **Endif**

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8.2.8 Step D – Kernel Activation

Requirement – Sequence of Execution	
R8.1	The next step is defined in section 8.2.9.

Requirement – Kernel Activation	
R8.2	The PURE contactless reader has to generate a new unpredictable number using a method equivalent to the method used in EMV contact transaction. This value is allocated to the terminal parameter: Unpredictable Number (tag '9F37') Note: An alternative implementation is to let the kernel to generate the Unpredictable Number (see K2.3)
R8.3	The PURE contactless reader must support the PURE contactless kernel. The support of additional contactless kernels by the PURE contactless reader is outside of scope of this specification.
R8.4	The contactless reader must communicate the following data elements to the PURE contactless kernel: **1-If** the value of 'Current Process Indicator' is set to 'Online Response', **1-Then** the minimum list of data elements that should be available to the PURE contactless kernel is defined in Table 27. **Note:** **2-If** ***Start*** parameter in the 'Last Kernel Outcome Message' is equal to the value '***Start = D***' **2-Then** SELECT Response Message Data Field (FCI) of the selected application shall <u>not</u> be communicated to the PURE contactless kernel. **2-Else** SELECT Response Message Data Field (FCI) of the selected application shall be communicated to the PURE contactless kernel. **2-Endif** **1-Else** the minimum list of data elements that should be available to the PURE contactless kernel is depending on the implementation options supported and on the type of transaction **2-If** Transaction Type = Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1) **2-And** (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli)) **2-Then** • no implementation option supported: minimum list of data elements is defined in Table 28, • Implementation option 1 supported (**IO_Option1** = 1): minimum list of data elements is defined in Table 29. This list is depending on the type of transaction • Implementation option 2 supported (**IO_Option2** = 1): minimum list of data elements is defined in Table 30. This list is depending on the type of transaction • When several implementation options are supported, the minimum list of data elements must comply with all the requirements associated to the implementation options supported. **2-Else**

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Requirement – Kernel Activation

- no implementation option supported: minimum list of data elements is defined in Table 24,
- Implementation option 1 supported (**IO_Option1** = 1): minimum list of data elements is defined in Table 25. This list is depending on the type of transaction
- Implementation option 2 supported (**IO_Option2** = 1): minimum list of data elements is defined in Table 26. This list is depending on the type of transaction
- When several implementation options are supported, the minimum list of data elements must comply with all the requirements associated to the implementation options supported.

2-Endif

Note:

SELECT Response Message Data Field (FCI) of the selected application shall be communicated to the PURE contactless kernel.

1-Endif

The origin of these data elements may be:

- Dynamic Transaction Parameters that the Terminal Application has communicated to the contactless reader
- Result of contactless reader processing (application selection response (FCI), 'Unpredictable Number'[23] and 'Current Contactless Pre-Processing Indicators')
- If the resulting buffer exceeds the maximum length supported by the kernel (see K20.6), the PURE contactless reader must stop the transaction and must communicate to the Terminal Application the "**End Application**" final outcome message with the following parameters:

  Start: N/A
  Online Response Data: N/A
  CVM: N/A
  UI Request on Outcome Present: No
  UI Request on Restart Present: No
  Data Record Present: No
  Discretionary Data Present: No
  Alternate Interface Preference: N/A
  Receipt: N/A
  Field Off Request: N/A
  Removal Timeout: Zero

The next step is defined in section 8.2.10

[23] An alternative implementation is to let the kernel to generate the Unpredictable Number.

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8.2.9 Kernel Processing

Requirement – Conditions of Execution	
R9.1	The PURE contactless reader must support the PURE contactless kernel. The support of additional contactless kernels by the PURE contactless reader is outside of scope of this specification.

Requirement – Sequence of Execution	
R9.2	The next step is defined in section 8.2.10.

Requirement – Kernel Processing	
R9.3	All requirements related to kernel processing are described in section 8.2.14.

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8.2.10 Kernel Outcome Message Management

Overview

This function is dedicated to the analysis of all the outcome messages communicated by the PURE contactless kernel or generated by the PURE contactless reader in case of issue.

Requirement – Sequence of Execution

R10.1 The next step is depending on the type of outcome message analyzed.

Requirement – Kernel Outcome Message Management Processing

R10.2

The PURE contactless reader has to check ***Field Off Request*** parameter in the outcome message to analyze.

1-If *Field Off Request* parameter has a value other than 'N/A'

1-Then

- the contactless field shall be turned off and shall remain off for the period indicated by the hold time (***Field Off Request*** parameter value in units of 100ms)

1-Else

- The PURE contactless reader has to check ***Start*** parameter in the outcome message to analyze.
- **2-If *Start*** parameter is not equal to '***Start C***' and is not equal to '***Start D'***

  2-Then the contactless field shall be turned off.

  2-Endif

1-Endif

R10.3 The PURE contactless reader has to check ***UI Request on Outcome Present*** parameter in the outcome message to analyze. If ***UI Request on Outcome Present*** parameter is equal to 'Yes' then the PURE contactless reader has to send a **User Interface Request** message using as parameter values, the sub-parameters associated to ***UI Request on Outcome Present*** parameter

R10.4 If the "***Removal Timeout***" parameter in the outcome message to analyze, is not equal to zero, then the PURE contactless reader must start a timeout function using the value of the parameter (value 1 means 100ms) and set the 'Contactless Reader Timeout Indicator' to '0'

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Requirement – Kernel Outcome Message Management Processing

The additional requirements related to the processing of outcome message to analyze are defined in Table 19.

	Kernel outcome Message	Contactless reader Processing
R10.5	***Online Request***	• The PURE contactless reader has to keep the message in contactless reader internal static parameter "Last Kernel Outcome Message". • **If** the "***Start***" parameter in the outcome message to analyze is not equal to ***N/A***, **Then** the PURE contactless reader must set 'Restart Flag" parameter to 'Yes'. **Else** the PURE contactless reader must set 'Restart Flag" parameter to 'No'. • The PURE contactless reader forwards this outcome message analyzed to the Terminal Application for processing. When the value of the 'Currently Selected Application (AID of the application currently selected)' is present in the outcome message, it must be also available to the Terminal Application. • For the PURE contactless reader, the transaction is completed. It is up to the Terminal Application to analyze the outcome message and to process the message (see section 7).
	Approved	
	Declined	
	End Application ("Start" = N/A)	
	Try Another Interface	
	End Application ("Start" ≠ N/A)	• The PURE contactless reader has to keep the message in contactless reader internal static parameter "Last Kernel Outcome Message". • The PURE contactless reader must set 'Restart Flag" parameter to 'Yes'. • After confirming that the card has been removed, the next step is the step defined in section 8.2.5 Note: 'Current Process Indicator' is not updated • Note: The PURE contactless reader does not need to check if the "Start" parameter in the outcome message is equal to "B".
	Select Next	• The PURE contactless reader has to remove the application currently selected from the 'Application Candidate List'. The next step is the step defined in section 8.2.7 and shall use the updated 'Application Candidate List'. Note: The PURE contactless reader does not need to check if the "***Start***" parameter in the outcome message is equal to "C".

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Requirement – Kernel Outcome Message Management Processing	
Try Again	• The PURE contactless reader must set 'Restart Flag" parameter to 'Yes'. • The next step is the step defined in section 8.2.5 Note: 'Current Process Indicator' is not updated Note: The PURE contactless reader does not need to check if the "***Start***" parameter in the outcome message is equal to "B".

Table 19 – PURE contactless kernel Outcome Message Processing

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8.2.11 Timeout handling

Requirement – Conditions of Execution	
R11.1	This function is processed when a timeout happens.

Requirement – Timeout handling Processing	
R11.2	When the timeout is reached, • the PURE contactless reader send a **User Interface Request** message with the following parameters: Message Identifier: '17' ("Card Read OK. Please Remove Card") Status: Card Read Successfully (Remove card) • The contactless field shall be turned off • Set the 'Contactless Reader Timeout Indicator' to '1'

8.2.12 Transaction Cancellation

Requirement – Conditions of Execution	
R12.1	At any time, the PURE contactless reader must be capable to receive a request from the Terminal Application, for cancelling a transaction in process.

Requirement – Transaction Cancellation	
R12.2	The PURE contactless reader has to communicate the cancellation request to the Kernel. Method used is specific to each Kernel implementation.

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8.2.13 General Requirements

Requirement – General Requirements	
R13.1	**If** Contactless reader encounters a data object from the contactless card payment application that is designated as terminal or reader sourced, **Then** it shall ignore the data object and continue the transaction as if the data object had not been present. **Endif**
R13.2	**If** Contactless reader encounters a data object from the contactless card payment application that is not recognized, **Then** the unrecognized data object shall be ignored and the transaction shall continue as if the data object had not been present. **Endif**
R13.3	The PURE contactless reader shall apply padding according to the rules defined in the EMV contact specifications.
R13.4	The PURE contactless reader shall accept TLV data elements in any order.
R13.5	The PURE contactless reader shall set any bytes or bits that are specified as Reserved for Future Use (RFU) to zero unless explicitly specified.
R13.6	The PURE contactless reader shall not examine received bytes or bits specified as RFU nor shall the reader behavior depend upon the coding of these bytes or bits, unless explicitly stated.

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8.2.14Reader/Terminal Interface

Requirement – General Requirements	
R14.1	Terminal to reader interface shall comply with section 7.5

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9 PURE contactless kernel Detailed Specifications

9.1 Transaction Flow

The following section and figure provides an overview of the transaction flow supported in PURE contactless kernel. The PURE contactless kernel must comply with the detailed descriptions and the reference requirements that are provided in the following sections.

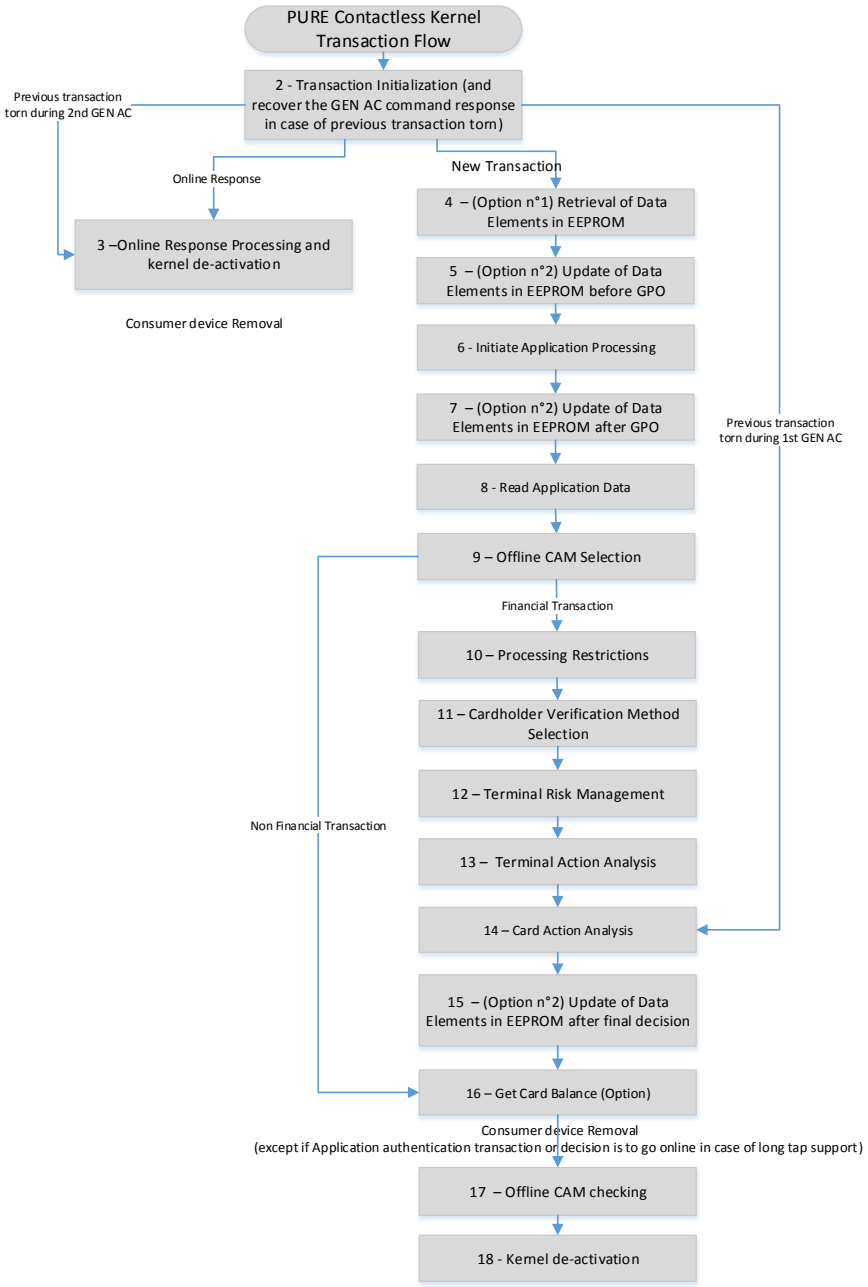


Figure 5 High-Level Sample Transaction Flow

The PURE contactless kernel may interface with a contactless reader compliant with either the PURE contactless reader Specifications or the EMV Entry Point specifications. During PURE contactless kernel activation, contactless reader hands over control to the PURE

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contactless kernel. Once the PURE contactless kernel is activated, all commands and responses with the contactless device are processed by the PURE contactless kernel.

2. Contactless reader activates the PURE contactless kernel to process the transaction. The contactless reader provides transaction data (Dynamic Transaction Parameters), and the relevant [POS, terminal, contactless reader, PURE contactless kernel] configuration parameters to the PURE contactless kernel.

   The PURE contactless kernel initializes some kernel specific parameters (TTPI, TVR, internal indicator...) using information communicated by the contactless reader.

   If the payment transaction was torn during the previous kernel call and torn transaction recovery is supported by the card payment application and the terminal (Acquirer option), the kernel checks if the kernel call is related to a tentative to restart a torn transaction. If it is the case, the kernel sends the ECHO command for retrieval of last GENERATE AC command response. If the transaction was torn during the 1st GENERATE AC command, the next step is step 14 (analysis of GENERATE AC command response). If the transaction was torn during the 2nd GENERATE AC command, the next step is step 3 (analysis of GENERATE AC command response). In case of ECHO command failure, the kernel uses the standard transaction flow

3. If the Acquirer has decided to support either the option “additional tap for Issuer Response communication (AC_Online_Tap)”or the option “long tap (AC_Long_Tap)”, and Issuer Script commands and/or Issuer Authentication Data are present in the parameters sent by the contactless reader, the PURE contactless kernel sends the script commands/Issuer Authentication Data to the Consumer contactless device. The transaction is completed. The PURE contactless kernel provides the outcome message that may contain Issuer Script Result when at least one script commands is processed.

4. If the PURE contactless kernel supports the Implementation option 1 “Retrieval of data element values stored in EEPROM using GET DATA command”

   a. If the type of transaction is “Retrieve data elements-78”, the PURE contactless kernel will try to retrieve the data element values requested by the contactless reader. The transaction is completed after retrieval of the value of each data elements. The PURE contactless kernel will provide the data element values read in the outcome message.

   b. For other types of transaction:

      i. If the contactless reader has requested no data element, the PURE contactless kernel continues the transaction flow.

      ii. If the contactless reader has requested some data elements, the PURE contactless kernel will try to retrieve the data element values requested by the contactless reader and will provide the data element values in the outcome message when the transaction will be completed.

5. If the PURE contactless kernel supports the Implementation option 2 “Update data element values stored in EEPROM using PUT DATA command”

   Note: The contactless reader may request the PURE contactless kernel to update data elements. It also indicates whether the update shall be processed *before* or *after* the GET

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PROCESSING OPTIONS (GPO) command processing or shall be processed after 1st GENERATE AC command processing.

a. If the type of transaction is "Update data elements-79", the PURE contactless kernel will try to update the data elements that need to be updated prior to the GPO command. In case of PUT DATA command error processing, the PURE contactless kernel will either continue the transaction or stop the transaction.
If there is no data element to update after the GPO command or if an error happens and has for consequence to stop the transaction, the transaction is completed. The PURE contactless kernel will provide PUT DATA command processing result in the outcome message.
In the other cases, the PURE contactless kernel continues the transaction flow.

b. For the other types of transaction, the PURE contactless kernel will try to update the data elements that need to be updated prior to the GPO command, if the contactless reader requests it and then continues the transaction flow. In case of PUT DATA command error processing, the PURE contactless kernel will apply either continue the transaction or stop the transaction. If the contactless reader has not requested any data element update, the PURE contactless kernel continues the transaction flow.

6. The PURE contactless kernel sends the GET PROCESSING OPTIONS command to the card to initialize the transaction. The card returns the Application Interchange Profile (AIP) and the Application File Locator (AFL) and optionally the Contactless Cryptogram Information Data (CCID).
When CCID is present in GPO command response,
   - if card decision is to switch to the contact interface, the transaction is completed.
   - if card decision is to request the Customer to provide a Confirmation Code using the contactless device, the transaction is completed.
   - If card preliminary decision is to decline the transaction, a PURE contactless kernel configuration parameter allows to stop the transaction

7. If the PURE contactless kernel supports the Implementation option 2 "Update data element values stored in EEPROM using PUT DATA command", the PURE contactless kernel will try to update the data elements that need to be updated after GPO command,.
If the type of transaction is "Update data elements-79", after updating the data elements, the transaction is completed. The PURE contactless kernel will provide PUT DATA command processing result in the outcome message.
For the other types of transaction, after updating the data elements, the PURE contactless kernel continues the transaction flow. In case of PUT DATA command error processing, the PURE contactless kernel will either continue the transaction or stop the transaction.

8. The PURE contactless kernel reads the card data as indicated by the AFL.

9. Using AIP and PURE contactless kernel configuration parameters, the PURE contactless kernel has to determine

   - For financial transaction: the strongest offline CAM mutually supported (SDA or CDA) and performs some preliminary verification.
   - For Application Authentication transaction: the PURE contactless kernel asks for the generation of a dynamic signature using INTERNAL AUTHENTICATE command.

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For transactions other than Application Authentication transaction, the next step is the step 10. For Application Authentication Transactions, the next step is the step 16.

10. If the PURE contactless kernel is setup for supporting "Processing Restrictions checks", the PURE contactless kernel performs the following verifications:
    a. Application Usage Control
    b. Application Version Number
    c. Application Expiration Date
    d. Application Effective Date
11. The PURE contactless kernel has to determine if cardholder verification is required for the current transaction. If it is the case, the PURE contactless kernel has to determine the method that will be used.
12. If the PURE contactless kernel is setup for supporting "Terminal Risk Management", the PURE contactless kernel performs Terminal Velocity Checking
13. Based on the result of the checks performed during the transaction (result memorized in TVR) and PURE contactless kernel configuration parameters Terminal Action Codes (TAC), the PURE contactless kernel computes a first transaction outcome. If Issuer Action Codes (IAC) parameters are present in the data elements read , the IACs participate also in the terminal decision
14. The PURE contactless kernel issues a GENERATE AC command including Combined Data Authentication (CDA) request when it is the offline CAM selected during step 9.
    If card decision is to switch to the contact interface, the transaction is completed. Card may request to switch to the contact interface either using CCID or by rejecting a terminal request to approve offline a transaction.
15. If the PURE contactless kernel supports the Implementation option 2 "Update data element values stored in EEPROM using PUT DATA command", the PURE contactless kernel will try to update the data elements that need to be updated after processing 1st GENERATE AC command, if the contactless reader requests it.
    After updating the data elements, the PURE contactless kernel continues the transaction flow. In case of PUT DATA command error processing, the PURE contactless kernel will apply contactless reader requirement (either continue the transaction or stop the transaction).
16. If the PURE contactless kernel is setup for card balance retrieval, the PURE contactless kernel sends the GET DATA for retrieval of the card balance and the card currency code. A GET DATA error processing has no impact on the transaction. After completion of the command, the card can be removed from the contactless field except if both the Acquirer has decided to support the 'long tap' option (AC_Long_Tap) and the card decision was to request an online authorization.
17. The PURE contactless kernel performs the offline CAM previously selected
    a. When offline CAM selected is CDA and the card has communicated a CDA signature, the PURE contactless kernel verifies the CDA signature. In case of CDA failure or no CDA signature available, the PURE contactless kernel forces the decision to declined

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b. When offline CAM selected is SDA, the PURE contactless kernel verifies the SDA signature. In case of failure, the PURE contactless kernel forces the decision to declined

c. When offline CAM selected is DDA, the PURE contactless kernel verifies the DDA signature. In case of failure, the PURE contactless kernel forces the decision to declined

18. The PURE contactless kernel provides to the contactless reader, the outcome message in relation with the card decision. When card decision is to request an online authorization and if the Acquirer has decided to support either the option “additional tap for Issuer Response communication (AC_Online_Tap)”or the option “long tap (AC_Long_Tap)”, the outcome message may request to re-activate the PURE contactless kernel after online processing.

At any time, the contactless reader may request the PURE contactless kernel to cancel the transaction in process.

At any time, communication error may happen with the Consumer device. The impact is depending on card payment application setting defined by the Issuer and on terminal setting defined by the Acquirer.
When both the card payment application and the terminal application are supporting torn transaction recovery, the impact of a communication problem with the Consumer device is specific to the command that was processed when the communication problem happens.

- When a communication problem with the Consumer device happens during the processing of the 1st GENERATE AC command or the 2nd GENERATE AC command, the kernel asks the reader to request the cardholder to present again his device for restarting the payment transaction from where the transaction was stopped (the terminal application is not involved)
- In all the other cases, the kernel asks the reader to request the cardholder to present again his device for restarting the payment transaction from the beginning.

In both cases, the terminal application is not involved.

When either the card payment application or the terminal application setting defined by the Acquirer are not supporting torn transaction recovery, the payment transaction is aborted in case of communication problem with the Consumer device. It is up to the terminal application to decide to restart the payment transaction.

The section 9.2 provides details about each step of the transaction.

Note: The following indicators are used for the management of communication problem:

- ‘ECHO Command supported’ (see section 11.2.19.2)
- ‘Communication Problem’ Indicator (see section 11.2.19.1)
- ‘Echo Transaction’ Indicator = 1 (see section 11.2.19.3)

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The following section 9.2 provides a detailed description of each step of the transaction flow.

Each section in section 9.2 contains the following sub-sections:

Requirement – Conditions of Execution	
Rx.y	When present, this sub-section provides the pre-requisite for processing the function described in the section. Example: This sub-section may indicate that the processing of this function is mandatory or may depend on the support of an implementation option

Requirement – Sequence of Execution	
Rx.y	This sub-section provides what will be the next step at the completion of the function.

Requirement – <Name of the function>	
Rx.y	This sub-section provides a detailed description of the function.

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9.2 Detailed description of the transaction flow

This chapter provides detailed transaction processing requirements for PURE contactless kernel including information related to EMV functions.

9.2.1 PURE Contactless Kernel Database

Requirement – PURE Contactless Kernel Database

K1.1

- For processing contactless transaction requested by the contactless reader, the PURE contactless kernel needs availability of the following types of information: 'Kernel Transaction Database' containing:
  - Dynamic Transaction Parameters communicated by the contactless reader for the current transaction (see section 9.3.1)[24]
  - information communicated by the card payment application in the response of the following commands (GET PROCESSING OPTIONS command, the READ RECORD command and the GET DATA command on data elements requested by the terminal application[24]
  - FCI and data elements present in the FCI of the application selected by the reader[24]
  - internal kernel data objects, internal kernel indicators and internal kernel flags (see Table 20)
    Note: These internal kernel parameters are present only in the 'kernel Transaction Database'. All references to these parameters in the document are referring to the parameter values present in the 'Kernel Transaction Database'
- Static Configuration Parameters associated to the AID selected (AID value communicated among the dynamic transaction parameters) (see Table 8)

Management of these data elements is specific to each kernel implementation.

K1.2

A PURE contactless kernel must keep in memory the contents of the 'Kernel Transaction Database' up to the completion of the transaction.

A payment transaction may be split in several taps: online additional tap for processing Issuer Host decision related to the transaction or tap for recovering card decision in case of torn transaction.

The 'Kernel Transaction Database' is reinitialized when the reader starts a new transaction or when the terminal is power off or reset. In this document NULL is used to refer to the empty value.

Management of the 'Kernel Transaction Database' is specific to each implementation.

[24] The kernel may support a list of known data elements (list of referenced tag values) that the Acquirer must be capable to update. It is an implementation option. When this list is supported, the kernel may store in the 'Kernel Transaction Database' only the value of data elements referenced in this list; the data elements not referenced in this list are ignored without any impact on the transaction processing. When this list is supported, all the data elements referenced in this document as either mandatory or optional, shall be referenced in the default list used for kernel certification. When the implementation supports no list of referenced data element, all the data elements are stored in the 'Kernel Transaction Database'.

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Requirement – PURE Contactless Kernel Database	
K1.3	The Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message when a data element is duplicated in the 'Kernel Transaction Database') before first card decision: • A data element is duplicated if a data element already present in the 'Kernel Transaction Database' is associated to the same tag value as the new data element communicated by the card payment application. • A data element is also considered as duplicated if it is present several times in a single command response • The commands that shall not contain in their response data element already present in the 'Kernel Transaction Database', are the GET PROCESSING OPTIONS command and the READ RECORD command. The GET DATA command may communicate information already present in the 'Kernel Transaction Database'. The new value replaces the value already stored in the 'Kernel Transaction Database'.

Data Element Name	Tag
'GDDOL Resulting Buffer'	'9F71'
'Data Elements Update Result'	'9F74'
CVM Result	'9F34'
Terminal Verification Results (TVR)	'95'
'Offline CAM Selected' Indicator	-
'Communication Problem' Indicator	-
'ECHO Command supported' Indicator	-
PDOL related data	-
CDOL1 related Data	-

Table 20: List of internal kernel parameters present in Kernel Transaction Database

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9.2.2 Transaction Initialization

During Transaction Initialization, PURE contactless kernel initializes internal variables and performs verifications.

Requirement – Conditions of Execution	
K2.1	This function is the first function processed, each time the contactless reader activates the PURE contactless kernel.

Requirement – Sequence of Execution	
K2.2	If this step is completed on "terminate", the next step is processed by the contactless reader. In the other case, the next step is depending on the processing of this step. It may be the step defined in one of the following sections: • section 9.2.3 • section 9.2.4 • section 9.2.5 • section 9.2.6 Note: At any time, the contactless reader may request the Kernel to cancel the transaction. In that case, the next step is defined in section 9.2.20

Requirement – Transaction Initialization Processing	
K2.3	When the Kernel is activated, except in the case of transaction cancellation request, the contactless reader shall provide Dynamic Transaction Parameters to the Kernel. The list of parameters is depending on the type of transaction to process and on the implementation options supported (see Table 24, Table 25, Table 26, Table 27, Table 28, Table 29 and Table 30). Note: When the Unpredictable Number is generated by the kernel, new unpredictable number must be generated using a method equivalent to the method used in EMV contact transaction. This value is allocated to the parameter: Unpredictable Number (tag '9F37') The contactless reader may provide additional transaction parameters. The kernel may ignore these parameters. Error = 0 'Echo Transaction' Indicator = 0 **1-If** Authorization Response Code (tag '8A') is present in Dynamic Transaction Parameters **1-Then** Error-Type = 2

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Requirement – Transaction Initialization Processing

2-If one of the following data elements is missing in Dynamic Transaction Parameters ('9F02'[25], '5F2A'[27] and '9F37'[26])
2-Then Error = Error-Type
2-Endif
1-Else
Error-Type = 1
2-If Transaction Type (tag '9C')[25] is not present in Dynamic Transaction Parameters
2-Then Error = Error-Type
2-Else
3-If Implementation option 1 (**IO_Option1** = 1) supported **And** Transaction Type = '78'
3-Then
4-If one of the following data elements is missing in Dynamic Transaction Parameters (Contactless Pre-Processing Indicators , '6F'and 'BF71')
4-Then Error = Error-Type
4-Endif
3-Else
4-If Implementation option 2 supported (**IO_Option2** = 1) **And** Transaction Type = '79'
4-Then
5-If one of the following data elements is missing in Dynamic Transaction Parameters (Contactless Pre-Processing Indicators , '6F'and 'BF70')
5-Then Error = Error-Type
5-Endif
4-Else
5-If Transaction Type **=** Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1)
5-Then
6-If (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli))
6-Then
7-If one of the following data elements is missing in Dynamic Transaction Parameters (Contactless Pre-Processing Indicators , '6F', and '9F37')
7-Then Error = Error-Type
7-Endif
6-Then Error = Error-Type
6-Endif
5-Else

[25] When the kernel design mandates the presence of this data element among the Dynamic Transaction Parameters, checking its presence is not relevant.

[26] When the kernel design mandates the presence of the Unpredictable Number among the Dynamic Transaction Parameters or when the Unpredictable Number is generated by the kernel, checking its presence is not relevant.

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Requirement – Transaction Initialization Processing	
	6-If one of the following data elements is missing in Dynamic Transaction Parameters (Contactless Pre-Processing Indicators , '6F', '9F02'[25], '5F2A'[27], '9F1A'[27], '9A'and '9F37'[26]) **6-Then** Error = Error-Type **6-Endif** **5-Endif** **4-Endif** **3-Endif** **2-Endif** **1-Endif** **1-If** Error ≠ 0 **1-Then** Apply Requirement K2.6 **1-Endif**
K2.4	**1-If** implementation option 2 is supported(**IO_Option2** = 1), **Then** set "Data Elements Update Result" to '0000' **1-Endif** **1-If** implementation option 1 is supported(**IO_Option1** = 1), **Then** set "GDDOL Resulting Buffer' to Empty **1-Endif** **1-If** "SELECT Response Message Data Field (FCI) of the selected application" is part of the dynamic transaction parameters **1-Then** 'FCI shall be structured as defined in the EMV specifications [EMV-1: Section 11.3]. **2-If** the FCI does not parse correctly **2-Then** Error = Error-Type **2-Else** **3-If** 'Echo Card Identifier'('9F75') is part of FCI **3-And** 'ECHO Command supported' Indicator in 'Kernel Transaction Database' is equal to '01' **3-And** the FCI in dynamic transaction parameters is equal to the FCI currently present in 'Kernel Transaction Database' **3-And**'Communication Problem' Indicator in 'Kernel Transaction Database' is not equal to '00' **3-And** Transaction Type ≠ Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1) **3-Then** **4-If** [(Authorization Response Code (tag '8A') is present in Dynamic Transaction Parameters) **And** ('Communication Problem' Indicator in 'Kernel Transaction Database' is equal to '02')]

[27] This data element may not be part of the Dynamic Transaction Parameters if it is part of the static PURE contactless kernel configuration data (see Table 8)

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Requirement – Transaction Initialization Processing

4-Or [(Authorization Response Code (tag '8A') is not present in Dynamic Transaction Parameters) **And** ('Communication Problem' Indicator in 'Kernel Transaction Database' is equal to '01')]

4-Then

Send the ECHO command using

- P1 = 'Communication Problem' Indicator in 'Kernel Transaction Database'
- The input data depends on P1 value
  - P1=01: Unpredictable value in 'Kernel Transaction Database'
  - P1=02:Issuer Authentication Data in 'Kernel Transaction Database'

5-If communication problem detected during ECHO command

5-Then

The Kernel shall terminate the transaction and shall provide the "Communication Error – Unknown Payment Result" outcome message.

5-Else

6-If ECHO command fails on SW not equal to '9000' (error other than communication problem)

6-Then

'Echo Transaction' Indicator = 0

Error = 'Communication Problem' Indicator in 'Kernel Transaction Database' '1 for ECHO on 1st GEN AC and 2 for ECHO on 2nd GEN AC

6-Else

'Echo Transaction' Indicator = 1

Memorize the ECHO command response for later usage

6-Endif

5-Endif

4-Endif

3-Endif

3-If Authorization Response Code (tag '8A') is not present in Dynamic Transaction Parameters

3-And 'Echo Transaction' Indicator = 0 *'normal transaction*

3-Then

The kernel has to initialize the following kernel internal data elements

- Terminal Transaction Processing Information - TTPI (see Table 21)
- Set 'Offline CAM Selected' Indicator to '00' (None)
- Reset the contents of the 'Kernel Transaction Database'
- Store in the 'Kernel Transaction Database', the Dynamic Transaction Parameters communicated by the reader (including the "SELECT Response Message Data Field (FCI) of the selected application" value (please refer to K1.1)
- Store in the 'Kernel Transaction Database', the data elements present in the FCI (if a data element is duplicated, the kernel shall terminate

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Requirement – Transaction Initialization Processing	
	the transaction and shall provide the "Transaction Completed on Error" outcome message) (please refer to K1.1) • The kernel has to initialize Terminal Verification Result (TVR) to '8000000000' *('TVR bit 'Offline CAM was not performed' set to '1')* • Set 'Communication Problem' Indicator to '00' **4-If** 'Echo Card Identifier'('9F75') present in Application Selection response **4-And** TTPI bit 'Support of torn transaction recovery using ECHO command – byte 3 bit 6' is set to '1'[28] **4-Then** *'Both card and terminal support the ECHO command for this transaction* Set 'ECHO Command supported' Indicator to '01' **4-Else** Set 'ECHO Command supported' Indicator to '00' **4-Endif** **4-If** Contactless Pre-Processing Indicators 'Reader Contactless Floor Limit Exceeded Indicator' is set to 1[29] **4-Then** Set TVR bit 'Transactions exceeds floor limit' to '1' **4-Endif** **3-Endif** **2-Endif** **1-Else** **2-If** none of the following tag values ('91', '71' or '72') is present in the dynamic transaction parameters **2-Or** (long tap option not supported (**AC_Long_Tap** = 0)) **2-Then** Error = Error-Type **2-Endif** **1-Endif**
K2.5	**1-If** Error = 0 **1-And** Authorization Response Code (tag '8A') is present in Dynamic Transaction Parameters **1-Then** *'Online authorization response processing* **2-If** (online tap option not supported (**AC_Online_Tap** = 0)) **2-and** (long tap option not supported (**AC_Long_Tap** = 0)) **2-Then** Error = 2 **2-Else** **3-If** "SELECT Response Message Data Field (FCI) of the selected application" is part of the dynamic transaction parameters **3-And** 'Echo Transaction' Indicator = 0 *'normal transaction*

[28] This is equal to (AC_ECHO = 1) in Contactless Application Capabilities

[29] For transaction type requiring the transaction to be always forced online, the Terminal Floor limit values should be set to 000000000000. It is the responsibility of the Acquirer to satisfy to this recommendation.

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Requirement – Transaction Initialization Processing	
	3-Then **4-If** the FCI in dynamic transaction parameters is not equal to the FCI currently present in 'Kernel Transaction Database' **4-Then** *'card in communication with the reader is not the card used before online authorization process* Set 'Communication Problem' Indicator to '03' **4-Else** Set 'Communication Problem' Indicator to '00' **4. Endif** **3-Endif** the next step is defined in section 9.2.3 **2-Endif** **1-Else** *'Card payment application decision is not yet known (new transaction or transaction replay due to torn transaction during 1st GENERATE AC command)* **2-If** Error = 0 **3-If** 'Echo Transaction' Indicator = 1 *'restart a transaction previously torn* **3-Then** the next step is defined in section 9.2.14 **3-Else** **4-If [**(implementation option 4 is not supported (**IO_Option4**= 0)) **Or** (Transaction Type ≠ Trans_Type_Auth_Appli)] **4-Then** The Dynamic Transactions parameters ('9F02', '5F2A', '9F1A', '9A','9C' and '9F37') are stored in 'Kernel Transaction Database **4-Endif** **4-If** implementation option 1 is supported (**IO_Option1** = 1) **4-Then** the next step is defined in section 9.2.4 **4-Else** **5-If** implementation option 2 is supported (**IO_Option2** = 1) **5-Then** the next step is defined in section 9.2.5 **5-Else** the next step is defined in section 9.2.6 **5-Endif** **4-Endif** **3-Endif** **2-Endif** **1-Endif**
	1-If Error = 1 **1-Then** the Kernel shall terminate the transaction and provide the "Transaction Completed on Error" outcome message. **1-Endif**
K2.6	**1-If** Error = 2 **1-Then** the Kernel shall terminate the transaction and provide the "Financial Transaction Completed - erroneous re-activation of the kernel after an Online Request" outcome message. **1-Endif**

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Requirement – Transaction Initialization Processing	
K2.7	Transaction Status Information (TSI) is not supported.

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Byte	Bit	TTPI bit meaning and category	Value
1	8-7	RFU (00)	Contactless Application Capability Byte 1 value (***AC_Mode, AC_Contact, AC_Offline, AC_OnlinePIN, AC_Signature***)
	6	EMV contactless Mode support:	
	5	EMV contact transaction support	
	4	Terminal capability to process a transaction online	
	3	Online PIN support	
	2	Signature support	
	1	RFU (0)	0
2	8	Initial terminal request related to transaction completion	Value of "Reader Contactless Floor Limit Exceeded" Contactless Pre-Processing Indicator
	7	Terminal request related to cardholder verification	Value of "CVM Required Limit Exceeded" Contactless Pre-Processing Indicator
	6	Single unit of currency (Status Check) Check result	Value of "Status Check Requested" Contactless Pre-Processing Indicator
	5	Zero Amount Check result	Value of "Zero Amount" Contactless Pre-Processing Indicator
	4	RFU (0)	0
	3	Terminal requirement related to offline CAM	Contactless Application Capability Byte 4 value (**AC_OfflineCAM**)
	2-1	RFU (00)	00
3	8	Support of additional tap for Issuer Response communication	Contactless Application Capability Byte 3 value (**AC_Online_Tap**)
	7	Support of Consumer Device CVM checking as a possible CVM method	Contactless Application Capability Byte 3 value (**AC_CC_Tap**)
	6	Support of torn transaction recovery using ECHO command	Contactless Application Capability Byte 3 value (**AC_ECHO**)
	5-1	RFU (000000)	000000
4	8-1	RFU (00000000)	00000000
5		**Terminal Interchange Profile (TIP)**	
	8	Reliability of TIP byte value in TTPI	Contactless Application Capability Byte 5 value (***AC_Reliab)***
	7	SDA support	Contactless Application Capability Byte 5 value (***AC_SDA)***
	6	DDA support 1= DDA is supported 0= DDA is not supported	If implementation 4 is supported, Contactless Application Capability Byte 5 value (***AC_DDA***) In the other cases, 0

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Byte	Bit	TTPI bit meaning and category	Value
	5	Cardholder verification method supported by the terminal	Contactless Application Capability Byte 5 value (***AC_CVM)***
	4	Support of complementary terminal checks and terminal action analysis	Contactless Application Capability Byte 5 value (***AC_TRM)***
	3-2	RFU (00)	
	1	CDA support	Contactless Application Capability Byte 5 value (***AC_CDA***)

Table 21: TTPI Initial Value

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9.2.3 Online Response Processing and Kernel De-Activation

Requirement – Conditions of Execution	
K3.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K3.2	The transaction is stopped after this step. The next step is processed by the contactless reader.

Requirement – Online Response Processing	
K3.3	**1-If** 'Echo Transaction' Indicator = 1 **1-Then** **T**he next step is the step K3.5 for analysis of ECHO command response **1-Else** **2-If** 'Communication Problem' Indicator = '03' **2-Then** Completion_ID = 1 ('Terminate on error') *'the card is not the correct one* the next step is the step K3.7 **2-Else** Completion_ID = 0 ('no information from card') Script_Result_Available =0 **3-If** Critical Script commands (tag 71) is present in the Dynamic Transaction Parameters communicated by the contactless reader **3-Then** • Script command template ('71') shall be processed as specified in EMV contact specifications [EMV-3: Section 10.10]. The kernel shall support management of EMV-defined Issuer Script Result parameter. *(communication problem has no impact on the transaction flow – script command processed is only considered as failed*) • Script_Result_Available =1 **3-Endif** **2-Endif** **1-Endif**
K3.4	**1-If** Issuer Authentication Data (tag '91') is present in the Dynamic Transaction Parameters **1-Then** **2-If** 'Kernel Transaction Database' is available (not NULL) **and** CDOL2 (tag '8D') is present in 'Kernel Transaction Database' **2-Then** 'Determine the decision to request to the card Payment Application **3-If** value of Authorization Response Code - ARC ('8A') = '00' **Decision** = 'Approve' **3-Else** **Decision** = 'Decline'

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Requirement – Online Response Processing	
	3-Endif Add the Dynamic Transaction Parameters communicated by the contactless reader in 'Kernel Transaction Database' (if a parameter is already present in 'Kernel Transaction Database' , overwrite it by the value communicated by the contactless reader) Send the 2nd GENERATE AC command using ○ as input data the list of data referenced in CDOL2 (For each data, the value to use shall be the value present in 'Kernel Transaction Database'. If the data is not present in 'Kernel Transaction Database', the kernel has to apply DOL rules defined in EMV contact specifications [EMV-3: Section 5.4]) ○ request a TC cryptogram if Decision is 'Approve' and AAC in the other case.[30] **3-If** communication problem prior or during 2nd GENERATE AC **3-Then** **4-If** 'ECHO Command supported' Indicator = '01' • Set 'Communication Problem' Indicator to '02' • The Kernel shall terminate the transaction and shall provide the "Communication Error – Unknown Payment Result" outcome message. **4-Endif** **3-Endif** **3-If** 2nd Generate AC returns '9000 **3-Then** The next step is the step K3.5 for analysis of GENERATE AC command response[31] **3-Else** The next step is the step K3.6 **3-Endif** **2-Else** The next step is the step K3.6 **2-Endif** **1-Else** The next step is the step K3.6 **1-Endif**
K3.5	**1-If** command response cannot be parsed correctly[31] **1-Or** at least one mandatory data field (see Table 64 and Table 65) is not present in the command Response message **1-Or** card decision (CID) is approved (the two CID most significant bits = '01') **and** Authorization Response Code (tag '8A') is not equal to "00" **1-Then** Completion_ID = 1 ('error')

[30] CDA signature request is not support with 2nd GENERATE AC command.

[31] Only format 2 is supported for GENERATE AC command and ECHO command

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Requirement – Online Response Processing	
	1-Else **2-If** card decision (CID) is approved **2-Then** Completion_ID = 2 ('approved') **2-Else** Completion_ID = 3 ('declined') **2-Endif** **1-Endif** The next step is the step K3.6
K3.6	**1-If** Critical Script commands (tag 72) is present in the Dynamic Transaction Parameters communicated by the contactless reader **1-Then** • Script command template ('72') shall be processed as specified in EMV contact specifications [EMV-3: Section 10.10]. The kernel shall support management of EMV-defined Issuer Script Result parameter. *(communication problem has no impact on the transaction flow – script command processed is only considered as failed*) • Script_Result_Available =1 **1-Endif**
K3.7	**1-Depending** on Completion_ID **1-Value** 0 : 'no information from card, let the terminal to decide' **2-If** Script_Result_Available is equal to 1 **2-Then** The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Erroneous re-activation of the kernel after an Online Request" outcome message. 'Issuer Script Results' (using PURE proprietary tag '9F73') is added to the outcome message field "Data Record". It is the only data added. **2-Else** The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Erroneous re-activation of the kernel after an Online Request" outcome message. The outcome message field "Data Record" is set to 'No'. **2-Endif** **1-Value** 1 : terminate on error **2-If** Script_Result_Available is equal to 1 **2-Then** The Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Transaction declined (AAC)" outcome message. 'Issuer Script Results' (using PURE proprietary tag '9F73') is added to the outcome message field "Data Record". It is the only data added. **2-Else** The Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Transaction declined (AAC)" outcome message. The outcome message field "Data Record" is set to 'No'. **2-Endif** **1-Value** 2 : transaction approved

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Requirement – Online Response Processing

The Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Transaction approved (TC)" outcome message

CVM information is defined based on CVM Results value in 'Kernel Transaction Database'

1-If byte 'CVM Result' is equal to '00' (Unknown) **or** to '02' (Successful)

1-Then

2-Depending on 6 least significant bits of byte 'CVM Performed'

2-Value '02': CVM is 'Online PIN'

2-Value '1E': CVM is 'Signature'

2-Value '1F': CVM is 'No CVM'

2-Value '24': CVM is 'Confirmation Code Verified'

2-Other Value: CVM is 'N/A'

2-End_Depending

1-Else

CVM is 'N/A'

1-Endif

The outcome message fields are:

- **CVM** field / **UI Request on Outcome Present** field

2-Depending on CVM Information

2-Value 'Signature'

CVM = Obtain Signature

UI Request on Outcome Present

Message Identifier: '1A' ("Approved – Please Sign")

Status: Card Read Successfully (Remove card)

2-Value 'No CVM'

CVM = No CVM

UI Request on Outcome Present

Message Identifier: '03' ("Approved")

Status: Card Read Successfully (Remove card)

2-Value 'Online PIN'

CVM = Online PIN

UI Request on Outcome Present

Message Identifier: '03' ("Approved")

Status: Card Read Successfully (Remove card)

2-Value 'Confirmation Code Verified'

CVM = 'Confirmation Code Verified'

UI Request on Outcome Present

Message Identifier: '03' ("Approved")

Status: Card Read Successfully (Remove card)

2- Any other Value

CVM = 'N/A'

UI Request on Outcome Present

Message Identifier: '03' ("Approved")

Status: Card Read Successfully (Remove card)

2-End_Depending

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Requirement – Online Response Processing

- "Value Qualifier: Balance" is not present
- The Kernel has to try to provide all the data elements that may be present in field "Data Record" (see Table 31)using information present in 'Kernel Transaction Database' and in the response of the 2nd GENERATE AC. A missing mandatory data element has no impact. 'Issuer Script Results' (using PURE proprietary tag '9F73') is added only when at least one script command is processed (Script_Result_Available is equal to 1).

1-Value 3 : transaction declined

The Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Transaction declined (AAC)" outcome message

- The Kernel has to try to provide all the data elements that may be present in field "Data Record" (see Table 31)using information present in 'Kernel Transaction Database' and in the response of the 2nd GENERATE AC. A missing mandatory data element has no impact. 'Issuer Script Results' (using PURE proprietary tag '9F73') is added only when at least one script command is processed (Script_Result_Available is equal to 1).

1-End_depending

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9.2.4 Retrieval of Data Element values in EEPROM

Requirement – Conditions of Execution	
K4.1	This function is supported only in PURE contactless kernel supporting the implementation option 1

Requirement – Sequence of Execution	
K4.2	After processing this step **1-If** the transaction type is '78', **1-Then** the Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Retrieval of data elements" outcome message. **1-Else** **2-If** implementation option 2(**IO_Option2** = 1) is supported **2-Then** the next step is defined in section 9.2.5 **2-Else** the next step is defined in section 9.2.6 **2-Endif** **1-Endif**

Requirement – Retrieval of Data Element values Processing	
K4.3	**1-If** Memory Slot Read Template parameter (tag 'BF71') is part of the Dynamic Transaction Parameters **and** Implementation option 1 supported (**IO_Option1** = 1) **1-Then** **2-If** GDDOL parameter (tag 9F70) is part of data field of Memory Slot Read Template **2-Then** The kernel has to retrieve the value of all the data elements listed in the GDDOL parameter using the GET DATA command, add the data element retrieved in the 'Kernel Transaction Database' (please refer to K1.1) and add also the data element retrieved to the kernel parameter "GDDOL Resulting Buffer – tag '9F71' by concatenating the data elements retrieved using the following rule: • The data element retrieved are concatenated using the TLV structure (Tag, length of the data element retrieved, value of the data element retrieved) • **If** the tag of any data object identified in the GDDOL results in an error when processing the GET DATA command and is therefore unknown to the kernel, **then** the terminal shall add in GDDOL Resulting Buffer the tag value of the data element concatenated with a byte set to '00'. • The length specified in the GDDOL entry is ignored (the data element value retrieved using GET DATA may have a length different than the length referenced in GDDOL. Note: The data element retrieved may be either a primitive data element or a constructed data element

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Requirement – Retrieval of Data Element values Processing

Note: if a data element retrieved using GET DATA command is already present in the 'Kernel Transaction Database', its value replaced the value currently present in the database. It has no impact on the transaction processing.

2-Endif

1-Endif

1-If Transaction Type (tag '9C') is equal to '78'

1-Then

The Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Retrieval of data elements" outcome message.

2-If GDDOL Resulting Buffer" not empty

2-Then

"GDDOL Resulting Buffer" (9F71) needs to be added to the outcome message field "Discretionary Data Record".

2-Else

The outcome message field "Discretionary Data Record" is set to '***No***'

2-Endif

1-Else

2-If communication problem during a GET DATA command processing

2-Then

The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message.

2-Else

3-If implementation option 2 is supported(**IO_Option2** = 1)

3-Then the next step is defined in section 9.2.5

3-Else the next step is defined in section 9.2.6

3-Endif

2-Endif

1-Endif

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9.2.5 Update of Data Element values in EEPROM

Requirement – Conditions of Execution	
K5.1	This function is supported only in PURE contactless kernel supporting the implementation option 2

Requirement – Sequence of Execution	
K5.2	If this step is not completed on "terminate", the next step is defined in section 9.2.6

Requirement – Update of Data Element values Processing	
K5.3	Update=0 **1-If** "Memory Slot Update template" parameter (tag 'BF70') is part of the Dynamic Transaction Parameters **and** Implementation option 2 supported (**IO_Option2** = 1) **1-Then** Update=1 The kernel has to analyze each entry of the "Memory Slot Update template" parameter. **For each entry** of the "Memory Slot Update template – 'BF70'" parameter **If** "Memory Slot Update Entry" - is of type 1 (tag 'A2') **Then** the Kernel shall process the request defined in the entry • Update 'Data Elements Update Result – '9F74'" o Increases by one the information "number of data elements updated" o Set the information "Data Elements Update Result" to '02' • Update using PUT DATA (no secure messaging) command the data element specified in the entry (identified by the tag value). The new value is the value specified in the entry. • In case of PUT DATA command error (including communication problem) o Update in 'Data Elements Update Result – '9F74'", the information "Result" to '01' o Check in the entry the value of "Memory Slot Update Entry Setting – '85'" and memorizes it for later use o Stops the analysis of the contents of "Memory Slot Update template – 'BF70'" parameter (**Exit Loop**) **Else** the Kernel has to memorize the presence of "Memory Slot Update Entry - type 2 – tag 'A3'" and the presence of "Memory Slot Update Entry - type 3 – Tag 'A4'" **Endif** **Loop** on next Entry **1-Endif**
K5.4	**1-If** No communication problem during a PUT DATA command processing **1-Then** Terminate = 0 **2-If** Update = 1 **2-Then**

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Requirement – Update of Data Element values Processing	
	3-If Transaction Type (tag '9C') is equal to '79', **3-Then** **4-If** [the information "Result" in 'Data Elements Update Result – '9F74'" is equal to '01' (*error when processing PUT DATA*) **And** value of least significant bit of "Memory Slot Update Entry Setting – '85'" memorized previously (see K5.3) is equal to '1' (*transaction must be stopped in case of PUT DATA error*) **]** **Or** The kernel has found no **"**Memory Slot Update Entry Type 2 – 'A3'" in the "Memory Slot Update template" parameter.(*no additional PUT DATA*)[32] **4-Then** Terminate = 1 **4-Endif** **3-Else** **4-If** the information "Result" in 'Data Elements Update Result – '9F74'" is equal to '01' (*error when processing PUT DATA*) **And** value of least significant bit of "Memory Slot Update Entry Setting – '85'" is equal to '1' (*transaction must be stopped in case of PUT DATA error*) **4-Then** Terminate = 1 **4-Endif** **3-Endif** **2-Else** **3-If** Transaction Type (tag '9C') is equal to '79', **3-Then** Terminate = 1 **3-Endif** **2-Endif** **1-Endif**
K5.5	**1-If** communication problem during a PUT DATA command processing **1-Then** **2-If** Transaction Type (tag '9C') is equal to '79', **2-Then** The Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Data elements update using PUT DATA" outcome message. **3-If** GDDOL Resulting Buffer" not empty **3-Then** "GDDOL Resulting Buffer" (9F71) needs to be added to the outcome message field "Discretionary Data Record". **3-Endif** **3-If** 'Data Elements Update Result ('9F74') not empty

[32] The "Memory Slot Update Entry - type 3 – Tag 'A4'" are ignored when the transaction type is '79'

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Requirement – Update of Data Element values Processing

3-Then
"Data Elements Update Result ('9F74') needs to be added to the outcome message field "Discretionary Data Record".
3-Endif
2-Else
The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message.
2-Endif
1-Else
2-If Terminate = 1
2-Then
The Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Data elements update using PUT DATA" outcome message.
3-If 'Data Elements Update Result – '9F74' not empty
3-Then
'Data Elements Update Result – '9F74'needs to be added to the outcome message field "Discretionary Data Record".
3-Endif
3-If GDDOL Resulting Buffer" not empty,
3-Then
"GDDOL Resulting Buffer" (9F71) needs to be added to the outcome message field "Discretionary Data Record".
3-Endif
2-Else
The next step is defined in section 9.2.6
2-Endif
1-Endif

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9.2.6 Initiate Application Processing

Requirement – Conditions of Execution	
K6.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K6.2	If this step is not completed on "terminate", **1-If** implementation option 2 is supported (**IO_Option2** = 1) **1-Then** the next step is defined in section 9.2.7 **1-Else** the next step is defined in section 9.2.8 **1-Endif**

Requirement – Initiate Application Processing Function Processing	
K6.3	The PURE contactless kernel has to process "Initiate Application Processing" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.1]. Note: If PDOL is part of the application selection response, the kernel has to apply DOL rules defined in EMV contact specifications response (EMV-3 section 5.4): Example: If Terminal Transaction Data (tag 9F76) is part of PDOL but with a length distinct from the length defined in PDOL: • If the length in PDOL is less than the length of Terminal Transaction Data, the rightmost bytes of Terminal Transaction Data shall be truncated • If the length in PDOL is greater than the length of Terminal Transaction Data, the Terminal Transaction Data shall be right padded with hexadecimal zeroes
K6.4	The kernel must accept the presence of data elements other than AIP and AFL in the GPO command response [only Format 2 is supported]".
K6.5	**1-If** communication problem during the GPO command processing **1-Then** The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message. **1-Else** **2-If** the GPO command fails on error **Or** card response cannot be parsed correctly **Or** AIP or AFL or both are not present in the GPO Response message **2-Then** The Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **2-Else** • Data elements present in GPO command response (including AIP value, AFL value and data objects that are not recognized by the terminal (that is, their tags are unknown by the terminal))) are stored in 'Kernel Transaction Database' (please refer to K1.1). If a data element is already present in the

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Requirement – Initiate Application Processing Function Processing

'Kernel Transaction Database', the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message

The Kernel shall check the presence of CCID (tag 'C5') in the GPO Response message

3- If CCID is present **and** CCID bit 6 'Switch to contact interface' is set to '1'

3-Then the Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Try other interface" outcome message.

3-Else

4-If CCID is present **and** CCID bit 3-1 are set to '100' 'Confirmation Code required'

4-Then the Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- On-device CVM Requested" outcome message.

- Field "Value Qualifier: Balance" is not present
- Field "Data Record Present" is set to 'No'
- When implementation 1 is supported and 'Retrieval of Data Element values in EEPROM" step was processed, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
- When implementation 2 is supported and 'Update of Data Element values in EEPROM" step was processed, "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

4-Else

5- If CCID is present **and** CCID bit 8-7 are set to '00' 'transaction declined' **and** Kernel Configuration 'stop transaction at GPO if CCID indicates AAC' (***AC_Decline***) is set to 1

5-Then the Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Transaction declined (AAC)" outcome message.

- Field "Value Qualifier: Balance" is not present
- Field "Data Record Present" is set to 'No'
- When implementation 1 is supported and 'Retrieval of Data Element values in EEPROM" step was processed, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
- When implementation 2 is supported and 'Update of Data Element values in EEPROM" step was processed, "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

5-Else

6-If implementation option 2 is supported(**IO_Option2** = 1)

6-Then the next step is defined in section 9.2.7

6-Else the next step is defined in section 9.2.8

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Requirement – Initiate Application Processing Function Processing
6-Endif **5-Endif** **4-Endif** **3-Endif** **2-Endif** **1-Endif**

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9.2.7 Update of Data Element values in EEPROM after GPO command

Requirement – Conditions of Execution	
K7.1	This function is supported only in PURE contactless kernel supporting the implementation option 2

Requirement – Sequence of Execution	
K7.2	If this step is not completed on "terminate", the next step is defined in section 9.2.8

Requirement – Update of Data Element values Processing	
K7.3	**1-If** the implementation option 2 is supported (**IO_Option2** = 1) **And** the following conditions are satisfied: o the contactless reader has communicated "Memory Slot Update template" parameter (tag 'BF70') in Dynamic Transaction Parameters **and** o the kernel has identified the presence of "Memory Slot Update Entry - type 2" in template 'BF70' during transaction step defined in section 9.2.5 **and** o "Result" in 'Data Elements Update Result – '9F74'" is not equal to '01' (No error happened during transaction step defined in section 9.2.5) **1-Then** *The kernel has to analyze each entry of the "Memory Slot Update template" parameter.* **For each entry** of the "Memory Slot Update template – 'BF70'" parameter **2-If** "Memory Slot Update Entry" - is of type 2 (tag 'A3') **2-Then** the Kernel shall process the request defined in the entry • Update 'Data Elements Update Result – '9F74'" o Increases by one the information "number of data elements updated" o Set the information "Data Elements Update Result" to '02' • Update using PUT DATA (no secure messaging) command the data element specified in the entry (identified by the tag value). The new value is the value specified in the entry. • In case of PUT DATA command error(including communication problem) o Update in 'Result – '9F74'", the information "Data Elements Update Result" to '01' o Check in the entry the value of "Memory Slot Update Entry Setting – '85'" and memorizes it for later use o Stops the analysis of the contents of "Memory Slot Update template – 'BF70'" parameter (**Exit Loop**) **2-Endif** **Loop** on next Entry
K7.4	**2-If (**communication problem during the PUT DATA command processing) **And** (the transaction type is not '79')

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Requirement – Update of Data Element values Processing

2-Then
The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message.
2-Else
3-If the transaction type is '79',
3-Then
The Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Data elements update using PUT DATA" outcome message.
4-If information available in 'Data Elements Update Result – '9F74'
4-Then
'Data Elements Update Result – '9F74'needs to be added to the outcome message field "Discretionary Data Record".
4-Endif
4-If implementation 1 is supported (**IO_Option1** = 1) and 'Retrieval of Data Element values in EEPROM" step was processed,
4-Then
"GDDOL Resulting Buffer" (9F71) needs to be added to the outcome message field "Discretionary Data Record".
4-Endif
3-Else
4-If the information "Result" in 'Data Elements Update Result – '9F74'" is equal to '01' (*error when processing PUT DATA*)
And value of least significant bit of "Memory Slot Update Entry Setting – '85'" memorized previously (see K7.3) is equal to '1' (*transaction must be stopped in case of PUT DATA error*)
4-Then
The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Transaction declined (AAC)" outcome message.
- Field "Value Qualifier: Balance" is not present
- Field "Data Record Present" is set to 'No'
- When implementation 1 is supported and 'Retrieval of Data Element values in EEPROM" step was processed, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
- "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

4-Else
The next step is defined in section 9.2.8
4-Endif
3-Endif
2-Endif
1-Else
The next step is defined in section 9.2.8

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Requirement – Update of Data Element values Processing
1-Endif

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9.2.8 Read Application Data

Requirement – Conditions of Execution	
K8.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K8.2	If this step is not completed on "terminate", the next step is defined in section 9.2.9 (note: steps described in section 9.2.9 up to section 9.2.12 may be performed in any order)

Requirement – Read Application Data Processing	
K8.3	The PURE contactless kernel has to process "Read Application Data" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.2]. Only the list of mandatory objects and the management of the unknown data object is different.
K8.4	Requirement removed
K8.5	The terminal shall read all the files and records indicated in the AFL using the READ RECORD command identifying the file by its SFI. **0-Loop on the records referenced in AFL**[33] Read the record using READ RECORD command **1-If** communication problem during the READ RECORD command processing **1-Then** The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message. **1-Else** *'All use cases that have for consequence in EMV contact specifications to terminate the transaction (including mandatory data missing), must have same consequence in contactless mode (checking the presence of the data elements required for offline CAM shall be performed during step 9.2.9 when necessary).* **2-If** the READ RECORD command fails on error (other than communication problem) **2-Or** card response cannot be parsed correctly **2-Or** for files with SFI in the range 1 to 10, the record contents is not TLV-coded with tag equal to '70' **2-Or** one or several mandatory data object (see K8.3) is missing **2-Or** at least one data element does not have the adequate format **2-Then**

[33] Two possible implementations:
- Implementation 1: Read all records and then loop on the records read for analyzing their contents
- Implementation 2: loop on all records but analyze the record content immediately after reading a record.

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Requirement – Read Application Data Processing	
	The Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **2-Else** Data elements present in the record read are stored in 'Kernel Transaction Database'(please refer to K1.1). If a data element is already present in the 'Kernel Transaction Database', the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message **2-Endif** **1-Endif** **0-End Loop**
K8.6	Verify the presence of the mandatory data elements in the 'Kernel Transaction Database': **1-If** Transaction Type = Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1) **1-And** (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli)) **1-Then** *'Application Authentication transaction* There is no mandatory data element. **1-Else** *'For other transaction* • '8C' – Card Risk Management Data Object List 1 (CDOL1) • '57' – Track2 Equivalent Data • List of data objects defined in Mandatory Object List (MTOL) (if list is not empty) All other data elements are optional. **1-Endif** **1-If** any mandatory data element are not present, the terminal shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message The next step is defined in section 9.2.9

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9.2.9 Offline CAM Selection

Requirement – Conditions of Execution	
K9.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K9.2	For transactions other than Application Authentication transaction, the next step is defined in section 9.2.10 (note: steps described in section 9.2.9 up to section 9.2.12 may be performed in any order) For Application Authentication Transactions, the next step is defined in section 9.2.16. Note: Detection of a communication problem has no impact during this step

Requirement – Offline CAM Selection Processing	
K9.3	Note: The kernel is using the information present in the 'Kernel Transaction Database' and the Static Configuration Parameters associated to the AID **1-If** the AIP value specifies that 'CDA is supported' [AIP byte 1 bit 1 set to '1'] **and** CDA supported in the PURE contactless kernel Capabilities (***AC_CDA*)** **and [**(implementation option 4 is not supported (**IO_Option4**= 0)) **Or** (Transaction Type ≠ Trans_Type_Auth_Appli)] **1-Then** check the presence of the data elements listed in Table 23. Set TVR bit 'Offline CAM was not performed' to '0' **2-If** a mandatory data element is missing **2- or** the kernel is not supporting the CA Public Key value referenced in Certification Authority Public Key Index **2-Then** set TVR bit 'CDA failed' to '1' **2-Else** Set 'Offline CAM Selected' Indicator to '02' (CDA) **2-Endif** **1-Else** **2-If** the AIP value specifies that 'DDA is supported' [AIP byte 1 bit 6 set to '1'] **and** DDA supported in the PURE contactless kernel Capabilities (***AC_DDA*)** **and** Transaction Type = Trans_Type_Auth_Appli **and** implementation option 4 is supported (**IO_Option4**= 1) **and** (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli)) **2-Then** check the presence of the data elements listed in Table 23. Set TVR bit 'Offline CAM was not performed' to '0' **3-If** a mandatory data element is missing **3- or** the kernel is not supporting the CA Public Key value referenced in Certification Authority Public Key Index **3-Then** set TVR bit 'DDA failed' to '1' **3-Else**

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Requirement – Offline CAM Selection Processing

- Set 'Offline CAM Selected' Indicator to '03' (DDA)
- The kernel has to ask the contactless device to generate an RSA dynamic signature as it is defined in EMV contact specifications [EMV-2: Section 6]. If DDOL (tag '9F49') is not present among the information read, Default DDOL must be used. If there is a communication problem during the processing of the INTERNAL AUTHENTICATE command[34] or if the command fails on error or for any other reasons defined in the EMV contact transaction, the kernel has to set the bit 'DDA failed' in TVR to '1'.

Note: The verification of the certificates and of the dynamic signature are performed later in the transaction flow

3-Endif

2-Else

3-If the AIP value specifies that 'SDA is supported' [AIP byte 1 bit 7 set to '1']

and SDA supported in the PURE contactless kernel Capabilities (***AC_SDA*)**

3-Then

Set TVR bit 'SDA was performed' (Byte 1 bit 2) to '1'

Set TVR bit 'Offline CAM was not performed' to '0'

Check the presence of the data elements listed in Table 22.

4-If a mandatory data element is missing

4- or the kernel is not supporting the CA Public Key value referenced in Certification Authority Public Key Index

4-Then

set TVR bit 'SDA failed' to '1'

4-Else

Set 'Offline CAM Selected' Indicator to '01' (SDA)

4-Endif

3-Endif

2-Endif

1-Endif

1-If Transaction Type = Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1)

1-And (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli))

1-Then

The next step is defined in section 9.2.16

1-Else

The next step is defined in section 9.2.10

1-Endif

[34] When the kernel is in charge of the generation of the Unpredictable Number, it is recommended to generate an unpredictable value dedicated to the INTERNAL AUTHENTICATE command.

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Tag	Description	Requirement
'8F'	Certification Authority Public Key Index	Mandatory
'90'	Issuer Public Key Certificate	Mandatory
'93'	Signed Static Application Data	Mandatory
'92'	Issuer Public Key Remainder	Optional
'9F32'	Issuer Public Key Exponent	Mandatory

Table 22 – Data required for SDA

Tag	Description	Requirement
'8F'	Certification Authority Public Key Index	Mandatory
'90'	Issuer Public Key Certificate	Mandatory
'92'	Issuer Public Key Remainder	Optional
'9F32'	Issuer Public Key Exponent	Mandatory
'9F46'	ICC Public Key Certificate	Mandatory
'9F47'	ICC Public Key Exponent	Mandatory
'9F48'	ICC Public Key Remainder	Optional

Table 23 – Data required for CDA and DDA

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9.2.10 Processing Restrictions

Requirement – Conditions of Execution	
K10.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K10.2	The next step is defined in section 9.2.11 (note: steps described in section 9.2.9 up to section 9.2.12 may be performed in any order) Note: Detection of a communication problem has no impact during this step

Requirement – Processing Restrictions function Processing	
K10.3	**1-If** (in Contactless Kernel Capabilities parameter, the 'EMV-defined Processing Restriction and Terminal Risk Management" bit is set to '1' (***AC_TRM***)) **1-and** (AIP value specifies that 'Terminal risk management is to be performed' [AIP byte 1 bit 4 set to '1']) **1-Then**
K10.4	The PURE contactless kernel has to process "Processing Restrictions" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.4]. It comprises the following compatibility checks: • Application Version Number • Application Usage Control • Application Effective/Expiration Dates Checking
K10.5	**If** "Application Version Number" is not present in the 'Kernel Transaction Database', **then** the "Application Version Number" check is not performed and it has no impact on TVR value.
K10.6	**If** "Application Usage Control" parameter is not present in the 'Kernel Transaction Database', **then** the "Application Usage Control" check is not performed and it has no impact on TVR value.
K10.7	**If** "Application Effective Date" parameter is not present in the 'Kernel Transaction Database', **then** the "Application Effective Date" check is not performed and it has no impact on TVR value.
K10.8	**If** "Application Expiration Date" parameter is not present in the 'Kernel Transaction Database', **then** the "Application Expiration Date" check is not performed and it has no impact on TVR value.
	1-Endif The next step is defined in section 9.2.11

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9.2.11 Cardholder Verification Method Selection

Requirement – Conditions of Execution	
K11.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K11.2	If the next step is defined in section 9.2.12 (note: steps described in section 9.2.9 up to section 9.2.12 may be performed in any order) Note: Detection of a communication problem has no impact during this step

Requirement – Cardholder Verification Method Processing	
K11.3	**Set** CVM Result to '3F0000' **1-If** CCID (tag 'C5') is present in the GPO Response message **1-then** **2-If** TTPI bit 'Terminal requests cardholder verification – byte 2 bit 7' is set to '1' **2-or** CCID (CVM Information – bits 3-1) value different from '000' (card requests to perform cardholder verification) **2-then** **3-Depending on** CCID (CVM Information – bits 3-1) value **3-Value** '000': 'No CVM Required **Set** CVM Result parameter to '1F0000' Set TVR 'Cardholder verification was not successful' to '1' **3-Value** '010': 'Online PIN required **4-If** online PIN is supported in TTPI value (byte 1 bit 3) **4-then** Set TVR 'Online PIN Entered' bit to '1' **Set** CVM Result parameter to '020000' **4-Else** Set CVM Result parameter to '020001' Set TVR 'Cardholder verification was not successful' to '1' **4-Endif** **3-Value** '001': 'Signature **4-If** signature is supported in TTPI value (byte 1 bit 2) **4-then Set** CVM Result parameter to '1E0000' **4-Else** Set CVM Result parameter to '1E0001' Set TVR 'Cardholder verification was not successful' to '1' **4-Endif** **3-Value** '101': 'Confirmation Code already checked using Customer device' **Set** CVM Result parameter to '240002' (see section 11.2.5) *Note: When CCID (CVM Information – bits 3-1) is equal to '100' the transaction is stopped during step 9.2.6.* *Note: Confirmation Code is always supported by the PURE kernel*

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Requirement – Cardholder Verification Method Processing	
	3-Value '011': 'using CVM List **4-If** Kernel Configuration indicates that EMV-defined cardholder verification method using CVM List is not supported (***AC_CVM***) **4-Or** if "CVM List" parameter is not present in the 'Kernel Transaction Database' **4-Then** Set TVR 'Cardholder verification was not successful' to '1' **4 -Else** Process cardholder verification as it is defined in the EMV contact specifications [EMV-3: Section 10.5] with the following differences: • Support of new Cardholder Verification Rule codes for Confirmation Code and for Confirmation Code plus signature(paper) (see section 11.2.5) • Offline PIN is always considered as not supported by the terminal • 'No CVM Required" is considered as supported only when TTPI bit 'Terminal requests cardholder verification – byte 2 bit 7' is set to '0' • Terminal capability related to online PIN support is defined in TTPI value (byte 1 bit 3) • Terminal capability related to signature support is defined in TTPI value (byte 1 bit 2) • Terminal capability related to Confirmation Code is defined in TTPI value (byte 3 bit 7) • When Online PIN is the method selected, • the kernel shall not ask the cardholder for the Online PIN value (only CVM Result is updated) • the kernel shall set TVR 'Online PIN Entered' bit to '1' • When 'Confirmation Code' or 'Confirmation Code and Signature' is the CVM selected, the value of byte 3 in CVM Result shall be set to '00-unknown' *Note: decision to request the cardholder to enter a confirmation code will be taken during set 9.2.14 based on GENERATE AC decision.* **4-Endif** Note: CVM Result parameter must reflect the result of CVM method selection using EMV-defined algorithm. **3-End_Depending** **2-Endif**
	1-Else **2-If** TTPI bit 'Terminal requests cardholder verification – byte 2 bit 7' is set to '1' **2-Then**
K11.4	**3- If** AIP value specifies that 'Cardholder verification is supported' [AIP byte 1 bit 5 set to '1'], **3-Then**

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Requirement – Cardholder Verification Method Processing

4-If Kernel Configuration indicates that EMV-defined cardholder verification method using CVM List is not supported (***AC_CVM***)
4-Or if "CVM List" parameter is not present in the 'Kernel Transaction Database'
4 - Then
Set TVR 'Cardholder verification was not successful' to '1'
4 - Else
Process cardholder verification as it is defined in the EMV contact specifications [EMV-3: Section 10.5] with the following differences:

- Support of new Cardholder Verification Rule codes for Confirmation Code and for Confirmation Code plus signature(paper) (see section 11.2.5)
- Offline PIN is always considered as not supported by the terminal
- 'No CVM Required" is considered as supported only when TTPI bit 'Terminal requests cardholder verification – byte 2 bit 7' is set to '0'
- Terminal capability related to online PIN support is defined in TTPI value (byte 1 bit 3)
- Terminal capability related to signature support is defined in TTPI value (byte 1 bit 2)
- Terminal capability related to Confirmation Code support is defined in TTPI (byte 3 bit 7)
- When Online PIN is the method selected,
  - the kernel shall not ask the cardholder for the Online PIN value (only CVM Result is updated),
  - the kernel shall set TVR 'Online PIN Entered' bit to '1'.
- When 'Confirmation Code' is the CVM selected, the value of byte 3 in CVM Result shall be set to '00-unknown'
  Note: decision to request the cardholder to enter a confirmation code will be taken during set 9.2.14 based on GENERATE AC decision.

4-Endif
Note: CVM Result parameter must reflect the result of CVM method selection using EMV-defined algorithm.
3-Else
Set to 1 the following TVR bit:

- Byte 3 bit 8: Cardholder verification was not successful

3-Endif
2-Endif
1-Endif
The next step is defined in section 9.2.12

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9.2.12 Terminal Risk Management

Requirement – Conditions of Execution	
K12.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K12.2	The next step is defined in section 9.2.13 (note: steps described in section 9.2.9 up to section 9.2.12 may be performed in any order). Note: Detection of a communication problem has no impact during this step

Requirement – Terminal Risk Management Processing	
K12.3	Note: The kernel is using the information present in the 'Kernel Transaction Database' **1-If** in Contactless Kernel Capabilities parameter, the 'EMV-defined Processing Restriction and Terminal Risk Management" bit is set to '1' (***AC_TRM***) **1-and** (AIP value specifies that 'Terminal risk management is to be performed' [AIP byte 1 bit 4 set to '1']) **1-Then**
K12.4	The PURE contactless kernel has to process "Terminal Risk Management" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.6] but limited to Terminal Velocity Checking[35]. Note: Any error on GET DATA command (including communication problem) during velocity checking must have same impact as the non-retrieval of ATC value or LOATC value using GET DATA command. **1-Endif** The next step is defined in section 9.2.13

[35] The support of an exception file is not requested. When an implementation supports it, it is outside the scope of this specification.

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9.2.13 Terminal Action Analysis

Requirement – Conditions of Execution	
K13.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K13.2	The next step is defined in section 9.2.14. Note: Detection of a communication problem has no impact during this step

Requirement – Terminal Action Analysis Processing	
K13.3	Note: The kernel is using the information present in the 'Kernel Transaction Database' and the Static Configuration Parameters associated to the AID **1-If** the Transaction Type indicates a Refund ('20') **1-then** the result of Terminal Action Analysis is to decline the transaction the kernel has to set 'Offline CAM Selected' Indicator to '00' (none) and set the bit for "CDA Signature Requested' in the GENERATE AC command to '0' **1-Else** the PURE contactless kernel has to process "Terminal Action Analysis" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.7] with the following differences (see also note below): • Terminal capability related to online capacity is defined in TTPI value (byte 1 bit 4) • If CDA is the offline CAM selected ('Offline CAM Selected' Indicator equal to 'CDA'), • the terminal shall set the bit for "CDA Signature Requested' in the GENERATE AC command to '1' only if at least one of the following conditions is satisfied ○ Terminal decision is to process the transaction offline or ○ TTPI bit 'Terminal requirement related to offline CAM – byte 2 bit 3' is set to '1' *'Offline CAM always requested* • If none of the conditions are satisfied, the kernel has to set 'Offline CAM Selected' Indicator to '00' (none) and set the bit for "CDA Signature Requested' in the GENERATE AC command to '0' **1-Endif** The next step is defined in section 9.2.14.

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Note: The following table provides complementary information related to Terminal Action Analysis according to the type of kernel configuration.

	Kernel supporting any type of terminal			Kernel dedicated to online-only terminal	
Online capability of the terminal	Online-only terminal	Offline with online capability	Offline only	Option 1: Online only	Option 2: Online only
Terminal configuration in ICS	Online capable [restricted to online-only not selected]	Online capable [restricted to online-only not selected]	Online capable [restricted to online-only not selected]	Online capable [restricted to online-only is selected]	Online capable [restricted to online-only not selected]
Contactless Application/Kernel Capabilities – byte 1 bit 4 [AC_Offline]	0 = Online capable reader	0 = Online capable reader	1 = Offline-only reader	0 = Online capable reader	0 = Online capable reader
TTPI Byte 1 bit 4	0 = Online capable reader	0 = Online capable reader	1 = Offline-only reader	0 = Online capable reader	0 = Online capable reader
Terminal Action Analysis	Same as in contact EMV algorithm dedicated to online capable terminal [TAC/IAC denial and TAC/IAC online are used] Note: Acquirer must set Terminal Floor limit to zero and TAC Online 'Terminal Floor limit exceeded' bit must be set to 1.	Same as in contact EMV algorithm dedicated to online capable terminal [TAC/IAC denial and TAC/IAC online are used]	Same as in contact EMV algorithm dedicated to offline-only terminal [TAC/IAC denial and TAC/IAC default are used]	Same as in contact EMV algorithm dedicated to online-only terminal [only TAC/IAC denial are used]	Same as in contact EMV algorithm dedicated to online capable terminal [TAC/IAC denial and TAC/IAC online are used] Note: Acquirer must set Terminal Floor limit to zero and TAC Online 'Terminal Floor limit exceeded' bit must be set to 1

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9.2.14 Card Action Analysis

Requirement – Conditions of Execution	
K14.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K14.2	If this step is not completed on "terminate", **1-If** implementation option 2 is supported(**IO_Option2** = 1) **1-Then** the next step is defined in section 9.2.15 **1-Else** the next step is defined in section 9.2.16

Requirement – Card Action Analysis Processing	
K14.3	**1-If** 'Echo Transaction' Indicator = 1 **1-Then** 'analysis of ECHO command response **2-If** ECHO command response cannot be parsed correctly[36] **2-Or** at least one mandatory data field (see Table 64 and Table 65) is not present in the ECHO command Response message **2-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **2-Else** **3-If [**CCID is not present in the ECHO command Response message **and** CCID was present in the GET PROCESSING OPTIONS command message] **3-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **3-Else** **4-If** CCID present in ECHO command Response and in GET PROCESSING OPTIONS **4-and** CVM Information (3 least significant bits) has a distinct value in both CCID values **4-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **4-Else** The next step is in K14.5 for analysis of ECHO command response **4-Endif** **3-Endif** **2-Endif** **1-Else** Prepare the data to send to 1st GENERATE AC command using the list of data referenced in CDOL1 and the data object values present in the 'Kernel Transaction Database' (If the data is unknown, the kernel has to apply DOL rules defined in EMV contact specifications). The result is CDOL1 Related Data.

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Requirement – Card Action Analysis Processing	
	The PURE contactless kernel has to process "Card Action Analysis" as it is defined in the EMV specifications for contact transactions [EMV-3: Section 10.8]. Send the 1st GENERATE AC command using as input data, CDOL1 related data. The bit "CDA Signature Requested" is set according to decision on previous step. **1-Endif**
	1-If communication problem detected during 1st GENERATE AC command **1-Then** **2-If** 'ECHO Command supported' Indicator = '00' **2-Then** The Kernel shall terminate the transaction and shall provide the "Communication Error – Payment Transaction not completed" outcome message. **2-Else** Set 'Communication Problem' Indicator to '01' The Kernel shall terminate the transaction and shall provide the "Communication Error – Unknown Payment Result" outcome message. **2-Endif** **1-Else** **2-If** the 1st GENERATE AC command fails on error (other than communication problem) **2-Or** card response cannot be parsed correctly[36] **2-Or** at least one mandatory data field (see Table 64 and Table 65) is not present in the 1st GENERATE AC Response message
K14.4	**2-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **2-Else** **3-If [**CCID is not present in the 1st GENERATE AC Response message **and** CCID was present in the GET PROCESSING OPTIONS command message] **3-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **3-Else** **4-If** CCID present in 1st GENERATE AC and in GET PROCESSING OPTIONS **4-and** CVM Information (3 least significant bits) has a distinct value in both CCID values **4-Then** the Kernel shall terminate the transaction and shall provide the "Transaction Completed on Error" outcome message. **4-Else** The next step is in K14.5 for analysis of GENERATE AC command response **4-Endif**

[36] Only format 2 is supported for GENERATE AC command and for ECHO command

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Requirement – Card Action Analysis Processing	
	3-Endif **2-Endif** **1-Endif**
	1-If [(in CVM Result ('CVM performed' byte is equal to one of the following Confirmation Code values ('24', or '64') and CVM Result byte equals to '00') **1-and (**card decision (CID) is declined (the two CID most significant bits = '00') **1-and** (CCID is not present in 1st GENERATE AC command response)] **1-Or [(**CCID is present in 1st GENERATE AC command response) **1-and** (CCID bit 3-1 are set to '100' 'Confirmation Code required')] **1-Then** • the kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- On-device CVM Requested" outcome message. • Field "Value Qualifier: Balance" is not present • Field "Data Record Present" is set to 'No' • When implementation 1 is supported and 'Retrieval of Data Element values in EEPROM" step was processed, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record". • When implementation 2 is supported and 'Update of Data Element values in EEPROM" step was processed, "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record". **1-Else**
K14.5	**2-If** 'Offline CAM Selected' Indicator is set to '02' (CDA) **2- and** card decision (CID) is declined (the two CID most significant bits = '00') **2-and** Signed Dynamic Application Data (9F4B) is not present in 1st GENERATE AC response **2-Then** 'Offline CAM Selected' Indicator is set to '00' (none) **2-Endif** **2-If** CCID is present **and** CCID bit 6 'Switch to contact interface' is set to '1' **2-Then** The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Try other interface" outcome message. **2-Else** **3-If** terminal has requested to approve offline the transaction **3-And** card decision (CID) is declined (the two CID most significant bits = '00') **3-Then** The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Try other interface" outcome message. **3-Else** **4-If** implementation option 2 is supported(**IO_Option2** = 1) **4-Then** the next step is defined in section 9.2.15

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Requirement – Card Action Analysis Processing
4-Else the next step is defined in section 9.2.16 **4-Endif** **3-Endif** **2-Endif** **1-Endif**

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9.2.15Update of Data Element values in EEPROM after Card Action Analysis

Requirement – Conditions of Execution	
K15.1	This function is supported only in PURE contactless kernel supporting the implementation option 2

Requirement – Sequence of Execution	
K15.2	If this step is not completed on "terminate", the next step is defined in section 9.2.16. Note: Detection of a communication problem has no impact during this step

Requirement – Update of Data Element values in EEPROM after Card Action Analysis Processing	
K15.3	**1-If** the implementation option 2 is supported (**IO_Option2** = 1) **And** the following conditions are satisfied: o the contactless reader has communicated "Memory Slot Update template" parameter (tag 'BF70') in Dynamic Transaction Parameters **and** o the kernel has identified the presence of "Memory Slot Update Entry - type 3" in template 'BF70' during transaction step defined in section 9.2.5 **and** o "Result" in 'Data Elements Update Result – '9F74'" is not equal to '01' (No error happened during transaction step defined in section 9.2.5 and in section 9.2.7) **1-Then** *The kernel has to analyze each entry of the "Memory Slot Update template" parameter.* **2- For each entry** of the "Memory Slot Update template – 'BF70'" parameter **3-If** "Memory Slot Update Entry" - is of type 3 (tag 'A4') **3-Then** the Kernel shall process the request defined in the entry **4-If** Conditions defined in "Memory Slot Update Entry Setting – '85'" associated to the entry are satisfied (condition depending on card decision (CID) communicated in 1st GENERATE AC response during step defined in section 9.2.14) **4-Then** o Update 'Data Elements Update Result – '9F74'" ▪ Increase by one the information "number of data elements updated" ▪ Set the information "Data Elements Update Result" to '02' o Update using PUT DATA (no secure messaging) command the data element specified in the entry (identified by the tag value). The new value is the value specified in the entry. o In case of PUT DATA command error (including communication problem) ▪ Update in 'Result – '9F74'", the information "Data Elements Update Result" to '01'

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Requirement – Update of Data Element values in EEPROM after Card Action Analysis Processing	
	○ Check in the entry the value of "Memory Slot Update Entry Setting – '85'" and memorizes it for later use ▪ Stops the analysis of the contents of "Memory Slot Update template – 'BF70'" parameter (**Exit Loop**) **4-Endif** **3-Endif** **2-Loop** on next Entry
K15.4	The next step is defined in section 9.2.16. **1-Else** The next step is defined in section 9.2.16. **1-Endif**

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9.2.16 Card Balance Retrieval

Requirement – Conditions of Execution	
K16.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K16.2	The next step is defined in section 9.2.17. Note: Detection of a communication problem has no impact during this step

Requirement – Get Card Balance Retrieval Processing	
K16.3	**1-If** in Contactless Kernel Capabilities parameter, the "retrieval of the card balance" bit is set to '1'(***AC_Balance***) **1-Then** The Kernel has to retrieve using GET DATA command, the values of the following card parameters in the following order: 1. '9F50' – Offline Balance 2. 'CD' - CRM Currency Code Failure during the processing of GET DATA command (including communication problem) has no impact on the transaction processing. When the command fails, the parameter that the kernel was trying to retrieve has to be considered as 'unknown'. **1-Endif**
K16.4	**1- If [(**Transaction Type = Trans_Type_Auth_Appli) **And** (implementation option 4 is supported (**IO_Option4**= 1)))] **1-Or** **[(**card decision (CID) present in the 1st GENERATE AC Response message is not online request (if the two CID most significant bits ≠ '10')) **Or** the Kernel is not setup for the support of long tap option (**AC_Long_Tap** = 0)] **1 – Then** The kernel shall send a **User Interface Request** message with the following parameters: • Message Identifier: '17' ("Card Read OK") • Status: Card Read Successfully Note: This will result in an indication to the cardholder that the card can be removed from the field. **1-Endif** The next step is defined in section 9.2.17.

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9.2.17 Offline CAM Checking

Requirement – Conditions of Execution	
K17.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K17.2	The next step is defined in section 9.2.18. Note: Detection of a communication problem has no impact during this step

Requirement – Offline CAM Checking Processing	
K17.3	**1-If** 'Offline CAM Selected' Indicator is set to '01' (SDA) **1-Then** **2-If** at least one of the following conditions is satisfied ○ Decision taken during "Card Action Analysis" is to process the transaction offline or ○ Kernel Configuration requests offline CAM checking in all circumstances(TTPI bit 'Terminal requirement related to offline CAM – byte 2 bit 3' is set to '1') ○ [(Application Authentication Transaction (Transaction Type = Trans_Type_Auth_Appli)) when Implementation option 4 is supported (**IO_Option4** = 1)) And (option selected by Acquirer in "Contactless Applicat Capabilities" parameter (AC_Auth_Appli))] **2-Then** The kernel has to perform SDA checking as it is defined in EMV contact specifications [EMV-2: Section 5]. The Kernel has to use the information present in the 'Kernel Transaction Database'. (exception: TVR shall not be updated in case of SDA failure) Note: For Issuer Identifier value checking, the kernel must use PAN data element value if present. When PAN data element is not present, the kernel must use PAN value written inside 'Track2 Equivalent Data' data element. **3-If** (SDA fails) **3-And [**(implementation option 4 is not supported (**IO_Option4**= 0)) **Or** (Transaction Type ≠ Trans_Type_Auth_Appli)] **3-Then** the kernel has to update decision communicated by the card to decline (CID set to '00' – AAC). **3-Else** The Data Authentication Code (DAC) value (tag '9F45') retrieved in the recovered Signed Static Application Data signature needs to be added to the 'Kernel Transaction Database'. **3-Endif** **2-Else** the kernel has to go to next transaction flow step. **2-Endif** **1-Else** **2-If** 'Offline CAM Selected' Indicator is set to '02' (CDA)

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Requirement – Offline CAM Checking Processing

2-Then

The kernel has to perform CDA checking as it is defined in EMV contact specifications [EMV-2: Section 6.6]. The Kernel has to use the information present in the 'Kernel Transaction Database' and during the "card action analysis" step (or get back from Kernel Transaction Database'). (note if Signed Dynamic Application Data ('9F4B') is not present in the 1st GENERATE AC command response, CDA is considered as failed)

Note: For Issuer Identifier value checking and PAN value checking, the kernel must use PAN data element value if present. When PAN data element is not present, the kernel must use PAN value written inside 'Track2 Equivalent Data' data element.

TVR shall not be updated in case of CDA failure.

3-If CDA fails

3-Then the kernel has to update decision communicated by the card to decline (CID set to '00' – AAC).

3-Else

The ICC Dynamic Number (IDN) value (tag '9F4C') retrieved in the recovered Signed Dynamic Application Data signature needs to be added to the 'Kernel Transaction Database'.

3-Endif

2-Else

3-If 'Offline CAM Selected' Indicator is set to '03' (DDA)

3-Then

Note: DDA may be used only in the case of Application Authentication Transaction. PAN value and Track2 Equivalent Data value may not be available to the kernel for this type of transaction.

The kernel has to perform DDA checking as it is defined in EMV contact specifications [EMV-2: Section 6] using the information present in the 'Kernel Transaction Database' and the INTERNAL AUTHENTICATE command response (step 9.2.9).

Exception: For Issuer Identifier value checking and PAN value checking, the kernel must use PAN data element value if present. When PAN data element is not present, the kernel must use PAN value written inside' Track2 Equivalent Data' data element. When both the PAN value and 'Track2 Equivalent Data' are not available, all the checks requiring the knowledge of PAN value are bypassed (they are not considered as failed). When the PAN value is available, failure of the checks involving PAN value is considered as DDA failure.

TVR shall be updated in case of DDA failure.

3-Else

The ICC Dynamic Number (IDN) value (tag '9F4C') retrieved in the recovered Signed Dynamic Application Data signature needs to be added to the 'Kernel Transaction Database'.

3-Endif

2-Endif

1-Endif

The next step is defined in section 9.2.18.

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9.2.18Kernel de-activation

Requirement – Conditions of Execution	
K18.1	Support of this function is mandatory

Requirement – Sequence of Execution	
K18.2	This is the last step of the transaction flow. Transaction is completed on "terminate". Note: Detection of a communication problem has no impact during this step

Requirement – Kernel De-activation Processing	
K18.3	Note: The kernel is using the information present in the 'Kernel Transaction Database' **1-If [**(Transaction Type = Trans_Type_Auth_Appli when Implementation option 4 is supported (**IO_Option4** = 1)) *'Application Authentication Transaction* **1-And** (option selected by Acquirer in "Contactless Application Capabilities" parameter (AC_Auth_Appli)) **1-Then** The Kernel shall terminate the transaction and shall provide the "Non-Financial Transaction Completed- Offline authentication" outcome message. • "Data Record": The Kernel must provide all the data elements referenced in Table 32. The Acquirer or the Payment Scheme may define additional data elements (parameter ATDTOL). If a data element is missing, it has no impact. • When implementation 1 is supported and conditions were satisfied for processing 'Retrieval of Data Element values in EEPROM" transaction step, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record". • When implementation 2 is supported and conditions were satisfied for processing at least one of the transaction steps described in sections (9.2.5 or 9.2.7), "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record". **1-Endif**
K18.4	**1-If** card decision (CID) is approved (the two CID most significant bits = '01') **1-Then** • The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Transaction approved (TC)" outcome message. • CVM information is fulfilled based on CVM Results value **1-If** byte 'CVM Result' is equal to '00' (Unknown) **or** to '02' (Successful) **1-Then** **2-Depending** on 6 least significant bits of byte 'CVM Performed' **2-Value** '02': CVM is 'Online PIN'

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Requirement – Kernel De-activation Processing

2-Value '1E': CVM is 'Signature'
2-Value '1F': CVM is 'No CVM'
2-Value '24': CVM is 'Confirmation Code Verified'
2-Other Value: CVM is 'N/A'
2-End_Depending
1-Else
CVM is 'N/A'
1-Endif

- Message Identifier is fulfilled based on CVM value
  - '03' ("Approved") if signature is not requested
  - '1A' ("Approved – Please Sign") if signature is requested
- "Value Qualifier: Balance" is added if the Kernel was able to retrieve the value of COTA Offline Balance and CRM Currency Code during step 9.2.16
- "Data Record": The Kernel must provide all the data elements referenced in Table 31. The Acquirer or the Payment Scheme may define additional mandatory data elements (parameter ATOL). If a mandatory data element is missing, it has no impact on kernel decision.
- When implementation 1 is supported and conditions were satisfied for processing 'Retrieval of Data Element values in EEPROM" transaction step, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
- When implementation 2 is supported and conditions were satisfied for processing at least one of the transaction steps described in sections (9.2.5, 9.2.7 or 9.2.15), "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

1-Endif

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Requirement – Kernel De-activation Processing

1-If card decision (CID) is declined (the two CID most significant bits = '00')

1-Then

- The Kernel shall terminate the transaction and shall provide the "Financial Transaction Completed- Transaction declined (AAC)" outcome message.
  - If transaction is not terminated at GET PROCESSING OPTIONS command level
    - "Value Qualifier: Balance" is added if the Kernel was capable to retrieve the value of COTA Offline Balance and CRM Currency Code during step 9.2.16
    - The Kernel has to try to provide all the data elements that may be present in field "Data Record". A missing mandatory data element has no impact.
  - When implementation 1 is supported and conditions were satisfied for processing 'Retrieval of Data Element values in EEPROM" transaction step, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
  - When implementation 2 is supported and conditions were satisfied for processing at least one of the transaction steps described in sections (9.2.5, 9.2.7 or 9.2.15), "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

K18.5

1-Endif

1-If card decision (CID) is online request (the two CID most significant bits = '10')

1-Then

2-If CCID is present in 1st GENERATE AC command response **and** 1st GENERATE AC CCID bit 4 is equal to '1'('Additional tap for Issuer Response supported') and online tap option is supported (**AC_Online_Tap** = 1)

2-Then the Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Online Request (2nd Tap for Issuer Response)" outcome message

2-Else

K18.6

3-If the long tap option is supported (**AC_Long_Tap** = 1)

3-Then the Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Online Request (Long Tap)" outcome message

3-Else the Kernel shall terminate the transaction and shall provide, the "Financial Transaction Completed- Online Request (No Additional Tap)" outcome message

3-Endif

2-Endif

For each outcome message, the outcome message fields are:

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Requirement – Kernel De-activation Processing

- CVM information is fulfilled based on CVM Results value
  1-If byte 'CVM Result' is equal to '00' (Unknown) **or** to '02' (Successful)
  1-Then
  2-Depending on 6 least significant bits of byte 'CVM Performed'
  2-Value '02': CVM is 'Online PIN'
  2-Value '1E': CVM is 'Signature'
  2-Value '1F': CVM is 'No CVM'
  2-Value '24': CVM is 'Confirmation Code Verified'
  2-Other Value: CVM is 'N/A'[37]
  2-End_Depending
  1-Else
  CVM is 'N/A'
  1-Endif
- Message Identifier is fulfilled based on CVM information value
  - 1B' ("Authorizing Please Wait") if online PIN is not requested
  - '09' ("Please Enter your PIN") if online PIN is requested
- "Value Qualifier: Balance" is added if the Kernel was able to retrieve the value of COTA Offline Balance and CRM Currency Code during step 9.2.16
- "Data Record": The Kernel must provide all the data elements referenced in Table 31. The Acquirer or the Payment Scheme may define additional mandatory data elements (parameter ATOL). If a mandatory data element is missing, it has no impact on kernel decision.
- When implementation 1 is supported and conditions were satisfied for processing 'Retrieval of Data Element values in EEPROM" transaction step, "GDDOL Resulting Buffer" (9F71) needs to be added to field "Discretionary Data Record".
- When implementation 2 is supported and conditions were satisfied for processing at least one of the transaction steps described in sections (9.2.5, 9.2.7 or 9.2.15), "Data Element Update Result" (9F74) needs to be added to field "Discretionary Data Record".

1-Endif

[37] It is recommended to use N/A. It is nevertheless permitted to set CVM information to 'No CVM'

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9.2.19 Communication Errors

Requirement – Communication Errors	
K19.1	**If** a Transmission, Protocol, or Timeout error as defined in Book D is reported to the Kernel, **Then** The impact is depending on the step of the transaction processed (see previous sections). **Endif**

9.2.20 Transaction Cancellation

The Kernel may receive at any time a transaction cancellation order initiated by the Merchant (attended terminal) or by the Cardholder (unattended terminal).

Requirement – Transaction Cancellation	
K19.2	The Kernel shall be capable of receiving a cancellation order from the Contactless reader at any time during transaction processing.
K19.3	**If** a cancellation order is received from the Contactless reader **Then** The Kernel shall: • Reset the contents of the 'Kernel Transaction Database' • Terminate the transaction and provides the "Transaction cancellation by Contactless reader" outcome message. **Endif**

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9.3 Interface Contactless Kernel / Contactless reader

9.3.1 Contactless reader Information to communicate at Kernel activation

9.3.1.1 Transactions other than Application Authentication transaction

Requirement – Information to communicate to PURE contactless kernel

When the PURE contactless kernel supports no implementation option, the contactless reader must provide the following list of dynamic transaction parameters (except when reactivating the PURE contactless kernel after an online request outcome message – see K20.4). The contactless reader may provide additional transaction parameters (please refer to K1.1).

K20.1

Data Element	Tag	Requirement
Contactless Pre-Processing Indicators associated to the application AID selected	-	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	‘6F’	Mandatory
Amount, Authorized (Numeric)	'9F02'	Mandatory
Amount, Other (Numeric)	'9F03'	Optional
Transaction Currency Code	‘5F2A’	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when contactless reader does not support multi-currencies feature)
Terminal Country Code	‘9F1A’	Mandatory (May not be present if it is a static kernel configuration parameter)
Transaction Date	'9A'	Mandatory
Transaction Time	‘9F21’	Optional
Transaction Type	'9C'	Mandatory
Terminal Transaction Data	‘9F76’	Optional
Unpredictable Number	'9F37'	Mandatory[23]

Table 24 - Transaction Parameters (no implementation option supported)

Requirement – Information to communicate to PURE contactless kernel

K20.2 When the PURE contactless kernel supports the implementation option 1, the contactless reader must provide the following list of dynamic transaction parameters (except when reactivating the PURE contactless kernel after an online request outcome message – see K20.4). The contactless reader may provide additional transaction parameters (please refer to K1.1).

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Requirement – Information to communicate to PURE contactless kernel

Data Element	Tag	Requirement when transaction type '78'	Requirement for any other transaction Type
Contactless Pre-Processing Indicators associated to the application AID selected	-	Mandatory	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	'6F'	Mandatory	Mandatory
Transaction Type	'9C'	Mandatory	Mandatory
Memory Slot Read Template (List of Data Object to retrieve)	'BF71'	Mandatory	Optional (if present, the kernel will use it)
Amount, Authorized (Numeric)	'9F02'	Optional (if present, the kernel will ignore it)	Mandatory
Amount, Other (Numeric)	'9F03'	Optional (if present, the kernel will ignore it)	Optional
Transaction Currency Code	'5F2A'	Optional (if present, the kernel will ignore it)	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when contactless reader does not support multi-currencies feature)
Terminal Country Code	'9F1A'	Optional (if present, the kernel will ignore it)	Mandatory (May not be present if it is a static kernel configuration parameter)
Transaction Date	'9A'	Optional (if present, the kernel will ignore it)	Mandatory
Transaction Time	'9F21'	Optional (if present, the kernel will ignore it)	Optional
Terminal Transaction Data	'9F76'	Optional	Optional
Unpredictable Number	'9F37'	Optional (if present, the kernel will ignore it)	Mandatory[23]

Table 25 - Transaction Parameters (implementation option 1 supported)

Requirement – Information to communicate to PURE contactless kernel

K20.3 When the PURE contactless kernel supports the implementation option 2, the contactless reader must provide the following list of dynamic transaction parameters (except when reactivating the PURE contactless kernel after an online request outcome message – see K20.4). The contactless reader may provide additional transaction parameters (please refer to K1.1).

Data Element	Tag	Requirement when transaction type '79'	Requirement for any other transaction Type

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Requirement – Information to communicate to PURE contactless kernel

Contactless Pre-Processing Indicators associated to the application AID selected	-	Mandatory	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	‘6F’	Mandatory	Mandatory
Transaction Type	'9C'	Mandatory	Mandatory
Memory Slot Update Template	‘BF70’	Mandatory	Optional (if present, the kernel will process it)
Amount, Authorized (Numeric)	'9F02'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory
Amount, Other (Numeric)	'9F03'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Optional
Transaction Currency Code	‘5F2A’	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory (May not be present if it is a static kernel configuration parameter. It may be the case when contactless reader does not support multi-currencies feature)
Terminal Country Code	‘9F1A’	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory (May not be present if it is a static kernel configuration parameter)
Transaction Date	'9A'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory
Transaction Time	‘9F21’	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Optional
Terminal Transaction Data	‘9F76’	Optional	Optional
Unpredictable Number	'9F37'	Optional (if present, the kernel may use it if the tag value is referenced in PDOL)	Mandatory[23]

Table 26 - Transaction Parameters (implementation option 2 supported)

Requirement – Information to communicate to PURE contactless kernel

K20.4 When the PURE contactless kernel is setup for the support of the long tap option or for the support of the online tap option, the contactless reader must provide the following list of dynamic transaction parameters when reactivating the PURE contactless kernel after an online request outcome message. According to the implementation, the contactless reader may provide some additional parameters (please refer to K1.1).

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Requirement – Information to communicate to PURE contactless kernel

Data Element	Tag	Requirement
Contactless Pre-Processing Indicators associated to the application AID selected	-	Optional (if present, the kernel will ignore it)
SELECT Response Message Data Field (FCI) of the selected application	'6F'	Shall not be present when the contactless reader has started at **Start D** (long tap)
Authorization Response Code Values supported: • "00" = Transaction approved • Other value = Transaction declined	'8A'	Mandatory
Amount, Authorized (Numeric)	'9F02'	Mandatory
Amount, Other (Numeric)	'9F03'	Optional
Transaction Currency Code	'5F2A'	Mandatory
Terminal Transaction Data	'9F76'	Optional
Unpredictable Number	'9F37'	Mandatory[23]
Issuer Authentication Data	'91'	Optional
Critical Issuer Script Command Template	'71'	Optional When both data elements are present, the kernel process first the commands declared in data element '71'.
Non-critical Issuer Script Command Template	'72'	

Table 27 - Transaction Parameters after completion of an online authorization

K20.5	When the PURE contactless kernel supports several implementation options, the contactless reader must comply with all the requirements associated to the implementation options supported.
K20.6	The kernel implementation must be capable to support at least 1000 bytes buffer for storage of the dynamic transaction parameters listed for each type of transaction in the following tables: Table 24 to Table 30

9.3.1.2 Application Authentication Transactions

Requirement – Information to communicate to PURE contactless kernel

K20.7 When the PURE contactless kernel supports no other implementation option, the contactless reader must provide the following list of dynamic transaction parameters for Application Authentication Transactions. The contactless reader may provide additional transaction parameters (please refer to K1.1).

Data Element	Tag	Requirement

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Requirement – Information to communicate to PURE contactless kernel

Contactless Pre-Processing Indicators associated to the application AID selected[38]	-	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	‘6F’	Mandatory
Transaction Type	'9C'	Mandatory
Unpredictable Number	'9F37'	Mandatory[23]

Table 28 - Transaction Parameters for Application Authentication Transaction (implementation option 4 supported)

When the PURE contactless kernel supports the implementation option 1, the contactless reader must provide the following list of dynamic transaction parameters for Application Authentication Transactions. The contactless reader may provide additional transaction parameters (please refer to K1.1).

K20.8

Data Element	Tag	Requirement
Contactless Pre-Processing Indicators associated to the application AID selected[38]	-	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	‘6F’	Mandatory
Transaction Type	'9C'	Mandatory
Memory Slot Read Template (List of Data Object to retrieve)	‘BF71’	Optional (if present, the kernel will use it)
Unpredictable Number	'9F37'	Mandatory[23]

Table 29 - Transaction Parameters for Application Authentication Transaction (implementation option 1 and option 4 supported)

Requirement – Information to communicate to PURE contactless kernel

K20.9 When the PURE contactless kernel supports the implementation option 2, the contactless reader must provide the following list of dynamic transaction parameters for Application Authentication Transaction. The contactless reader may provide additional transaction parameters (please refer to K1.1).

[38] When the contactless reader is compliant with section 9, all Contactless Pre-Processing Indicators values are set to ‘0’. A reader compliant with another specification (for example EMV Entry Point) need to be setup to get same result (either Reader Contactless limits associated to transaction type “Application Authentication Transactions” set to a high value or omitted).

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Requirement – Information to communicate to PURE contactless kernel

Data Element	Tag	Requirement
Contactless Pre-Processing Indicators associated to the application AID selected[18]	-	Mandatory
SELECT Response Message Data Field (FCI) of the selected application	‘6F’	Mandatory
Transaction Type	'9C'	Mandatory
Memory Slot Update Template	‘BF70’	Optional (if present, the kernel will process it)
Unpredictable Number	'9F37'	Mandatory[23]

Table 30 - Transaction Parameters for Application Authentication Transaction (implementation option 2 and option 4 supported)

K20.10	When the PURE contactless kernel supports several implementation options, the contactless reader must comply with all the requirements associated to the implementation options supported.

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9.3.2 Kernel Outcome messages to communicate to the contactless reader

9.3.2.1 Kernel Outcome message List

A PURE contactless kernel has to support the following outcome message.

Step	Use case	Outcome message to support
1	No use case	
2	Either a mandatory dynamic transaction parameter or mandatory static configuration parameters is missing	Transaction Completed on Error
	The Application Selection response cannot be parsed correctly as defined for the SELECT command	Transaction Completed on Error
	No data to process after an online authorization	Financial Transaction Completed - erroneous re-activation of the kernel after an Online Request
3	Condition not satisfied for processing chip data present in online response message	Financial Transaction Completed- Erroneous re-activation of the kernel after an Online Request
	No chip data to process after an online authorization	Financial Transaction Completed - erroneous re-activation of the kernel after an Online Request
	Only script commands in chip data received in online response message	Financial Transaction Completed - erroneous re-activation of the kernel after an Online Request
	Terminal request not consistent with Issuer Authorization Host decision	Financial Transaction Completed- Transaction declined (AAC)
	Transaction approved based on chip data received in online response message	Financial Transaction Completed- Transaction approved (TC)
	Transaction rejected based on chip data received in online response message	Financial Transaction Completed- Transaction declined (AAC)
4	If implementation option 1 is supported and transaction type is ‘78’	Non-Financial Transaction Completed– Retrieval of data elements
	If implementation option 1 is supported and transaction type is not ‘78’ and GDDOL present	-
5	If implementation option 2 is supported and transaction type is ‘79’	Non-Financial Transaction Completed- Data elements update using PUT DATA
	If implementation option 2 is supported, transaction type is not ‘79’ and contactless reader requests to update some data elements. A PUT DATA command has failed. Contactless reader requirement is to stop the transaction in case of failure	Non-Financial Transaction Completed- Data elements update using PUT DATA
6	GET PROCESSING OPTIONS command processing • Error during command processing • Card response cannot be parsed correctly	Transaction Completed on Error

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Step	Use case	Outcome message to support
	• Card requests to switch to contact interface	Financial Transaction Completed– Try other interface
	• Card requests a Confirmation Code	Financial Transaction Completed- On-device CVM Requested
	• Transaction declined	Financial Transaction Completed– transaction declined (AAC)
7	If implementation option 2 is supported and transaction type is '79'	Non-Financial Transaction Completed- Data elements update
	If implementation option 2 is supported, transaction type is not '79' and contactless reader requests to update some data elements. A PUT DATA command has failed. Contactless reader requirement is to stop the transaction in case of failure	Non-Financial Transaction Completed- Data elements update
8	Error during READ RECORD command on records referenced in AFL READ RECORD response cannot be parsed correctly Record read is not TLV structured	Transaction Completed on Error
9	No use case	-
10	No use case	-
11	No use case	-
12	No use case	-
13	No use case	-
14	1st GENERATE AC command processing • Error during command processing • Card response cannot be parsed correctly	Transaction Completed on Error
15	If implementation option 2 is supported contactless reader requests to update some data elements. A PUT DATA command has failed. Contactless reader requirement is to stop the transaction in case of failure	Non-Financial Transaction Completed- Data elements update
16	No use case	
17	No use case	
18	For Application Authentication Transactions	Non-Financial Transaction Completed – Offline Authentication
	For transactions other than Application Authentication transaction, based on card decision in 1st GENERATE AC and on offline CAM checking result (if required), the decision may be either Transaction approved	Financial Transaction Completed– Transaction approved (TC)
	Transaction declined	Financial Transaction Completed– Transaction declined (AAC)
	Transaction to be processed online	Financial Transaction Completed– Online Request (No Additional Tap)

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Step	Use case	Outcome message to support
		Financial Transaction Completed– Online Request (Long Tap)
		Financial Transaction Completed– Online Request (2nd Tap for Issuer Response)
19	Communication error	Communication Error
20	Transaction cancellation	Transaction cancellation by Contactless reader

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9.3.2.2 Transaction Completed on Error

The contactless reader should try another contactless application mutually supported by the card and the terminal.

Requirement – Transaction completed on error	
K21.1	The Kernel shall provide a ***Select Next*** Outcome with the following parameters: **Start:** C **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** No **UI Request on Restart Present:** No **Data Record Present:** No **Discretionary Data Present:** No **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero

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9.3.2.3 Non-Financial Transaction Completed

Requirement – Non-Financial Transaction completed – Retrieval of data elements

K21.2	The Kernel shall provide an ***End Application*** Outcome with the following parameters: **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** Yes Message Identifier: '10' ("Please remove card") Status: Card Read Successfully (Remove card) Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present. **UI Request on Restart Present:** No **Data Record Present:** No **Discretionary Data Present:** Yes TLV structured buffer containing: ▪ Tag '9F71' – GDDOL Resulting Buffer containing concatenated value of the data elements requested in GDDOL **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero

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Requirement – Non-Financial Transaction completed – Retrieval of data elements
Requirement – Non-Financial Transaction completed – Data elements update using PUT DATA

K21.3	The Kernel shall provide an ***End Application*** Outcome with the following parameters: **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** Yes Message Identifier: '10' ("Please remove card") Status: Card Read Successfully (Remove card) Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present. **UI Request on Restart Present:** No **Data Record Present:** No **Discretionary Data Present:** Yes TLV structured buffer containing when available: ▪ Tag '9F71' – GDDOL Resulting Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported) ▪ Tag '9F74' – Data Elements Update Result **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero

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Requirement – Non-Financial Transaction completed – Offline authentication

The Kernel shall provide an ***End Application*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '10' ("Please remove card")

Status: Card Read Successfully (Remove card)

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part of the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

UI Request on Restart Present: No

Data Record Present: Yes

K21.3 TLV structured buffer containing at least the data elements listed Table 32. The Issuer may request the presence of additional data elements listed in parameter ATDTOL (see section 11.2.29). When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: N/A

Removal Timeout: Zero

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9.3.2.4 Financial Transaction Completed

Note: Implementation must support at least 1024 bytes for outcome message. When the maximum size is exceeded, the information to remove first are the values of the additional data elements referenced in ATOL. If it is not sufficient, GDOL Resulting Buffer value must be removed from the outcome message.

Requirement – Financial Transaction Completed – Transaction approved (TC)

The Kernel shall provide an ***Approved*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: According to the CVM that the kernel has selected:

No CVM/ Obtain Signature/ Confirmation Code Verified

UI Request on Outcome Present: Yes

Message Identifier: as applicable

'03' ("Approved") if No CVM requested

'1A' ("Approved – Please Sign") if signature requested

Status: Card Read Successfully (Remove card)

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part of the data elements available in 'Kernel Transaction Database'. In the other cases this field is not present.

[*Value Qualifier: "Balance"****]***[39]

[*Value: COTA Offline Balance (Tag '9F50') returned by card* ***]***

K21.4 ***[****Currency Code: CRM Currency Code (Tag 'CD') returned by card****]***

UI Request on Restart Present: No

Data Record Present: Yes

TLV structured buffer containing at least the data elements listed in Table 31. Acquirer or Payment Scheme may require the presence of additional data elements listed in parameter ATOL (see section 11.2.4).

Note: When at least one script command is processed during the online response analysis, Data Record field must be present in the outcome message and must at least contain Issuer Script Result.

When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)

[39] Parameters in brackets ***[ ]*** are provided only if the kernel was capable to retrieve the COTA Offline Balance (Tag '9F50') value and the CRM Currency Code (tag CD).

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Requirement – Financial Transaction Completed – Transaction approved (TC)
▪ Tag ‘9F74’ – Data Element Update Result (only if implementation option 2 is supported) **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero

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Requirement – Financial Transaction Completed – Transaction declined (AAC)

The Kernel shall provide a ***Declined*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '07' ("Not Authorized")

Status: Card Read Successfully (Remove card)

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part of the data elements available in 'Kernel Transaction Database'. In the other cases this field is not present.

[*Value Qualifier: "Balance"****]***[40]

[*Value: COTA Offline Balance (Tag '9F50') returned by card* ***]***

[*Currency Code: CRM Currency Code (Tag 'CD') returned by card****]***

UI Request on Restart Present: No

Data Record Present: Yes (except if transaction declined at GET PROCESSING OPTIONS command level or declined due to error during additional online tap)

K21.5

TLV structured buffer containing at least the data elements listed in Table 31. Acquirer or Payment Scheme may require the presence of additional data elements listed in parameter ATOL (see section 11.2.4).

Note: When at least one script command is processed during the online response analysis, Data Record field must be present in the outcome message and must at least contain Issuer Script Result.

When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: N/A

Receipt: N/A

[40] Parameters in brackets ***[ ]*** are provided only if the kernel was capable to retrieve the COTA Offline Balance (Tag '9F50') value and the CRM Currency Code (tag CD).

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Requirement – Financial Transaction Completed – Transaction declined (AAC)

Field Off Request: N/A

Removal Timeout: Zero

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Requirement – Financial Transaction Completed – Try other interface

The Kernel shall provide a ***Try Another Interface*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '18' ("Please insert or swipe card")

Status: Ready to Read

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

K21.6 **UI Request on Restart Present:** No

Data Record Present: No

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: Contact Chip

Receipt: N/A

Field Off Request: N/A

Removal Timeout: Zero

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Requirement – Financial Transaction Completed – Online Request (No Additional Tap)

The Kernel shall provide an ***Online Request*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: According to the CVM that the kernel has selected:

'N/A'[41]/No CVM/ Obtain Signature/ Online PIN/ Confirmation Code Verified

UI Request on Outcome Present: Yes

Message Identifier: as applicable

'1B' ("Authorizing Please Wait") if No CVM or Signature

'09' ("Please Enter your PIN") if online PIN requested

Status: Card Read Successfully (Remove card)

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

[*Value Qualifier: "Balance"****]***[42]

[*Value: COTA Offline Balance (Tag '9F50') returned by card* ***]***

[*Currency Code: CRM Currency Code (Tag 'CD') returned by card****]***

K21.7 **UI Request on Restart Present:** No

Data Record Present: Yes

TLV structured buffer containing at least the data elements listed in Table 31. Acquirer or Payment Scheme may require the presence of additional data elements listed in parameter ATOL (see section 11.2.4).

When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: N/A

[41] It is recommended to support 'N/A' but it is allowed to replace 'N/A' by 'No CVM'.

[42] Parameters in brackets ***[ ]*** are provided only if the kernel was capable to retrieve the COTA Offline Balance (Tag '9F50') value and the CRM Currency Code (tag CD).

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Requirement – Financial Transaction Completed – Online Request (No Additional Tap)

Removal Timeout: Zero

Requirement – Financial Transaction Completed – Online Request (Long Tap)

The Kernel shall provide an ***Online Request*** Outcome with the following parameters:

Start: D

Online Response Data: Any

CVM: According to the CVM that the kernel has selected:

'N/A'[41]/No CVM/ Obtain Signature/ Online PIN/ Confirmation Code Verified

UI Request on Outcome Present: Yes

Message Identifier: as applicable

'1B' ("Authorizing Please Wait") if No CVM or Signature

'09' ("Please Enter your PIN") if online PIN requested

Status: Processing

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

[*Value Qualifier: "Balance"****]***[43]

K21.8 ***[****Value: COTA Offline Balance (Tag '9F50') returned by card* ***]***

[*Currency Code: CRM Currency Code (Tag 'CD') returned by card****]***

UI Request on Restart Present: Yes

Message Identifier: '16' ("Processing")

Status: Processing

Data Record Present: Yes

TLV structured buffer containing at least the data elements listed in Table 31. Acquirer or Payment Scheme may require the presence of additional data elements listed in parameter ATOL (see section 11.2.4).

When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)

[43] Parameters in brackets ***[ ]*** are provided only if the kernel was capable to retrieve the COTA Offline Balance (Tag '9F50') value and the CRM Currency Code (tag CD).

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Requirement – Financial Transaction Completed – Online Request (Long Tap)
▪ Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported) **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Value of POS parameter "POS Timeout (Long tap)"

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Requirement – Financial Transaction Completed – Online Request (2nd Tap for Issuer Response)

The Kernel shall provide an ***Online Request*** Outcome with the following parameters:

Start: B

Online Response Data: EMV Data

CVM: According to the CVM that the kernel has selected:

No CVM/ Obtain Signature/ Online PIN/ Confirmation Code Verified

UI Request on Outcome Present: Yes

Message Identifier: as applicable

'1B' ("Authorizing Please Wait") if No CVM or Signature

'09' ("Please Enter your PIN") if online PIN requested

Status: Card Read Successfully

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

[*Value Qualifier: "Balance"****]***[44]

[*Value: COTA Offline Balance (Tag '9F50') returned by card* ***]***

[*Currency Code: CRM Currency Code (Tag 'CD') returned by card****]***

UI Request on Restart Present: Yes

K21.9 Message Identifier: '21' ("Present Card Again")

Status: Ready to Read

Data Record Present: Yes

TLV structured buffer containing at least the data elements listed in Table 31. Acquirer or Payment Scheme may require the presence of additional data elements listed in parameter ATOL (see section 11.2.4)

When a data element is not present in the list of data elements known by the kernel, the data element is ignored without any impact on transaction processing.

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: N/A

[44] Parameters in brackets ***[ ]*** are provided only if the kernel was capable to retrieve the COTA Offline Balance (Tag '9F50') value and the CRM Currency Code (tag CD).

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Requirement – Financial Transaction Completed – Online Request (2nd Tap for Issuer Response)

Removal Timeout: Zero

Data Element Name	Tag	Source
Amount, Authorized (Numeric)	'9F02'	POS
Amount, Other (Numeric)	'9F03'	POS (if present)
Application Cryptogram (AC)	'9F26'	Card
Application Interchange Profile (AIP)	'82'	Card
Application Transaction Counter (ATC)	'9F36'	Card
Cryptogram Information Data (CID)	'9F27'	Card
Issuer Application Data (IAD)	'9F10'	Card
Terminal Country Code	'9F1A'	POS
Terminal Verification Results (TVR)	'95'	Kernel
Transaction Currency Code	'5F2A'	POS
Transaction Date	'9A'	POS
Transaction Type	'9C'	POS
Unpredictable Number (UN)	'9F37'	POS
Terminal Type	'9F35'	POS
Track 2 Equivalent Data	'57'	Card
CVM Result	'9F34'	Kernel
Dedicated File Name (AID) of the application selected	'84'	Card
Application PAN Sequence Number	'5F34'	Card (if present in the card)
PAN	'5A'	Card (if present in the card)
Track 1 Discretionary Data	'9F1F'	Card (if present in the card)
Cardholder Name	'5F20'	Card (if present in the card)
Terminal Dedicated Data	'9F77'	Card (if present in the card)

Table 31 – Mandatory Data Elements to return in Data Record for transaction other than Application Authentication transaction

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Data Element Name	Tag	Source
Application Interchange Profile (AIP)	‘82’	Card
Terminal Verification Results (TVR)	‘95’	Kernel
Terminal Dedicated Data	‘9F77’	Card (if present in the card)
Dedicated File Name (AID) of the application selected	‘84’	Card

Table 32 –Data Elements to return in Data Record for Application Authentication Transaction

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Requirement – Financial Transaction Completed – On-device CVM Requested

The Kernel shall provide an ***End Application*** Outcome with the following parameters:

Start: B

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '20' ("See Phone for Instructions")

Status: Processing Error

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

Hold Time: 13

K21.10 **UI Request on Restart Present:** Yes

Message Identifier: '21' ("Present Card Again")

Status: Ready to Read

Data Record Present: No

Discretionary Data Present: Yes if data available

TLV structured buffer containing when available:

- Tag '9F71' – GDDOL Resulting Buffer - Buffer containing concatenated value of the data elements requested in GDDOL (only if implementation option 1 is supported)
- Tag '9F74' – Data Element Update Result (only if implementation option 2 is supported)

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: 13

Removal Timeout: Zero

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Requirement – Financial Transaction Completed – Erroneous re-activation of the kernel after an Online Request	
K21.11	The Kernel shall provide an ***End Application*** Outcome with the following parameters: **Start:** N/A **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** No **UI Request on Restart Present:** No **Data Record Present:** No (except if script command processed) Note: When at least one script command is processed during the online response analysis, Data Record field must be present in the outcome message and must at least contain Issuer Script Result. **Discretionary Data Present:** No **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** N/A **Removal Timeout:** Zero

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9.3.2.5 Communication Error

Requirement – Communication Errors – Payment Transaction not completed

K21.12

The Kernel shall provide an ***Try Again*** Outcome with the following parameters:

Start: B

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '21' ("Present Card Again")

Status: Processing Error

Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

Hold Time: 13

UI Request on Restart Present: Yes

Message Identifier: '21' ("Present Card Again")

Status: Ready to Read

Data Record Present: No

Discretionary Data Present: No

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: 13

Removal Timeout: Zero

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Requirement – Communication Errors – Unknown Payment Result	
K21.13	The Kernel shall provide an ***Try Again*** Outcome with the following parameters: **Start:** B **Online Response Data:** N/A **CVM:** N/A **UI Request on Outcome Present:** Yes Message Identifier: '21' ("Present Card Again") Status: Processing Error Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present. Hold Time: 13 **UI Request on Restart Present:** Yes Message Identifier: '21' ("Present Card Again") Status: Ready to Read **Data Record Present:** No **Discretionary Data Present:** No **Alternate Interface Preference:** N/A **Receipt:** N/A **Field Off Request:** 13 **Removal Timeout:** Zero

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9.3.2.6 Transaction cancellation by Contactless reader

Requirement – Transaction cancellation by Contactless reader

The Kernel shall provide an ***End Application*** Outcome with the following parameters:

Start: N/A

Online Response Data: N/A

CVM: N/A

UI Request on Outcome Present: Yes

Message Identifier: '1C' ("Insert, Swipe or Try another card")

Status: Processing Error

K21.14 Language Preference: Value of data element 'Language Preference' (tag '5F2D') if it is part if the data elements available in Data Field FCI of the SELECT command response communicated to the kernel. In the other case this field is not present.

UI Request on Restart Present: No

Data Record Present: No

Discretionary Data Present: No

Alternate Interface Preference: N/A

Receipt: N/A

Field Off Request: N/A

Removal Timeout: Zero

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10 User Interface Management Detailed Specifications

Any module of the POS architecture (PURE contactless kernel, Contactless reader and Terminal Application) may have need of user messaging. They may request messages and status to be presented and POS Systems must be able to service these requests. The exact configuration of message display or status indication will vary according to design and implementation.

Requirement – User Interface Management	
U1.1	The POS System must comply with the requirements defined in the EMVco document "EMV Contactless Specifications for Payment Systems - Book A" [Reference EMV-5] • The User Interface Request messages and their parameters are defined in the EMV specifications [EMV-5: Section 7]. • When processing the User Interface Request messages, the requirements defined in the EMV specifications [EMV-5: Section 8 – requirements 8.1.1.15 to 8.1.1.18] shall be satisfied.
U1.2	Depending on the POS architecture, the **User Interface Request** messages are processed either by the Terminal Application or by the contactless reader or by both.

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Requirement – User Interface Management

The **User Interface Request** messages are using the following parameters:

Message Identifier: Indicates the text string to be displayed

The values currently used in PURE contactless kernel are:

- '03' ("Approved") [45]
- '07' ("Not Authorized")
- '09' ("Please Enter your PIN")
- '10' ("Please remove card")
- '14' ("Welcome")
- '15' ("Present Card")
- '16' ("Processing")
- '17' ("Card Read OK. Please Remove Card")
- '18' ("Please insert or swipe card")
- '19' ("Please Present One Card Only")
- '1A' ("Approved – Please Sign")
- '1B' ("Authorizing Please Wait")
- '1C' ("Insert, Swipe or Try another card")
- '20' ("See Phone for Instructions")
- '21' ("Present Card Again")

U1.3

Status: Identifies the status of the transaction (for example when the card can be removed) to be indicated through the setting of lights or LEDs and/or the production of an audio signal.

The values currently used in PURE contactless kernel are:

- Ready to Read
- Card Read Successfully (Remove card)
- Processing
- Processing Error
- Idle

Language Preference: If Language Preference (as per EMV Tag '5F2D' according to ISO 639-1) is present and if the language is supported by the contactless reader or Terminal Application, then the message identified and all subsequent messages to the cardholder are displayed in this language until the transaction concludes.

Hold Time: indicates that the contactless reader is to delay the processing of the Message Identifier in the next User Interface Request until the Hold Time has elapsed. The Hold Time is an integer in units of 100ms;

Value Qualifier / Value / Currency Code: when used, the 3 parameters are present in a User Interface Request message. The type of value currently used in PURE contactless kernel is Balance. It communicates to the Terminal Application the current application balance.

- ***Value Qualifier:*** value 'Balance'
- ***Value:*** Application Balance value

[45] Following an online processing, it is also allowed to use « Online Approved ».

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Requirement – User Interface Management	
	• ***Currency Code:*** Currency code associated to the application balance value.
U1.4	It is recommended to support a set of four clearly visual single color status indicators (for example LEDs). Each indicator should be associated to a 'Status' value and should be one when the 'Status' has the value associated. When the 'Status' value is '***Idle***' all the indicators should be off. Note: Payment Scheme may mandate to comply with this recommendation.
U1.5	It is recommended to activate an audio signal when the 'Status' value is equal to '***Card Read Successfully (Remove card)***'. Note: Payment Scheme may mandate to comply with this recommendation.
U1.6	Messages shall be displayed to the Merchant. They may be also displayed to the Cardholder

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11 Data Elements Dictionary

11.1 List of Data Elements

The following table provides list of the data elements that may be used during a PURE contactless payment transaction. The ‘Origin’ column indicates where the data element value is stored or where it is generated.

The possible origins are:

- Host = Issuer Authorization host
- T = Terminal Application module
- R = contactless reader module
- K = PURE contactless kernel
- Card = Card payment application in contactless device

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Tag	Reference Specification	Origin	Name
-	PURE	R	AID Contactless Pre-Processing Indicators
-	PURE	R	Application Candidate List
-	PURE	K	Additional Tag Object List (ATOL)
-	PURE	R	Current Contactless Pre-Processing Indicators
-	PURE	R	Reader internal Indicators • Current Process Indicator • Contactless Reader Timeout Indicator • Restart Flag
-	PURE	R	Currently Selected Application
-	EMV	K	Certification Authority Public Keys Information
-	PURE	R	Contactless Application Capabilities
-	PURE	K	Contactless Kernel Capabilities
-	PURE	R - K	Contactless POS Implementation Options
-	PURE	R	Last Kernel Outcome Message
-	PURE	K	Mandatory Tag Object List (MTOL)
-	PURE	K	Kernel internal Indicators • Communication Problem Indicator • ECHO command support Indicator • Echo Transaction Indicator • Offline CAM Selected Indicator
-	PURE (Option 4)	K	Authentication Transaction Data Tag Object List (ATDTOL)
-	PURE	K	Kernel Transaction Database
-	PURE	K	POS Timeout (Long tap)
-	PURE	T	POS Timeout (Transaction Result Display)
-	EMV	O	Reader Contactless Transaction Limit per AID and financial transaction type
-	EMV	O	Reader Contactless Floor Limit per AID and financial transaction type
-	EMV	O	Reader CVM Required Limit per AID and financial transaction type
-	EMV	O	Status Check Support flag per AID and financial transaction type

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Tag	Reference Specification	Origin	Name
-	EMV	O	Zero Amount Allowed flag per AID and financial transaction type
-	EMV	O	Terminal Floor Limit per AID and financial transaction type
-	EMV	K	Terminal Action Code – Default
-	EMV	K	Terminal Action Code – Denial
-	EMV	K	Terminal Action Code – Online
-	PURE	T	Transaction Type value for Application Authentication Transactions
‘50’	EMV	Card	Application Label
‘57’	EMV	Card	Track 2 Equivalent Data
‘5A’	EMV	Card	PAN
‘5F20’	EMV	Card	Cardholder Name
‘5F24’	EMV	Card	Application Expiration Date
‘5F25’	EMV	Card	Application Effective Date
‘5F28’	EMV	Card	Issuer Country Code
'5F2A'	EMV	T	Transaction Currency Code
'5F2D'	EMV	Card	Language Preference
‘5F34’	EMV	Card	Application PAN Sequence Number
‘6F’	EMV	Card	SELECT Response Message Data Field (FCI)
‘71’	EMV	Host / T	Critical Issuer Script Command
‘72’	EMV	Host / T	Non Critical Issuer Script Command
‘82’	EMV	Card	Application Interchange Profile (AIP)
‘84’	EMV	Card	Dedicated File Name (AID)
‘85’	PURE (option 2)	T	Memory Slot Update Entry Setting
‘8A’	EMV	Host / T	Authorization Response Code
‘8C’	EMV	Card	Card Risk Management Data Object List 1 (CDOL1)
‘8D’	EMV	Card	Card Risk Management Data Object List 2(CDOL2)
‘8E’	EMV	Card	Cardholder Verification Method (CVM) List

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Tag	Reference Specification	Origin	Name
‘8F’	EMV	Card	Certification Authority Public Key Index
‘90’	EMV	Card	Issuer Public Key Certificate
‘91’	EMV	Host / T	Issuer Authentication Data
‘92’	EMV	Card	Issuer Public Key Remainder
‘93’	EMV	Card	Signed Static Application Data
‘94’	EMV	Card	Application File Locator (AFL)
‘95’	EMV	K	Terminal Verification Result (TVR)
‘9A’	EMV	T	Transaction Date
‘9C’	EMV	T	Transaction Type
‘9F01’	EMV	K	Acquirer Identifier
‘9F02’	EMV	T	Amount, Authorized (Numeric)
‘9F03’	EMV	T	Amount, Other (Numeric)
‘9F07’	EMV	Card	Application Usage Control
‘9F08’	EMV	Card	Application Version Number
‘9F09’	EMV	K	Terminal Application Version Number
‘9F0D’	EMV	Card	Issuer Action Code - Default
‘9F0E’	EMV	Card	Issuer Action Code - Denial
‘9F0F’	EMV	Card	Issuer Action Code - Online
‘9F10’	EMV	Card	Issuer Application Data (IAD)
‘9F11’	EMV	Card	Issuer Code Table Index
‘9F12’	EMV	Card	Application Preferred Name
‘9F13’	EMV	Card	Last Online Application Transaction Counter (ATC) Register
‘9F14’	EMV	Card	Lower Consecutive Offline Limit
‘9F15’	EMV	K	Merchant Category Code
‘9F1A’	EMV	T	Terminal Country Code
‘9F1F’	EMV	Card	Track 1 Discretionary Data
‘9F21’	EMV	T	Transaction Time
‘9F23’	EMV	Card	Upper Consecutive Offline Limit

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Tag	Reference Specification	Origin	Name
'9F26'	EMV	Card	Application Cryptogram (AC)
'9F27'	EMV	Card	Cryptogram Information Data (CID)
'9F2A'	EMV	Card	Kernel Identifier
'9F32'	EMV	Card	Issuer Public Key Exponent
'9F33'	EMV	K	Terminal Capabilities
'9F34'	EMV	K	CVM Result
'9F35'	EMV	K	Terminal Type
'9F36'	EMV	Card	Application Transaction Counter (ATC)
'9F37'	EMV	R	Unpredictable Number
'9F38'	EMV	Card	Processing Options Data Object List (PDOL)
'9F42'	EMV	Card	Application Currency Code
'9F46'	EMV	Card	ICC Public Key Certificate
'9F47'	EMV	Card	ICC Public Key Exponent
'9F48'	EMV	Card	ICC Public Key Remainder
'9F49'	EMV	Card	Dynamic Data Authentication Data Object List (DDOL)
'9F4A'	EMV	Card	Static Data Authentication Tag List
'9F4B'	EMV	Card	Signed Dynamic Application Data
'9F4D'	EMV	Card	Log Entry
'9F4E'	EMV	K	Merchant Name and Location
'9F4F'	EMV	Card	Log Format
'9F50'	PURE	Card	COTA Offline Balance
'9F70'	PURE (option 1)	T	GDDOL – List of data elements to retrieve using GET DATA
'9F71'	PURE (option 1)	K	GDDOL Resulting Buffer
'9F72'	PURE (Option 1)	T	Memory Slot Identifier
-	EMV	K	Issuer Script Results
'9F73'	PURE	K	Issuer Script Results
'9F74'	PURE	K	Data Elements Update Result

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Tag	Reference Specification	Origin	Name
'9F75'	PURE	Card	Echo Card Identifier
'9F76'	PURE	T	Terminal Transaction Data
'9F77'	PURE	T	Terminal Dedicated Data
'A2'	PURE (option 2)	T	Memory Slot Update Entry Type 1
'A3'	PURE (option 2)	T	Memory Slot Update Entry Type 2
'A4'	PURE (option 2)	T	Memory Slot Update Entry Type 3
'A5'	EMV	Card	File Control Information (FCI) Proprietary Template
'BF0C'	EMV	Card	File Control Information (FCI) Issuer Discretionary Data
'BF70'	PURE (option 2)	T	Memory Slot Update template
'BF71'	PURE (Option 1)	T	Memory Slot Read Template
'C5'	PURE	Card	Contactless Cryptogram Information Data (CCID)
'C7'	PURE	K	Terminal Transaction Processing Information (TTPI)
'CD'	PURE	Card	CRM Currency Code
'DF01' to 'DF0A'	PURE	T	Proprietary Data Element (memory slot)

Table 33 List of Data Elements

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11.2 PURE Proprietary Data Elements

11.2.1 AID Contactless Pre-Processing Indicators

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

AID Contactless Pre-Processing Indicators is a contactless reader internal transaction-transient parameter containing the result of the preliminary transaction processing checks performed on each AID supported.

This parameter is specific to each implementation. It may be structured as defined in the following table.

Byte	Bit	AID Contactless Pre-Processing Indicators
1	8	Status Check Requested Indicator 1= Status Check requested
	7	Zero Amount Indicator 1= Zero amount transaction
	6	CVM Required Limit Exceeded Indicator 1= CVM Required Limit exceeded
	5	Reader Contactless Floor Limit Exceeded Indicator 1= Reader Contactless Floor Limit Exceeded
	4	Contactless Application Not Allowed 1 = the application AID cannot be selected for contactless transaction
	3-1	RFU (0)

Table 34: AID Contactless Pre-Processing Indicators

11.2.2 Application Candidate List

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

The 'Application Candidate List' is a contactless reader internal static parameter containing list of the contactless card payment applications mutually supported by the contactless device and the contactless reader.

The information to memorize for each application, are:

- ADF Name (AID)
- Application Priority Indicator(if present)
- Kernel ID (if present)

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Note: NULL is used in this document to refer to the value of an empty Candidate List (specific to each Terminal Software Provider).

11.2.3 Authorization Response Code

Tag : '8A'

Length : 2 byte

Format : an

Description:

This EMV-defined data element defines the disposition of an online authorization request. The PURE contactless kernel supports only the following values:

- "00" = Transaction approved
- "Z3"= Unable to go online – decline"Y3"= Unable to go online - approved
- other value = Transaction Declined

Prior to communicating the information to the contactless reader, the Terminal Application needs to convert the authorization code value communicated by the Acquirer to one of the values supported by the PURE contactless kernel.

11.2.4 Additional Tag Object List (ATOL)

Tag : -

Length : Variable (implementation must support at least 30 bytes length)

Format : Binary

Description:

In case of offline approval or online authorization request, the PURE contactless kernel must provide a minimum list of data element to the Terminal Application (see Table 31) in the final outcome message.

The PURE proprietary data element ATOL allows an Acquirer or a Payment Scheme to add additional data elements to the list of mandatory data elements.

ATOL value is the concatenation of the tag values[46] associated to each additional mandatory object.

Note: When the value is set to '00', the list is empty.

[46] Any tag value compliant with the BER TLV rules defined in the EMV specification, may be requested in ATOL. The management of tag value not compliant with BER TLV rules is specific to each implementation. It is recommended to ignore the non-compliant tag value but depending on the wrong value used, the ATOL may be considered as empty if, for example, it is impossible to extract the other tag values present in the list.

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11.2.5 Cardholder Verification Method code in CVM List (PURE specific)

Tag : -

Length : -

Format : -

Description:

The PURE contactless kernel is supporting a new cardholder verification method not defined in the EMV contact specification: Confirmation Code checked using the Customer contactless device.

b8	b7	b6	b5	b4	b3	b2	b1	Cardholder Verification Rule
	0							Fail cardholder verification if this CVM is unsuccessful
	1							Apply succeeding CV Rule if this CVM is unsuccessful
0	-	1	0	0	1	0	0	Confirmation Code checked using the Customer contactless device

Table 35 – CVR for Confirmation Code checked using the Customer contactless device

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11.2.6 Cryptogram Information Data (CID)

Tag : ‘'9F27'

Length : 1 byte

Format : binary

Description:

CID is part of the parameters available in the GENERATE AC command response. It provides information related to card decision.

In case of offline CAM failure after the GENERATE AC command processing, the PURE contactless kernel forces CID value to ’00 – Transaction declined’.

Note: During troubleshooting, for identifying a transaction declined due to Offline CAM failure, the Issuer may check if the CID value is consistent with the most significant nibble of CVR value. CVR is part of Issuer Application Data. The most significant nibble of CVR provides information related to GENERATE AC command decision. This check can be performed only by the Issuer. A Terminal Application ignores the structure of the Issuer Application data contents.

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Card decision related to transaction completion** 00 : Transaction declined 01 : Transaction approved offline 10 : Request to process the transaction online 11 : RFU
		x	x	x	x	x	x	**RFU**

Table 36 - Cryptogram Information Data - CID

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11.2.7 Contactless Cryptogram Information Data (CCID)

Tag : 'C5'

Length : 1 byte

Format : binary

Description:

Contactless Cryptogram Information Data is a PURE proprietary contactless parameter that indicates the type of cryptogram generated by the card and the actions to be performed decided by the card.

In contactless mode, Contactless Cryptogram Information Data may be present in both the GET PROCESSING OPTIONS command and in the GENERATE AC response.

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Preliminary Card decision related to transaction completion** 00 : Transaction declined 01 : Transaction approved offline 10 : Request to process the transaction online 11 : RFU
-	-	x	x	x	-	-	-	**Transaction processing information** 1xx = Switch to contact interface RFU(0) RFU (0)
-	-	-	-	-	x	x	x	**CVM information** 000 = No CVM Required 010 = Online PIN required 001 = Signature required 100 = Confirmation Code required 101 = Confirmation Code verified 011 = Cardholder verification required (it is up to the terminal to decide the CVM to use)

Table 37 - GET PROCESSING OPTIONS Contactless Cryptogram Information Data

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b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Final Card decision related to transaction completion** (same value as in CID bits 8-7) 00 : Transaction declined 01 : Transaction approved offline 10 : Request to process the transaction online 11 : RFU
-	-	x	x	x	-	-	-	**Transaction processing information** 1xx= Switch to contact interface x1x = Error when processing 1st GENERATE AC (inconsistency between GPO command and 1st GENERATE AC command) xx1 = Additional tap for Issuer Response supported
-	-	-	-	-	x	x	x	**CVM information** 000 = No CVM Required 010 = Online PIN required 001 = Signature required 100 = Confirmation Code required 101 = Confirmation Code verified 011 = Cardholder verification required (it is up to the terminal to decide the CVM to use)

Table 38 - GENERATE AC Contactless Cryptogram Information Data

11.2.8CRM Currency Code

Tag : 'CD'

Length : 2 bytes

Format : n4

Description:

This PURE proprietary data element specifies the currency associated to the application offline balance value

11.2.9COTA Offline Balance

Tag : '9F50'

Length : 6 bytes

Format : Binary

Description:

This PURE proprietary data element specifies the remaining offline payment capacity of the selected application.

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11.2.10 Contactless Application/Kernel Capabilities

Tag : -

Length : 5 bytes

Format : Binary

Description:

This PURE proprietary terminal data element provides list of the options supported and the terminal capabilities for each payment application (AID) supported by the terminal.

Both the PURE contactless reader and the PURE contactless kernel are using this parameter. It may be either shared by both POS modules or duplicated in each module.

This parameter has two distinct names:

- Contactless Application Capabilities is the name of the parameter when it is used by the contactless reader
- Contactless Kernel Capabilities is the name of the parameter when it is used by the PURE contactless kernel

Byte	Bit	Contactless Application/Kernel Capabilities	Possible value	Abbreviation[47]
1	8-7	RFU (00)	00	-
	6	EMV contactless Mode support: 1 = EMV contactless mode supported 0 = EMV contactless mode not supported	1	AC_Mode
	5	EMV contact transaction support : 1 = EMV contact transaction supported 0 = EMV contact transaction not supported	0 / 1	AC_Contact
	4	Terminal capability to process a contactless transaction online 1 = Offline-only reader 0 = Online capable reader	0 / 1	AC_Offline
	3	Online PIN support 1= Online-PIN supported 0= Online PIN not supported or cardholder verification method selection using CVM List method	0 / 1	AC_OnlinePIN
	2	Signature support 1= signature supported 0= signature not supported or cardholder verification method selection using CVM List method	0 / 1	AC_Signature
	1	RFU (0)	0	-
2		RFU	'00'	
3	8	When implementation option 7 (**IO_Option7**) is not supported: RFU	0	AC_Online_Tap
		When implementation option 7 (**IO_Option7**) is supported: Support of additional tap for Issuer Response communication 1= terminal supports this option 0= terminal does not support this option	0 / 1	
	7	Terminal capability: Support of Consumer Device CVM checking 1= terminal supports the Confirmation Code as a possible CVM 0= terminal does not consider Confirmation Code as a possible CVM	0 / 1	AC_CC_Tap

[47] Abbreviation used in the document when referring to the option associated to the bit

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Byte	Bit	Contactless Application/Kernel Capabilities	Possible value	Abbreviation[47]
	6	When implementation option 8 (**IO_Option8**) is not supported: RFU	0	AC_ECHO
		When implementation option 8 (**IO_Option8**) is supported: Support of torn transaction recovery using ECHO command 1= terminal supports this option 0= terminal does not support this option	0 / 1	
	5	When implementation option 6 (**IO_Option6**) is not supported: RFU	0	AC_Long_Tap
		When implementation option 6 (**IO_Option6**) is supported: Support of long tap for Issuer Response communication 1= terminal supports this option 0= terminal does not support this option	0 / 1	
	4-1	Bit values ignored by the PURE dual-interface application	00000	
4	8	RFU	0	
	7	When implementation option 4 (**IO_Option4**) is not supported: RFU	0	AC_Auth_Appli
		When implementation option 4 (**IO_Option4**) is supported: Application Authentication Transaction support 1= Application Authentication Transaction support is selected by the Acquirer 0= Application Authentication Transaction support is not selected by the Acquirer	0 / 1	
	6-5	RFU	00	
	4	Retrieval of card Balance and CRM Currency Code supported	0 / 1	AC_Balance
	3	RFU	0	
	2	Stop transaction at GET PROCESSING OPTIONS command level if CCID indicates that the transaction is declined	0 / 1	AC_Decline
	1	Offline CAM checking requirements 1= Offline CAM required in all circumstances (transaction approved, declined or processed online) 0= Offline CAM required only when terminal decision is to process the transaction offline	0 / 1	AC_OfflineCAM
5		**Terminal Interchange Profile (TIP)**		
	8	Reliability of TIP byte value in TTPI (always 1 “Reliable information”)	1	AC_Reliab
	7	SDA support 1= SDA is supported 0= SDA is not supported	0 / 1	AC_SDA
	6	DDA support 1= DDA is supported 0= DDA is not supported	0 / 1	AC_DDA
	5	Cardholder verification method supported by the terminal 1= Terminal supports cardholder verification using EMV-defined method based on CVM List 0= Terminal does not support cardholder verification using EMV-defined method based on CVM List. The terminal lets the card to decide the cardholder verification method to use	0 / 1	AC_CVM
	4	Support of the EMV contact defined Processing Restriction checks and Terminal Risk Management checks 1= Terminal supports terminal checks (Processing restriction and terminal risk management) in complement of the PURE contactless reader Pre-processing function 0= terminal supports only checks defined in the PURE contactless reader Pre-processing function	0 / 1	AC_TRM
	3-2	RFU (00)	00	-

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Byte	Bit	Contactless Application/Kernel Capabilities	Possible value	Abbreviation[47]
	1	CDA support 1= CDA is supported 0= CDA is not supported	0 / 1	AC_CDA

Table 39: Contactless Application/Kernel Capabilities setting

11.2.11 Contactless POS Implementation Options

Tag : -

Length : 1 byte

Format : Binary

Description:

This PURE proprietary data element provides list of the implementation options supported by the POS. This information must be available to the PURE contactless reader module and to the PURE contactless kernel.

Byte	Bit	Contactless POS Implementation Options	Possible value	Abbreviation [48]
1		**Implementation Options**		
	8	Implementation option 1: Retrieval of data element values stored in EEPROM using GET DATA command 1 = Implementation option 1 is supported and is selected by the Acquirer 0= Implementation option 1 is either not supported or not selected by the Acquirer	0 / 1	IO_Option1
	7	Implementation option 2: Update of data element values stored in EEPROM data slots using PUT DATA command (no secure messaging) 1 = Implementation option 2 is supported and is selected by the Acquirer 0= Implementation option 2 is either not supported or not selected by the Acquirer	0 / 1	IO_Option2
	6	Implementation option 3: terminal supporting only fixed Amount transaction 1 = Implementation option 3 is supported and is selected by the Acquirer 0= Implementation option 3 is either not supported or not selected by the Acquirer	0 / 1	IO_Option3
	5	Implementation option 4: Application Authentication Transaction support 1= Implementation option 4 is supported and is selected by the Acquirer 0= Implementation option 4 is either not supported or not selected by the Acquirer	0/1	IO_Option4
	4	Implementation option 5: Capacity to restrict list of supported applications 1= Implementation option 5 is supported and is selected by the Acquirer 0= Implementation option 5 is either not supported or not selected by the Acquirer	0/1	IO_Option5
	3	Implementation option 6: Long Tap support 1= Implementation option 6 is supported and is selected by the Acquirer 0= Implementation option 6 is either not supported or not selected by the Acquirer	0/1	IO_Option6
	2	Implementation option 7: Online Additional Tap support 1= Implementation option 7 is supported and is selected by the Acquirer 0= Implementation option 7 is either not supported or not selected by the Acquirer	0/1	IO_Option7

[48] Abbreviation used in the document when referring to the option associated to the bit

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Byte	Bit	Contactless POS Implementation Options	Possible value	Abbreviation [48]
	1	Implementation option 8: ECHO command support 1= Implementation option 8 is supported and is selected by the Acquirer 0= Implementation option 8 is either not supported or not selected by the Acquirer	0/1	IO_Option8

Table 40: Contactless POS Implementation Options

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11.2.12 Current Contactless Pre-Processing Indicators

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

Current Contactless Pre-Processing Indicators is a transaction-transient parameter containing the value of the 'AID Contactless Pre-Processing Indicators' associated to the AID that the contactless reader has selected.

11.2.13 Currently Selected Application

Tag : -

Length : Variable (up to 16 bytes)

Format : Binary

Description:

The Currently Selected Application is a contactless reader internal static parameter containing the application AID of the last contactless card payment application selected.

Note: NULL is used in this document to refer to no application currently selected.

11.2.14 Data Elements Update Result

Tag : '9F74'

Length : 2

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

The PURE kernel uses this PURE proprietary terminal data element for returning the result of the data element update requested by the reader using Memory Slot Update Template.

The structure is:

Byte	Description
1	Result '00' = No data element updated '01' = Failure when updating a data element '02' = Successful update of all data elements
2	N = Number of data elements processed Note: • When the result is '02': Number of successful data elements updated = N • When the result is '01': Number of successful data elements updated = N - 1

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11.2.15 Echo Card Identifier

Tag : '9F75'

Length : Variable (up to 20 bytes)

Format : Binary

Description:

This PURE proprietary data element may be present in the response of the selection of a Card Payment Application (inside FCI Issuer Discretionary Data – 'BF0C'). Its presence indicates to the terminal that the card payment application is supporting the ECHO command allowing the retrieval of the response of the last GENERATE AC command processed by the card payment application.

11.2.16 GDDOL

Tag : '9F70'

Length : Variable (implementation must support at least 50 bytes length)

Format : Binary

Note: Required only when implementation option 1 is supported

Description:

This PURE proprietary terminal data element provides to the PURE kernel, the list of data elements to retrieve using the GET DATA command.
It is an EMV-defined DOL object (concatenation of tag[49]/Length values).

11.2.17 GDDOL Resulting Buffer

Tag : '9F71'

Length : Variable (implementation must support at least 512 bytes length)

Format : Binary

Note: Required only when implementation option 1 is supported

Description:

GDDOL Resulting Buffer is a PURE proprietary terminal data element. Its content is the result of the concatenation of the data elements listed in GDDOL. This buffer is TLV structured (Tag – length – value).

[49] Any tag value compliant with the BER TLV rules defined in the EMV specification, may be requested in GDDOL. The management of tag value not compliant with BER TLV rules is specific to each implementation. It is recommended to ignore the non-compliant tag value but depending on the wrong value used, the GDDOL may be considered as empty if, for example, it is impossible to extract the other tag values present in the list.

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Example:
When GDDOL has the following value: **DF01**05**C9**01**DF03**0A**CD**02
And the information retrieved using GET DATA during step are as defined below
GET DATA (DF01) = '0102030405060708'
GET DATA (C9) = error
GET DATA (DF03) = error
GET DATA (CD) = '0978'
The value of GDDOL Resulting Buffer is: '**DF01**08010203040506070**C9**00**DF03**00**CD**020978'

11.2.18 Issuer Script Results

Tag : '9F73'

Length : Variable (up to 10 bytes)

Format : Binary

Description:

This PURE proprietary kernel data element has exactly same structure as the equivalent EMV-defined parameter. EMV has allocated no tag value to Issuer Script Results.

11.2.19 Kernel Internal Indicators

This section describes the kernel internal indicators used in this document. Their usage is specific to each implementation.

11.2.19.1 Communication Problem Indicator

Tag : '-'

Length : 1 byte

Format : binary

Description:

Contactless PURE Kernel internal indicator specifying detection of a communication problem with the contactless device during the last GENERATE AC command processing.

The possible values are:

- '00' : No communication problem
- '01' : communication problem during 1st GENERATE AC
- '02' : communication problem during 2nd GENERATE AC
- '03': card in communication with the contactless reader after an online authorization (2nd tap) is distinct from the card used at the beginning of the transaction or the card in communication with the contactless reader after a torn transaction is not the same card.

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11.2.19.2 ECHO command supported Indicator

Tag : '-'

Length : 1 byte

Format : binary

Description:

Contactless PURE Kernel internal indicator specifying the support of the ECHO command for the current transaction or the previous transaction.

The possible values are:

- '00': Either the terminal or the contactless device are not supporting the ECHO command
- '01' = Both the terminal and the contactless device are supporting the ECHO command

11.2.19.3 ECHO Transaction Indicator

Tag : '-'

Length : 1 byte

Format : binary

Description:

Contactless PURE Kernel internal indicator specifying if the current transaction is the transaction is for the recovery of a previous torn transaction.

The possible values are:

- '00': 'normal' transaction
- '01' = transaction for recovery of a previous torn transaction (ECHO command is used)

11.2.19.4 Offline CAM Selected Indicator

Tag : '-'

Length : 1 byte

Format : binary

Description:

Contactless PURE Kernel internal indicator specifying the offline CAM selected for the current transaction or previous transaction.

The possible values are:

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- '00': None
- '01': SDA selected
- '02': CDA selected
- '03': DDA selected (only possible with Application Authentication Transaction)

11.2.20 Last Kernel Outcome Message

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

The 'Last Kernel Outcome Message' is a contactless reader internal static parameter containing the last outcome message that the contactless reader has received from the PURE contactless kernel.

Note: NULL is used in this document to refer to the value of 'Last Kernel Outcome Message' when no information is memorized (specific to each Terminal Software Provider).

11.2.21 Mandatory Tag Object List (MTOL)

Tag : -

Length : Variable (implementation must support at least 50 bytes length)

Format : Binary

Description:

The PURE contactless kernel mandates the presence of the following data elements in the kernel database at completion of "read Application data" function.

- '8C' – Card Risk Management Data Object List 1 (CDOL1)
- '57' – Track2 Equivalent Data

The PURE proprietary data element MTOL allows an Acquirer or a Payment Scheme to add additional data elements to the list of mandatory data elements.

MTOL value is the concatenation of the tag values[50] associated to each additional mandatory object.

Note: When the value is set to '00', the list is empty.

[50] Any tag value compliant with the BER TLV rules defined in the EMV specification, may be requested in MTOL. The management of non-compliant tag value with BER TLV rules is specific to each implementation. It is recommended to ignore the non-compliant tag value but depending on the wrong value used, the MTOL may be considered as empty if, for example, it is impossible to extract the other tag values present in the list.

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11.2.22 Memory Slot Identifier

Tag : '9F72'

Length : Variable (up to 16 bytes)

Format : Binary

Note: Required only when implementation option 1 or implementation 2 are supported

Description:

This PURE proprietary data element containing the identifier value associated to a memory slot.

11.2.23 Memory Slot Read Template

Tag : 'BF71'

Length : Variable (implementation must support at least 80 bytes length)

Format : Binary

Note: Required only when implementation option 1 is supported

Description:

This PURE proprietary terminal data element is a constructed data object composed of one or two entries. This data element provides to the PURE kernel, the list of data elements to read. In the current version, only one type of data element may be requested:

- Data elements that may be read using the GET DATA command. The list of data elements is provided using GDDOL (tag 9F70) parameter.

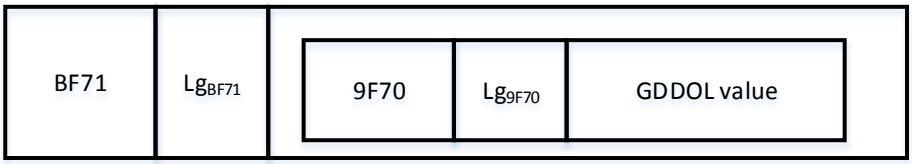


Figure 6 Memory Slot Read Template Structure

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11.2.24 Memory Slot Update Template

Tag : ‘BF70’

Length : Variable (implementation must support at least 800 bytes length. The accurate maximum length if depending on length of the other Dynamic Transaction Parameters)

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

This PURE proprietary terminal data element is a constructed data object composed of one or several entries. This data element provides to the PURE kernel, the list of data elements to update using the PUT DATA (without secure messaging) command and the impact on transaction processing in case of update failure. Each entry is using either tag ‘A2’, tag ‘A3’or tag ‘A4’.

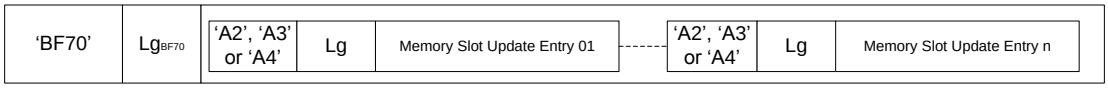


Figure 7 Memory Slot Update Template Structure

11.2.25 Memory Slot Update Entry type 1

Tag : ‘A2’

Length : Variable (implementation must support at least 260 bytes length)

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

Each entry of the Memory Slot Update Template provides to the PURE kernel, the information related to a single data element to update using the PUT DATA(without secure messaging) command and the impact on transaction processing in case of update failure. These update must be performed prior processing the GET PROCESSING OPTIONS command.

Note: The data element to update may be either a primitive data element or a constructed data element. In both cases, the kernel has to use the PUT DATA command with the tag value and the data element value communicated.

Each entry must contain

- Data element to update: tag[51] – length – value
- Memory Slot Update Entry Setting: tag ‘85’ – length = 01 – value

[51] Any tag value compliant with the BER TLV rules defined in the EMV specification, may be used. When the tag value is not compliant with BER TLV rules, the Memory Slot Update Entry is ignored.

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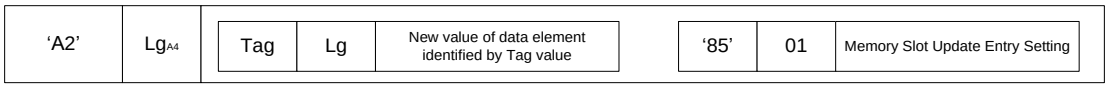


Figure 8 Memory Slot Update Entry type 1 Structure

11.2.26 Memory Slot Update Entry type 2

Tag : 'A3'

Length : Variable (implementation must support at least 260 bytes length)

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

Each entry of the Memory Slot Update Template provides to the PURE kernel, the information related to a single data element to update using the PUT DATA(without secure messaging) command and the impact on transaction processing in case of update failure. These update must be performed after processing successfully the GET PROCESSING OPTIONS command.

Note: The data element to update may be either a primitive data element or a constructed data element. In both cases, the kernel has to use the PUT DATA command with the tag value and the data element value communicated.

Each entry must contain

- Data element to update: tag[51] – length – value
- Memory Slot Update Entry Setting: tag '85' – length = 01 – value

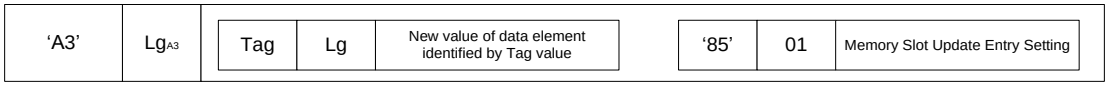


Figure 9 Memory Slot Update Entry type 2 Structure

11.2.27 Memory Slot Update Entry type 3

Tag : 'A4'

Length : Variable (implementation must support at least 260 bytes length)

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

Each entry of the Memory Slot Update Template provides to the PURE kernel, the information related to a single data element to update using the PUT DATA(without secure messaging) command and the impact on transaction processing in case of update failure. These update must be performed after processing 1st GENERATE AC command.

This type of Memory Slot Update Template entry is ignored when the transaction type is '79'.

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Note: The data element to update may be either a primitive data element or a constructed data element. In both cases, the kernel has to use the PUT DATA command with the tag value and the data element value communicated.

Each entry must contain

- Data element to update: tag[51] – length – value
- Memory Slot Update Entry Setting: tag '85' – length = 01 – value

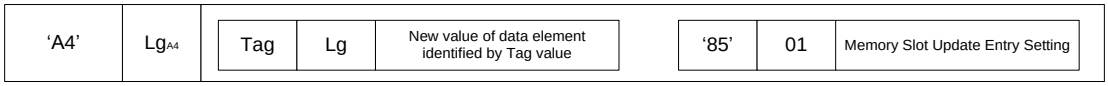


Figure 10 Memory Slot Update Entry type 3 Structure

11.2.28 Memory Slot Update Entry Setting

Tag : '85'

Length : 1

Format : Binary

Note: Required only when implementation option 2 is supported

Description:

This PURE proprietary data element must be present in each entry of the Memory Slot Update Template. It provides to the PURE kernel, information related to the processing of the PUT DATA command (condition required for processing the command, impact in case of failure).

Bit	Impact
8	**Additional conditions required for sending the PUT DATA command depending on card decision communicated in GENERATE AC command response** 0- Data element is updated using PUT DATA when card decision is ARQC 1- Data element is not updated using PUT DATA when card decision is ARQC
7	0- Data element is updated using PUT DATA when card decision is TC 1- Data element is not updated using PUT DATA when card decision is TC
6	0- Data element is updated using PUT DATA when card decision is AAC 1- Data element is not updated using PUT DATA when card decision is AAC
5-2	RFU (0000)
1	**Impact in case of PUT DATA command error** 0= Stop to process any additional Memory Slot Update Template entry but continue the transaction

Table 41 – Memory Slot Update Entry Setting (Memory Slot Update Template type 3)

Bit	Impact
8 - 6	**Additional conditions required for sending the PUT DATA command** Value should be '000' (not checked by the kernel)
5-2	RFU (0000)
1	**Impact in case of PUT DATA command error** 1= Stop the transaction

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	0= Stop to process any additional Memory Slot Update Template entry but continue the transaction

Table 42 – Memory Slot Update Entry Setting (Memory Slot Update Template type 1 and type 2)

11.2.29 Authentication Transaction Data Tag Object List (ATDTOL)

Tag : -

Length : Variable (implementation must support at least 40 bytes length)

Format : Binary

Description:

The PURE contactless kernel must provide a minimum list of data element to the Terminal Application (see Table 32) in the final outcome message of an Application Authentication Transaction.

The PURE proprietary data element ATDTOL allows an Acquirer or a Payment Scheme to add additional data elements to the list of mandatory data elements.

ATDTOL value is the concatenation of the tag values[52] associated to each data elements to communicate to the Terminal Application.

Note: When the value is set to '00', the list is empty.

11.2.30 Proprietary Data Elements (PDE)

Tag : 'DF01' – 'DF0A'

Length : Variable (implementation must support at least 250 bytes length)

Format : Binary

Description:

The Issuer has the option to support a function allowing the storage of proprietary data elements in the EEPROM of the application. This option allows the support of 10 PURE Application Proprietary Data Elements (PDE).

The proprietary data elements ('DF01' – 'DF0A') are managed using:

- GET DATA command for retrieval of the value and
- PUT DATA for updating the value.

Size and access conditions associated to each Proprietary Data Element are defined during the card personalization. The possible access conditions that may be used are:

[52] Any tag value compliant with the BER TLV rules defined in the EMV specification, may be requested in ATDTOL. The management of tag value not compliant with BER TLV rules is specific to each implementation. It is recommended to ignore the non-compliant tag value but depending on the wrong value used, the ATDTOL may be considered as empty if, for example, it is impossible to extract the other tag values present in the list.

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- Reading access condition using GET DATA command
  - Free access
  - Prior offline PIN presentation required
- Updating access condition using PUT DATA command
  - Free access
  - Secure messaging (integrity) required
  - Prior offline PIN presentation required
  - Prior offline PIN presentation required and Secure messaging (integrity) required

The access condition using offline PIN presentation, does not allow an access in contactless mode.

11.2.31 POS Timeout (Long tap)

Tag : -

Length : 2

Format : n4

Description:

In case of the "Financial Transaction Completed - Online Request (Long Tap)" outcome message, this parameter corresponds to the time after which cardholder is asked to remove the card.

Value is given in units of 100ms.

11.2.32 POS Timeout (Transaction Result Display)

Tag : -

Length : 2

Format : n4

Description:

This terminal parameter specifies during how much time the result of transaction completion needs to be displayed to the cardholder.

Value is given in units of 100ms.

11.2.33 Proximity Payment System Environment - PPSE

Tag : -

Length : -

Format : -

Description:

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PPSE is an application present in a contactless device. A contactless terminal uses this application for the retrieval of the contactless applications available in the contactless device.

The AID associated to the PPSE is: '2PAY.SYS.DDF01'.

11.2.34 PPSE Selection Response

Tag : -

Length : -

Format : -

Description:

The response of the selection of the PPSE application is as defined in the following table:

Data Element					Presence
'6F'	FCI template				M
	'84'	DF Name ('2PAY.SYS.DDF01')			M
	'A5'	FCI Proprietary Template			M
		'BF0C'	FCI Issuer Discretionary Data		M
			'61'	Directory Entry 1	M
				'4F' ADF Name	M
				'50' Application Label	O
				'87' Application Priority Indicator	C[53]
				'9F2A' Kernel Identifier	C[54]
				Other [55]tag: The PURE contactless reader supports additional data elements that may be either primitive data element or constructed data elements containing additional data	O
			…		
			'61'	Directory Entry n	C[56]
				'4F' ADF Name	M[56]
				'50' Application Label	O[56]
				'87' Application Priority Indicator	C[56]
				'9F2A' Kernel Identifier	C[56]

[53] Should be present when contactless device supports several applications in contactless mode

[54] If the application requires a specific kernel to be used

[55] Management of Extended Selection parameter (tag '9F29') is not supported in current version

[56] Only relevant if the directory entry is present

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	Other tag	The PURE contactless reader supports additional data elements that may be either primitive data element or constructed data elements containing additional data	O[56]

11.2.35 Reader Internal Indicators

This section describes the reader internal indicators used in this document. Their usage is specific to each implementation.

11.2.35.1 Contactless Reader Timeout Indicator

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

Contactless Reader Timeout Indicator is a contactless reader internal static parameter containing the status of timeout function previously launched by the contactless reader when processing a kernel outcome message with "***Removal Timeout***" parameter different from zero. In the current specifications, this case corresponds to only the decision to use the "long" tap. The value '1' indicates that the timeout limit was reached before the reception of the Issuer authorization host response.

11.2.35.2 Current Process Indicator

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

The Current Process Indicator is a contactless reader internal transaction-transient parameter containing the type of request communicated by the Terminal Application. The possible values are:

- New: new transaction
- Online Response: Terminal Application has communicated the online response.

11.2.35.3 Restart Flag

Tag : -

Length : specific to each implementation

Format : specific to each implementation

Description:

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Restart Flag is a contactless reader internal static parameter indicating whether the kernel is being started for a new transaction or in order to complete online processing or a transaction restart requested in previous outcome message.

The values to memorize are:

- 'No': New transaction
- 'Yes': A restart was requested for either restarting the transaction or for communication of the Issuer Host response.

11.2.36 Terminal AID value supported

Tag : 'DF7F'

Length : Variable (variable: from one byte up to 16 bytes)

Format : Binary

Description:

This optional PURE proprietary terminal data element allows the terminal application to communicate to the PURE Contactless Reader the list of AID that may be used for a single transaction.

11.2.37 Terminal Transaction Data

Tag : '9F76'

Length : Variable (implementation must support at least 250 bytes length. The accurate maximum length if depending on length of the other Dynamic Transaction Parameters)

Format : Binary

Description:

This PURE proprietary terminal data element allows the Terminal Application to communicate transaction-related information to the Card Payment application. Depending on card payment application setting, this information may participate in the card payment application risk management or may be inserted inside the card payment application transaction log.

Its usage and contents have to be defined by each Payment Scheme.

The PURE reader and the PURE kernel must be capable to support it among the Dynamic Transaction Parameters sent by the Terminal Application when it is used by the terminal application.

11.2.38 Terminal Dedicated Data

Tag : '9F77'

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Length : Variable (implementation must support at least 250 bytes length[57])

Format : Binary

Description:

This PURE proprietary card payment application data element allows the card to communicate card payment application-related information to the Terminal Application.

Its usage and contents have to be defined by each Payment Scheme.

Terminal Dedicated Data may be present either in the records referenced by the AFL value, in the response of Application Selection or in the response of GET PROCESSING OPTIONS command.

When Terminal Dedicated Data is present in the information read in the card payment application, the PURE reader and the PURE kernel must be capable to communicate its value to the terminal application using parameter "Data Record Present" field of the outcome message.

[57] This maximum length is possible only when Terminal Dedicated Data is stored in a readable record.

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11.2.39 Terminal Transaction Processing Information (TTPI)

Tag : 'C7'

Length : 5 bytes

Format : Binary

Description:

The Terminal Transaction Processing Information (TTPI) is a proprietary PURE contactless terminal data element.

TTPI may be part of the mandatory data element values requested to the terminal using PDOL.

Byte	Bit	TTPI bit meaning and category
1		**Terminal Capabilities**
	8-7	Bit values ignored by the PURE dual-interface application
	6	Terminal capability: Capability of the terminal to perform a contactless chip transaction 1 = EMV contactless mode supported 0 = EMV contactless mode not supported
	5	Terminal capability: Capability of the terminal to perform a contact chip transaction 1 = EMV contact transaction supported 0 = EMV contact transaction not supported
	4	Terminal capability: Capability of the terminal to process a transaction online 1 = Offline-only reader 0 = Online capable reader
	3	Terminal capability: Support of online-PIN 1= Online-PIN supported 0= Online PIN not supported or cardholder verification method selection using CVM List method
	2	Terminal capability: Support of signature 1= signature supported 0= signature not supported or cardholder verification method selection using CVM List method
	1	RFU (0) : Bit value ignored by the PURE dual-interface application
2		**Transaction processing request**
	8	Initial terminal request related to transaction completion 1 = Online authorization requested 0 = Online authorization not required
	7	Terminal request related to cardholder verification 1 = CVM required 0 = no CVM required or card not involved in the decision to perform cardholder verification

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Byte	Bit	TTPI bit meaning and category
	6	Single unit of currency (Status Check) Check result 1 = Status Check requested 0 = Status Check not requested or check not supported by the reader
	5	Zero Amount Check result 1 = Zero Amount Transaction 0 = Non-zero Amount Transaction or check not supported by the reader
	4	RFU (0)
	3	Terminal requirement related to offline CAM 0 = no specific requirement 1 = Offline CAM requested independently the completion of the transaction
	2-1	Bit values ignored by the PURE dual-interface application
3		**Terminal Options**
	8	Support of additional tap for Issuer Response communication 1= terminal supports this option 0= terminal does not support this option
	7	Terminal capability: Support of Consumer Device CVM checking as a possible CVM method 1= terminal supports the Confirmation Code as a possible CVM 0= terminal does not consider Confirmation Code as a possible CVM
	6	Support of torn transaction recovery using ECHO command 1= terminal supports this option 0= terminal does not support this option
	5-1	Bit values ignored by the PURE dual-interface application
4	8-1	Bit values ignored by the PURE dual-interface application
5		**Terminal Interchange Profile**
	8	Reliability of Terminal Interchange Profile byte value 0 = TIP information may not be reliable[58] 1 = TIP information reliable
	7	Terminal Capabilities: SDA support 1= Terminal supports SDA 0= terminal does not support SDA
	6	Terminal Capabilities: DDA support 1= Terminal supports DDA 0= terminal does not support DDA
	5	Terminal Capabilities: EMV-defined Cardholder verification method supported by the terminal 1= Terminal supports cardholder verification using EMV-defined method based on CVM List 0= Terminal lets the card to decide the cardholder verification method to use

[58] TIP byte is not reliable when its value is '00'.

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Byte	Bit	TTPI bit meaning and category
	4	Terminal Capabilities: Support of complementary terminal checks and terminal action analysis 1= Terminal supports terminal checks and terminal action analysis, in complement of the PURE contactless reader Pre-processing function 0= Terminal supports only checks defined in the PURE contactless reader Pre-processing function
	3-2	RFU (00): Bit values ignored by the PURE dual-interface application
	1	Terminal Capabilities: CDA support 1= Terminal supports CDA 0= Terminal does not support CDA

Table 43 – Terminal Transaction Processing Information

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11.2.40 Transaction Type for Application Authentication Transaction

Tag : -

Length : 1 byte

Format : 2n

Description:

This parameter defines the transaction type value associated to Application Authentication Transaction. Acquirers and Issuers shall use the same value. The value should be defined by the Payment Scheme.

The value shall be different from the transaction type values used by the PURE contactless kernel defined in the following table.

Type of transaction	Transaction Type Codification
Purchase transaction	'00'
Cash transaction (cash withdrawn or manual cash)	'01'
Purchase with cashback	'09'
Refund transaction	'20'
Retrieve application data elements using GET DATA command	'78' (codification specific to PURE)
Update application data elements using PUT DATA command	'79' (codification specific to PURE)

Table 44 –Types of transactions in PURE contactless reader

Example: The value '77' may be a possible value.

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12 PURE Application Commands

12.1 List of Application Commands

The PURE Application supports the following set of commands in contactless mode.

Command Name	CLA	INS	P1	P2	Standard Usage	Exceptional Usage
SELECT	'00'	'A4'	'04'	var	Yes	-
GET PROCESSING OPTIONS	'80'	'A8'	'00'	'00'	Yes	-
READ RECORD	'00'	'B2'	var	var	Yes	-
GET DATA	'80'	'CA'	var	var	-	Yes
GENERATE APPLICATION CRYPTOGRAM	'80'	'AE'	var	'00'	Yes	-
APPLICATION BLOCK	'8C'	'1E'	'00'	'00'	-	Yes
APPLICATION UNBLOCK	'8C'	'18'	'00'	'00'	-	Yes
CARD BLOCK	'8C'	'16'	'00'	'00'	-	Yes
PUT DATA (with Secure Messaging)	'0C'	'DA'	var	var	-	Yes
PUT DATA (without Secure Messaging)	'00'	'DA'	var	var	-	Yes
UPDATE RECORD	'0C'	'DC'	var	var	-	Yes
PIN CHANGE/UNBLOCK	'8C'	'24'	'00'	var	-	Yes
ECHO	'80'	var	'00'	'08'	-	Yes

Table 45 – PURE set commands available in contactless mode

The PURE contactless kernel has to support only the following set of commands:

- GET PROCESSING OPTIONS
- READ RECORD
- 1st GENERATE APPLICATION CRYPTOGRAM
- 2nd GENERATE APPLICATION CRYPTOGRAM
- GET DATA for terminal velocity checking

The PURE contactless reader software sends the SELECT command. The kernel gets only the application selection response.

When the PURE contactless kernel supports the implementation options related to EEPROM memory Management, the following additional commands are supported:

- GET DATA for any parameter
- PUT DATA (without secure messaging)

Other commands may be sent to the card through the kernel, but the kernel will not be in charge of the definition of the command parameters. The role of the kernel will be only to send the commands prepared by the reader and communicate back to the reader, the result of the command processing.

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12.2 Detailed Description of the commands

12.2.1 SELECT

12.2.1.1 Command Description

The SELECT command is used to select the card payment application. The card payment application may be selected using either its main Application Identifier (AID) value or using one of the alternate AID values associated with the application.

The PURE Application supports selection using partial AID value. When the partial AID is used and the AID value matches more than one application, the application selected depends on the sequence in which the applications are personalized onto the card.

12.2.1.2 Input APDU parameters

The input APDU parameters of the SELECT command are coded as defined in following table.

Code	Value
CLA	'00'
INS	'A4'
P1	'04'
P2	'00' – First or only occurrence '02' – Next occurrence
Lc	Var. (between '05' and '10')
Data	File name (Full AID value or Partial AID value)
Le	'00'

Table 46 – SELECT command

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12.2.1.3 Data Field Returned in the Response Message

The data field of the response message of any successful SELECT command contains the FCI template (tag '6F') of the selected application.

Tag				Description	Presence
'6F'				**FCI Template**	Mandatory
	'84'			DF Name	Mandatory
	'A5'			**FCI Proprietary Template**	Mandatory
		'50'		Application Label	Optional
		'87'		Application Priority Indicator	Optional
		'9F38'		PDOL	Optional
		'5F2D'		Language Preference	Optional
		'9F11'		Issuer Code Table Index	Optional
		'9F12'		Application Preferred Name	Optional
		'BF0C'		**FCI Issuer Discretionary Data**	Optional
			'9F75'	Echo Card Identifier	Optional
			'9F4D'	Log Entry	Optional
			'5F54'	Bank Identifier Code (BIC)	Optional
			'5F55'	Issuer Country Code (2 alpha format)	Optional
			'5F56'	Issuer Country Code (3 alpha format)	Optional
			'42'	Industry Identification Number (IIN)	Optional
			'xx xx'	1 or more additional proprietary data elements defined either by a Payment Scheme or by an Issuer or EMV-defined tag that are specifically allocated to 'BF0C'	Optional

Table 47 – SELECT Command Response Message (Application selection)

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12.2.1.4 Processing State returned in the Response Message

SW1 SW2	Value
‘90 00’	Command completed successfully
’62 83’	Application Blocked (warning)
'6A 81’	Card Blocked Note: In some card platform, ‘card blocked’ is managed directly by the card platform
'6A 82'	Application not found
'6A 86'	Incorrect parameters P1-P2 (Value not equal to either ‘0400’ or ‘0402’)

Table 48 - SELECT -Status words and Application Final State

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12.2.2GET PROCESSING OPTIONS

12.2.2.1 Command Description

The GET PROCESSING OPTIONS command is used to inform the PURE Application that the processing of a new transaction is beginning.

The terminal may be requested to communicate transaction-specific information to the ICC (as specified in the PDOL). When PDOL data is requested, a PDOL list with tag 9F38 must be placed in the FCI returned to the terminal in the application selection response. The PDOL list is personalized with the tags of data items that the card expects to receive in GPO command data. The terminal then uses this PDOL list to send transaction-specific information to the ICC.

In response to the GPO command, the ICC communicates to the terminal the Application Interchange Profile (AIP) value and the Application File Locator (AFL) value to use in processing the transaction. Additional information may be communicated to the terminal: Contactless Cryptogram Information Data (CCID) providing card preliminary decision.

12.2.2.2 Input APDU parameters

The input APDU parameters of the GET PROCESSING OPTIONS command are coded as defined in following table.

Code	Value	
CLA	‘80’	
INS	‘A8’	
P1	‘00’	
P2	‘00’	
Lc	○ When no PDOL personalized : ‘02’ ○ When PDOL list found in Application Selection Response: PDOL related data length + 2 or PDOL related data length + 3 (if PDOL related data length> 127)	
Data	Tag	‘83’
	Length	○ When no PDOL personalized : ‘00’ ○ When PDOL present in Application Selection Response: PDOL related data length Note: When PDOL related data length is greater than 127 bytes, the ‘Length’ is on 2 bytes (‘81’ ‘length value’). In the other case, the ‘Length’ is on 1 byte.
	Value	○ When no PDOL personalized : Not Present ○ When PDOL in Application Selection Response: PDOL related data
Le	‘00’	

Table 49 – GET PROCESSING OPTIONS command

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12.2.2.3 Data Field Returned in the Response Message

The data field of the response message is TLV structured as defined below (according to EMV format 2).

Tag		Description	Presence
'77'		Response Message Template Format 2	Mandatory
	'82'	Application Interchange Profile (AIP)	Mandatory
	'94'	Application File Locator (AFL)	Mandatory
	'C5'	Contactless Cryptogram Information Data (CCID)	Optional

Table 50 – GET PROCESSING OPTIONS command Response Message (Contactless)

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Preliminary Card decision related to transaction completion** 00 : Transaction declined 01 : Transaction approved offline 10 : Request to process the transaction online 11 : RFU
-	-	x	x	x	-	-	-	**Transaction processing information** 1xx = Switch to contact interface RFU(0) RFU (0)
-	-	-	-	-	x	x	x	**CVM information** 000 = No CVM Required 010 = Online PIN required 001 = Signature required 100 = Confirmation Code required 101 = Confirmation Code verified 011 = Cardholder verification required (it is up to the terminal to decide the CVM to use)

Table 51 - GET PROCESSING OPTIONS Contactless Cryptogram Information Data

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12.2.2.4 Processing State returned in the Response Message

SW1 SW2	Value
‘90 00’	Command completed successfully
'69 85'	Conditions of use not satisfied Note: In contactless mode, the SW is used when the terminal does not support EMV contactless mode. Contactless Magstripe is not supported in current version of this specification.
	When Application State rules are not satisfied
'6A 86'	Incorrect parameters P1-P2
‘67 00’	Invalid length of data (Lc)
’65 81’	Personalization issue
’69 85’	Format checks on the PDOL related data has failed

Table 52 - GET PROCESSING OPTIONS-Status words and Application Final State

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12.2.3READ RECORD

12.2.3.1 Command Description

The READ RECORD command is used to read the contents of the linear file records present in the PURE application.

It is possible to read only the records referenced in the AFL value communicated in the response of the GET PROCESSING OPTIONS command.

12.2.3.2 Input APDU parameters

The input APDU parameters of the READ RECORD command are coded as defined in following table.

Code	Value
CLA	'00'
INS	'B2'
P1	Record Number
P2	Reference Control Parameter
Lc	Not present
Data	Not present
Le	'00'

Table 53 – READ RECORD command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	x	x	x	-	-	-	SFI in the range 1 – 30
-	-	-	-	-	1	0	0	P1 is a record number

Table 54 – READ RECORD command: Reference Control Parameter

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12.2.3.3 Data Field Returned in the Response Message

The data field of the response message of any successful READ RECORD command contains the record data.

For SFIs in the range 1-10, the record is a BER-TLV constructed data object coded using tag '70'.

'70'	Length	Record Template

Table 55 – READ RECORD Response Message Data Field (SFI in the range 1-10)

For other SFI, the structure of the record template is outside the scope of these specifications.

12.2.3.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
'6C xx'	SW returned only in T=0 protocol: Wrong Le field; SW2 (xx) encodes the exact number of available data bytes
'6A 82'	File not found (SFI value referenced in P2 parameter does not exist)
'6A 83'	Record not found (SFI value referenced in P2 parameter exists but does not contain record referenced in parameter P1)
'6A 86'	Incorrect parameters P1-P2 (either P1='00' or wrong structure of parameter P2 or SFI value defined in parameter P2 not in the range 1-30)
'69 85'	Conditions of use not satisfied (only in contactless mode, when the record to read exists but is not present in AFL)

Table 56 - READ RECORD -Status words and Application Final State

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12.2.4GET DATA

12.2.4.1 Command Description

The GET DATA command is used to read a data object not contained in a Terminal File record personalized within the current application.

12.2.4.2 Input APDU parameters

The input APDU parameters of the GET DATA command are coded as defined in following table.

Code	Value
CLA	'80'
INS	'CA'
P1	Data object tag **If** tag of the data element to retrieve is on 2 bytes **then** • P1 = most significant byte of the tag value • P2 = least significant byte of the tag value **Else** tag of the data element to retrieve is on 1 byte • P1 = '00' • P2 = Tag value
P2	
Lc	Not Present
Data	Not Present
Le	'00'

Table 57 – GET DATA command

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Data elements that may be retrieved using GET DATA are:

P1P2	Data Element	Length
'9F36'	Application Transaction Counter (ATC)	2
'9F13'	Last Online ATC Register	2
'9F17'	PIN Try Counter (PTC)	1
'9F4F'	Log Format	26
'00 C8'	Default Terminal Transaction Processing Information - DTTPI	5
'00 82'	Application Interchange Profile [G][59]	2
'00 D8'	Consolidated Application File Locator - CAFL [G]	Var.
'00 94'	Application File Locator (retrieve the value communicated in GET PROCESSING OPTIONS command response) [G]	Var.
'00 C1'	Application Control	4
'9F 7E'	Issuer Life Cycle Data	48
'DF 11'	Currency Conversion Code 1	5
'DF 12'	Currency Conversion Code 2	5
'DF 20'	Card Issuer Action Code – CVM [G]	3
'DF 15'	Card Issuer Action Code – Denial [G]	3
'DF 16'	Card Issuer Action Code – Default [G]	3
'DF 17'	Card Issuer Action Code – Online [G]	3
'DF 1D'	Issuer Application Data Object List (IADOL)[G]	Var.
'00 CB'	CRM Country Code	2
'00 CD'	CRM Currency Code	2
'DF 0x'	Proprietary Data Element (PDE) with x in the range '1' to 'A'.[60]	Var.
'DF 10'	Profile Selection Table (PST)	Var.
'BF2x'	Profile Resource Object (PRO) associated to profile 'x' (with x in the range '0' to 'F')[61]	Var.
'00 D0'	Current Profile Information (CPI) [G]	1

Table 58 – GET DATA - PURE defined data elements

[59] [G] indicates the data element that can be retrieved using GET DATA only after GET PROCESSING OPTIONS command. Prior to the GPO command, the application must return an error response code of '6985'

[60] Retrieval of the value of a PDE may fail for several reasons:

- PDE usage not allowed: SW '6985'
- The PDE element requested is empty (no value currently defined): SW '6A88'
- Read access condition not satisfied: SW '6982'

[61] GET DATA on a tag related to a PRO value not personalized in the application, fails on SW '6A88'.

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Depending on application personalization setting, additional PURE Application proprietary data elements may be retrieved using GET DATA.

P1P2	Data Element	Length
‘DF 1A’	Maximum Single Transaction Amount (MTAL) limit [P][62]	6
‘DF 14’	Cumulative Offline Transaction Amount Counter (COTA) [P]	6
‘DF 18’	Lower Cumulative Offline Transaction Amount (LCOTA) [P]	6
‘DF 19’	Upper Cumulative Offline Transaction Amount (UCOTA) [P]	6
‘9F 50’	Offline Balance (UCOTA – COTA)	6
‘DF 13’	Consecutive Offline Transactions Counter (COTN) [P]	1
‘DF 1B’	Lower Consecutive Offline Limit (LCOL) [P]	1
‘DF 1C’	Upper Consecutive Offline Limit (UCOL) [P]	1

Table 59 – GET DATA – Conditional PURE defined data elements

12.2.4.3 Data Field Returned in the Response Message

The data field of the response message contains the TLV structured value of the data object identified by P1P2 of the command.

12.2.4.4 Processing State returned in the Response Message

SW1 SW2	Value
‘90 00’	Command completed successfully
‘6C xx’	SW returned only in T=0 protocol: Wrong Le field; SW2 (xx) encodes the exact number of available data bytes
’69 82’	Security status not satisfied
'69 85'	Conditions of use not satisfied (or reference data not initialized)
'6A 88'	Reference data not found

Table 60 – GET DATA - Status words and Application Final State

[62] [P] indicates the data element is profile dependant

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12.2.5GENERATE APPLICATION CRYPTOGRAM

12.2.5.1 Command Description

The GENERATE APPLICATION CRYPTOGRAM (GENERATE AC) command is the command used by a terminal when requesting confirmation of transaction processing from the ICC. The ICC decision may differ from the terminal request. The card application may either approve the terminal request or be more restrictive than the terminal.

The card application decision is authenticated by an Application Cryptogram (AC). The cryptogram returned may be any one of the cryptograms specified in the following table.

Cryptogram Type	Meaning
Application Authentication Cryptogram (AAC)	Transaction declined
Transaction Certificate (TC)	Transaction approved
Authorization Request Cryptogram (ARQC)	Online authorization requested.

Table 61 - GENERATE AC Cryptogram Types

In a contactless transaction, the terminal may request that the card application confirms terminal decision only one time per transaction. If decision is to process the transaction online, final decision is taken only by the terminal based on Issuer host decision.

When the terminal supports either the “long tap” option or the “online tap” option, the terminal may request that the card application confirms terminal decision after online processing.

The list of transaction-specific information the terminal must communicate to the card are defined in an EMV-defined data element:

- CDOL1: List of transaction-specific information to communicate in 1st GENERATE AC command
- CDOL2: List of transaction-specific information to communicate in 2nd GENERATE AC command

This command is also used when performing the Combined DDA/Application Cryptogram Generation (CDA) function as described in reference document [EMV-2]. In contactless mode, CDA is supported only in 1st GENERATE AC command.

Note: Support of the Combined DDA/Application Cryptogram Generation (CDA) in GENERATE AC command is dependent on the card platform used. An RSA capable card is required.

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12.2.5.2 Input APDU parameters

The input APDU parameters of the GENERATE AC command are coded as defined in following table.

Code	Value
CLA	'80'
INS	'AE'
P1	Reference Control Parameter
P2	'00'
Lc	var
Data	Transaction-related Data: ○ 1st GENERATE AC: CDOL1 related data ○ 2nd GENERATE AC: CDOL2 related data
Le	'00'

Table 62 – GENERATE AC command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
0	0	-	-	-	-	-	-	AAC
0	1	-	-	-	-	-	-	TC
1	0	-	-	-	-	-	-	ARQC
1	1	-	-	-	-	-	-	Value not allowed (SW='6A86')
-	-	x	-	-	-	-	-	RFU('0') Note: The PURE application ignores the value of this bit; Value '1' has no impact.
-	-	-	0	-	-	-	-	CDA signature not requested
-	-	-	1	-	-	-	-	CDA signature requested (supported only with 1st GENERATE AC)
-	-	-	-	x	x	x	x	RFU('0000') Note: The PURE application ignores the value of these bits; Value other than '0000' has no impact.

Table 63 – GENERATE AC command: Reference Control Parameter

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12.2.5.3 Data Field Returned in the Response Message

The data field returned in the response message is constructed as per EMV format 2: a constructed data object with BER-TLV tag equal to '77' containing several (BER-TLV) coded objects.

Tag		Description	Presence
'77'		Response Message Template Format 2	Mandatory
	'9F27'	Cryptogram Information Data (CID)	Mandatory
	'9F36'	Application Transaction Counter (ATC)	Mandatory
	'9F26'	Application Cryptogram (AC)	Mandatory
	'9F10'	Issuer Application Data (IAD)	Mandatory
	'C5'	Contactless Cryptogram Information Data (CCID)	Optional

Table 64 - GENERATE AC Response Data Field, CDA not requested or AAC card decision with CDA requested

Tag		Description	Presence
'77'		Response Message Template Format 2	Mandatory
	'9F27'	Cryptogram Information Data (CID)	Mandatory
	'9F36'	Application Transaction Counter (ATC)	Mandatory
	'9F4B'	Signed Dynamic Application Data	Mandatory
	'9F10'	Issuer Application Data (IAD)	Mandatory
	'C5'	Contactless Cryptogram Information Data (CCID)	Optional

Table 65 - GENERATE AC Response Data Field for TC card decision or ARQC card decision or AAC card decision with CDA requested

Note: Depending on card setting:

- the card generates a Signed Dynamic Application Data signature when CDA is requested and card decision is AAC or.
- the card generates no Signed Dynamic Application Data signature when CDA is requested and card decision is AAC.

Therefore when card decision is AAC, both 1st GENERATE AC response defined in Table 64and in Table 65 are valid.

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The Cryptogram Information Data is coded as specified in reference document [EMV-2] in section related to CCD.

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Cryptogram generated** (based on card decision) 00 : AAC (transaction declined) 01 : TC (transaction approved) 10 : ARQC (request an online authorization) 11 : RFU
-	-	0	0	-	-	-	-	Payment System-specific cryptogram
-	-	-	-	0	-	-	-	No advice required
-	-	-	-	-	0	0	0	No information given

Table 66 - GENERATE AC Cryptogram Information Data

The Contactless Cryptogram Information Data is coded as defined below.

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	-	-	-	-	-	-	**Final Card decision related to transaction completion** (same value as in CID bits 8-7) 00 : Transaction declined 01 : Transaction approved offline 10 : Request to process the transaction online 11 : RFU
-	-	x	x	x	-	-	-	**Transaction processing information** 1xx= Switch to contact interface x1x = Error when processing 1st GENERATE AC (inconsistency between GPO command and 1st GENERATE AC command) xx1 = Additional tap for Issuer Response supported
-	-	-	-	-	x	x	x	**CVM information** 000 = No CVM Required 010 = Online PIN required 001 = Signature required 100 = Confirmation Code required 101 = Confirmation Code verified 011 = Cardholder verification required (it is up to the terminal to decide the CVM to use)

Table 67 - GENERATE AC Contactless Cryptogram Information Data

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12.2.5.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully (ARQC returned during 1st GENERATE AC)
	Command completed successfully (TC/AAC returned during either 1st GENERATE AC or 2nd GENERATE AC)
’67 00’	Wrong Lc (Wrong CDOL data length)
'69 85'	Conditions of use not satisfied
	When format checks on the CDOL related data fail
	When Application State rules are not satisfied
'6A 86'	Incorrect P1-P2 parameters
‘6A 83’	Error when processing CSU bit ‘card block’ or CSU bit ‘application block’ (This error is depending on the card platform and may be due to wrong personalization)

Table 68 - GENERATE AC: Status words and Application Final State

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12.2.6ECHO

12.2.6.1 Command Description

The ECHO command allows a contactless reader to retrieve the last GENERATE AC command response in case of torn contactless transaction.

The ECHO command response corresponds to the last GENERATE AC response.

The ECHO command provides an error in the following cases:

- The previous transaction was not torn
- A new payment transaction was initialized using GET PROCESSING OPTIONS command
- The previous transaction was torn but not on the type of GENERATE AC command requested by the reader
- The reader provides wrong information value (Unpredictable Number or Issuer Authentication Data) related to the torn transaction

The support of this command is optional. When the card payment application supports the ECHO command, it is mandatory to personalize the parameter 'Echo Card Identifier' (tag '9F75') in the response of the selection of the card payment application.

12.2.6.2 Input APDU parameters

The input APDU parameters of the ECHO command are coded as defined in following table.

Code	Value
CLA	'80'
INS	'DF'
P1	Information requested '01' : last 1st GENERATE AC command response '02' : last 2nd GENERATE AC command response
P2	'00'
Lc	Depending on P1 value: P1 = '01' : 4 bytes P1 = '02' : 8 bytes
Data	Depending on P1 value: '01' : Unpredictable Number sent to the last 1st GENERATE AC '02' : 8 most significant bytes of Issuer Authentication Data sent to last 2nd GENERATE AC
Le	'00'

Table 69 – ECHO command

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12.2.6.3 Data Field Returned in the Response Message

The data field returned in the response message is constructed as per EMV format 2: a constructed data object with BER-TLV tag equal to ‘77’ containing several (BER-TLV) coded objects. The coded objects are the same as in the response of the GENERATE AC command (see section 12.2.5.3)

12.2.6.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc
'69 85'	Conditions of use not satisfied
	No information available
	When Application State rules are not satisfied
'6A 86'	Incorrect P1-P2 parameters

Table 70 - ECHO: Status words

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12.2.7PUT DATA (without Secure Messaging)

12.2.7.1 Command Description

The PUT DATA (without Secure Messaging) command is a command that provides the ability to update values of the Proprietary Data Element Tags for which the requirement for Secure Messaging or Offline PIN checking is specifically omitted at personalization time.

12.2.7.2 Input APDU parameters

The input APDU parameters of the PUT DATA command are coded as defined in following table.

Code	Value
CLA	'00'
INS	'DA'
P1	Data object tag **If** tag of the data element to retrieve is on 2 bytes **then** • P1 = most significant byte of the tag value • P2 = least significant byte of the tag value **Else** tag of the data element to retrieve is on 1 byte • P1 = '00' • P2 = Tag value
P2	
Lc	Var. (Data object length)
Data	Data object value
Le	Not Present

Table 71 – PUT DATA (without Secure Messaging) command

PURE Application data elements that may be updated using PUT DATA without Secure Messaging are:

P1P2	Data Element	Length
'DF 0x'	Proprietary Data Element (PDE) with x in the range '1' to 'A'. (only data element for which the requirement for Secure Messaging is specifically omitted at personalization time)	Var

Table 72 – PUT DATA (without Secure Messaging) – PURE defined data elements

There is no format check performed on these data elements.

12.2.7.3 Data Field Returned in the Response Message

No data filed is returned in the response message

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12.2.7.4 Processing State returned in the Response Message

SW1 SW2	Value
‘90 00’	Command completed successfully
’67 00’	Wrong Lc
’69 82’	Invalid security status
‘69 85’	Conditions of use not satisfied
‘6A 88’	Referenced data not found

Table 73 – PUT DATA (without Secure Messaging) – Status words and Application Final State

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12.2.8 APPLICATION BLOCK

12.2.8.1 Command Description

The APPLICATION BLOCK command is an issuer script command that invalidates the currently selected application, as well as any Alias applications linked to the currently selected application.

12.2.8.2 Input APDU parameters

The input APDU parameters of the APPLICATION BLOCK command are coded as defined in following table.

Code	Value	
CLA	'8C'	
INS	'1E'	
P1	'00'	
P2	'00'	
Lc	'06' or '0A' depending on MAC length	
Data	Tag	'8E'
	Length	'04' or '08' depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 74 - APPLICATION BLOCK command

12.2.8.3 Data Field Returned in the Response Message

No data field is returned in the response message

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12.2.8.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc
'69 85'	Conditions of use not satisfied
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
'6A 86'	Incorrect parameters P1-P2

Table 75 - APPLICATION BLOCK: Status words and Application Final State

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12.2.9 APPLICATION UNBLOCK

12.2.9.1 Command Description

The APPLICATION UNBLOCK command is an issuer script command that allows rehabilitation of the currently selected application, as well as any Alias applications linked to the currently selected application.

12.2.9.2 Input APDU parameters

The input APDU parameters of the APPLICATION UNBLOCK command are coded as defined in following table.

Code	Value	
CLA	'8C'	
INS	'18'	
P1	'00'	
P2	'00'	
Lc	'06' or '0A' depending on MAC length	
Data	Tag	'8E'
	Length	'04' or '08' depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 76 - APPLICATION UNBLOCK command

12.2.9.3 Data Field Returned in the Response Message

No data field is returned in the response message

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12.2.9.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc
'69 85'	Conditions of use not satisfied
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
'6A 86'	Incorrect parameters P1-P2

Table 77 - APPLICATION UNBLOCK: Status words and Application Final State

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12.2.10 CARD BLOCK

12.2.10.1 Command Description

The CARD BLOCK command is an issuer script command that allows to permanently disable all the applications in the ICC.

12.2.10.2 Input APDU parameters

The input APDU parameters of the CARD BLOCK command are coded as defined in the following table.

Code	Value	
CLA	‘8C’	
INS	‘16’	
P1	‘00’	
P2	‘00’	
Lc	‘06’ or ‘0A’ depending on MAC length	
Data	Tag	‘8E’
	Length	‘04’ or ‘08’ depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 78 – CARD BLOCK command

12.2.10.3 Data Field Returned in the Response Message

No data field is returned in the response message

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12.2.10.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc
'69 85'	Conditions of use not satisfied
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
'6A 86'	Incorrect parameters P1-P2

Table 79 - CARD BLOCK: Status words and Application Final State

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12.2.11 PIN CHANGE/UNBLOCK

12.2.11.1 Command Description

The PIN CHANGE/UNBLOCK command is an issuer script command that provides the issuer with the ability to either unblock the PIN or to simultaneously change and unblock the reference PIN.

12.2.11.2 Input APDU parameters

The input APDU parameters of the PIN CHANGE/UNBLOCK command are coded as defined in following table.

a) In the case of only unblocking the Offline PIN

Code	Value	
CLA	'8C'	
INS	'24'	
P1	'00'	
P2	'00'	
Lc	'06' or '0A' depending on MAC length	
Data	Tag	'8E'
	Length	'04' or '08' depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 80 - PIN CHANGE/UNBLOCK command – Unblock PIN

If P2 is '00', the reference PIN is unblocked and the PIN Try Counter is reset to the PIN Try Limit. The existing PIN value is retained as the PIN data component is not present.

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b) In the case of changing the Offline PIN value

Code	Value	
CLA	‘8C’	
INS	‘24’	
P1	‘00’	
P2	‘02’	
Lc	‘19’ or ‘1D’ depending on MAC length	
Data	Tag	‘87’
	Length	‘11’
	Value	Padding Indicator (1 byte - value = ‘01’)
		Enciphered PIN Block value (16 bytes)
	Tag	‘8E’
	Length	‘04’ or ‘08’ depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 81 - PIN CHANGE/UNBLOCK command – Change PIN

If P2 is ‘02’, the reference PIN is updated to the new PIN value and the PIN Try Counter is reset to the PIN Try Limit value.

The PIN block format before encryption uses the following structure.

1		2		3		4		5		6		7		8	
C	N	P	P	P	P	P/’F’	P/’F’	P/’F’	P/’F’	P/’F’	P/’F’	P/’F’	P/’F’	‘F’	‘F’

Table 82 – PIN CHANGE/UNBLOCK command - PIN Block Format before encryption

With

Abbreviation	Name	Value
C	Control field	4 bit binary number with value of 0010 (Hex '2')
N	PIN length	4 bit binary number with permissible values of 0100 to 1100 (Hex '4' to 'C')
P	PIN digit	4 bit binary number with permissible values of 0000 to 1001 (Hex '0' to '9')
P/’F’	PIN/filler	Determined by PIN length
‘F’	Filler	4 bit binary number with a value of 1111 (Hex 'F')

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12.2.11.3 Data Field Returned in the Response Message

No data filed is returned in the response message

12.2.11.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc
’69 82’	Invalid decrypted PIN block format [Security Status not satisfied]
'69 85'	Conditions of use not satisfied
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
'6A 86'	Incorrect parameters P1-P2
‘6A 88’	Error code when offline PIN was not personalized

Table 83 - PIN CHANGE/UNBLOCK - Status words and Application Final State

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12.2.12 PUT DATA (with Secure Messaging)

12.2.12.1 Command Description

The PUT DATA (with Secure Messaging) command is an issuer script command that provides the issuer with the ability to update the values of Card Risk Management and other internal data items.

12.2.12.2 Input APDU parameters

The input APDU parameters of the PUT DATA command are coded as defined in following table.

Code	Value	
CLA	'0C'	
INS	'DA'	
P1	Data object tag (see Table 85)	
P2		
Lc	Var. (Data object length + MAC length + 04) (see note)	
Data	Tag	'81'
	Length	Var. (Data object length) Note: When data object length is greater than 127 bytes, the 'Length' is on 2 bytes ('81' 'length value'). In that case Lc = Data object length + MAC length + 05. In the other case, the 'Length' is on 1 byte.
	Data	Data object value
	Tag	'8E'
	Length	'04' or '08' depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 84 – PUT DATA (with Secure Messaging) command

The following table provides the PURE Application data elements that may be updated using PUT DATA. On some data elements, the PUT DATA command performs a data format check.

P1P2	Data Element	Length	Format Check
'00 C9'	Application Contactless Usage	1	binary no format check
'00 82'	Application Interchange Profile *	2	binary no format check
'00 D8'	Consolidated Application File Locator *	Var.	binary no format check

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P1P2	Data Element	Length	Format Check
'00 C1'	Application Control[63]	4	binary no format check
'DF 11'	Currency Conversion Code 1	5	binary no format check
'DF 12'	Currency Conversion Code 2	5	binary no format check
'DF 20'	Card Issuer Action Code – CVM [P]	3	binary no format check
'DF 15'	Card Issuer Action Code – Denial *	3	binary no format check
'DF 16'	Card Issuer Action Code – Default *	3	binary no format check
'DF 17'	Card Issuer Action Code – Online *	3	binary no format check
'DF 1D'	Issuer Application Data Object List *	Var.	binary no format check
'00 CB'	CRM Country Code	2	bcd (n3 left padded with '0') no format check
'00 CD'	CRM Currency Code	2	bcd (n3 left padded with '0') no format check
'DF 0x'	Proprietary Data Element (PDE) with x in the range '1' to 'A'. (only data element for which the requirement for Secure Messaging is specifically selected at personalization time) Note: New length shall not exceed the length defined during card personalization	Var	binary no format check
'DF 10'	Profile Selection Table (PST) Note: New length shall not exceed the maximum length	Var.	binary no format check
'BF2x'	Profile Resource Object (PRO) associated to profile 'x' (with x in the range '0' to 'F') ** Note: New length shall not exceed the maximum length	Var.	binary no format check
'DF 18'	Lower Cumulative Offline Transaction Amount (LCOTA) *	6	bcd (n12) format check
'DF 19'	Upper Cumulative Offline Transaction Amount (UCOTA) *	6	bcd (n12) format check
'DF 1A'	Maximum Single Transaction Amount (MTAL) limit *	6	bcd (n12) format check
'DF 1B'	Lower Consecutive Offline Limit (LCOL) *	1	binary no format check
'DF 1C'	Upper Consecutive Offline Limit (UCOL) *	1	binary

[63] The value of Application Control Byte 4 bit 1 (TTPI presence in PDOL) cannot be updated (SW '6A80').

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P1P2	Data Element	Length	Format Check
			no format check

Table 85 – PUT DATA (with Secure Messaging) – PURE defined data elements

12.2.12.3 Data Field Returned in the Response Message

No data filed is returned in the response message.

12.2.12.4 Processing State returned in the Response Message

SW1 SW2	Value
'90 00'	Command completed successfully
’67 00’	Wrong Lc or length of tagged value
’69 82’	Invalid security status
'69 85'	Conditions of use not satisfied
	When format check is performed and fails
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
‘6A 80’	Wrong value
‘6A 88’	Referenced data not found

Table 86 – PUT DATA (with Secure Messaging) – Status words and Application Final State

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12.2.13 UPDATE RECORD

12.2.13.1 Command Description

The UPDATE RECORD command is an issuer script command that provides the issuer with the ability to update the contents of the linear file records present in the PURE application.

All the PURE application linear files (SFI in the range 1 – 30) that were personalized in the card, may be updated except the transaction log file.

12.2.13.2 Input APDU parameters

The input APDU parameters of the UPDATE RECORD command are coded as defined in following table.

Code	Value	
CLA	'0C'	
INS	'DC'	
P1	Record Number	
P2	Reference Control Parameter	
Lc	Var. (record length + MAC length + 04) (see note)	
Data	Tag	'81'
	Length	Var. (record length) Note: When data object length is greater than 127 bytes, the 'Length' is on 2 bytes ('81' 'length value'). In that case Lc = record length + MAC length + 05. In the other case, the 'Length' is on 1 byte.
	Data	New value of the record
	Tag	'8E'
	Length	'04' or '08' depending on MAC length defined in Application Control parameter
	Value	MAC value
Le	Not Present	

Table 87 – UPDATE RECORD command

b8	b7	b6	b5	b4	b3	b2	b1	Meaning
x	x	x	x	x	-	-	-	SFI in the range 1 – 30
-	-	-	-	-	1	0	0	P1 is a record number

Table 88 – UPDATE RECORD command: Reference Control Parameter

12.2.13.3 Data Field Returned in the Response Message

No data field is returned in the response message

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12.2.13.4 Processing State returned in the Response Message

SW1 SW2	Value
‘90 00’	Command completed successfully
’67 00’	Wrong length for record data
‘69 85’	Conditions of use not satisfied
	When Application State rules are not satisfied
’69 87’	Secure messaging data objects missing
’69 88’	Incorrect secure messaging data objects
‘6A 82’	File not found
‘6A 83’	Record not found
‘6A 86’	Incorrect P1-P2 parameters

Table 89 – UPDATE RECORD - Status words and Application Final State

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